EM411

Project Management

EM411 PROJECT MANAGEMENT

ABET eCOURSEBOOK AY2026

TABLE OF CONTENTS

SECTION I

COURSE ASSESSMENTS

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A AY2026 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-SE 19 December 2025

MEMORANDUM FOR COL Brandon Thompson, Program Director, Department of Systems Engineering, United States Military Academy

SUBJECT: AT26-1 Course Assessment – EM411: Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term (AT) 26-1.

2. POC for this action is the undersigned at daniel.finch@westpoint.edu.

DANIEL FINCH
Senior Instructor
EM411 Course Director

Encls
1. Course Assessment
2. Course Syllabus
3. Course End Feedback

DISTRIBUTION:

1 each EM411 Instructor, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. **Historic Course Offerings and Enrollments by Academic Year Term (AYT).**

a. **Offered.**

AYT 18		AYT 19		AYT 20		AYT 21		AYT 22		AYT 23		AYT 24		AYT 25		AYT 26	
18-1	18-2	19-1	19-1	20-2	18-1	23-1	23-2	22-1	22-2	23-1	23-2	24-1	24-2	25-1	25-2	26-1	26-2
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

b. **Enrollments.**

AYT	Enrollment
26-1	91
25-2	129
25-1	64
24-2	148
24-1	42
23-2	127
23-1	50
22-2	107
22-1	88
21-2	79
21-1	36
20-2	157
20-1	60
19-2	118
19-1	85
18-2	140
18-1	143

c. **Historical Term End Examination (TEE) Grade Averages.**

Academic Term	TEE Average
26-1	79.82%
25-2	83.83%
25-1	81.21%
24-2	79.76%
24-1	79.73%
23-2	83.13%
23-1	82.97%
22-2	84.39%
22-1	82.58%
19-2	86.51%
19-1	87.93%

18-2	86.77%
18-1	86.80%
17-1	85.36%
16-2	83.64%
16-1	85.82%
15-1	85.75%

d. **Historical Course Final Grade Averages.**

Academic Term	EM411 Average
26-1	88.13%
25-2	87.72%
25-1	85.77%
24-2	86.53%
24-1	85.59%
23-2	86.40%
23-1	85.89%
22-2	87.60%
22-1	87.11%
21-2	87.47%
21-1	86.27%
20-2	88.51%
20-1	84.63%
19-2	86.51%
19-1	87.93%
19-1	87.93%
18-1	85.65%
18-2	86.59%
18-1	85.65%

Course Overview.

a. **Applicability.** Project Management (PM) is a vital skill for any leader serving in the military or civilian industry, and greatly enhances an organization's ability to successfully achieve its project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

b. **Scope.** Each cadet will learn the skills necessary to successfully initiate, plan, execute, monitor, and control a project. Topics include Project Selection, Project Manager Roles and Responsibilities, Organizational Structure, Project Planning, Budgeting and Scheduling, Resource Allocation, Monitoring and Controlling, Risk Assessment and Response Management, and Evaluation and Termination. A semester-long project will reinforce course concepts and provide cadets the opportunity to integrate PM software into the development of a project plan. Cadets will spend several lessons with project management software to ensure they are aware of technology that can be applied.

Course Strategy.

1. EM411 is divided into four major blocks that align with the lifecycle of a successful project.

i) Block 1: *Project Initiation* covers project management organization, roles and responsibilities, project selection methods, and stakeholder analysis and management.

ii) Block 2: *Project Planning* focuses on techniques used to define the project's scope and required work, create a project budget and schedule, improve cost and duration estimates, and explore the issues related to project scheduling, resourcing, and risk management.

iii) Block 3: *Project Execution* introduces project monitoring, controlling, auditing, and termination.

iv) Block 4: The *Agile Project Management* block will teach how to apply agile approaches, specifically Scrum, to projects.

2. A group project is assigned to reinforce and apply learned concepts and to bridge the gap between theory and practical application. Microsoft Project will be used to develop project management skills and increase Cadet productivity.

3. All lesson objectives are referenced with the Project Management Body of Knowledge (PMBOK).

d. **Endstate.** Each Cadet understands the application of project management and the complex interrelated tasks associated with completing projects on time, within budget, and to specification.

Section II – Course Assessment

Achievement of Course Objectives. The course continues to accomplish the six course objectives included in the table below. Emphasis remains the ability to successfully achieve project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

1. Identify project management practices used in current organizations.
2. Identify and describe the necessary qualities and skills required of successful project managers and their teams.
3. Successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.
4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the PMBOK.

5. Utilize Project Management software in planning and managing successful projects.
6. Exercise oral and written communication skills.

1. **Course Content.**

a. **Body of Knowledge.** The course achieved the initially stated desires and intent in a holistic analysis. The course objectives covered the disciplined-accepted body of knowledge (Project Management Institute or PMI) is relevant to an undergraduate course in project management.

b. **Effective Assessment.** The combination of problem sets, quizzes, WPRs, problem sets, and Stage Gates worked well for assessing student understanding of the material. Frequent checks on learning were insightful in allowing instructors to validate assumptions about students' ability to apply fundamental course concepts. As such, I highly recommend retaining regular checks on learning, quizzes, and problem sets, even if for minimal point values, to lead students to become familiar with the material and to validate instructor assumptions about student retention.

2. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

b. **Software.** Cadets use Microsoft Project 2021

c. **Graded Requirements.** The course was composed of 11 graded events totaling 1000 points. With the exclusion of instructor points, 800 of the course's 950 points (84%) were assessed using individual events.

Graded Event	***Type***	***Points***	***% of Course***
Problem Set 1	Individual	75	7.5%
Problem Set 2	Individual	75	7.5%
Quiz 1	Individual	50	5.0%
Quiz 2	Individual	50	5.0%
Project Deliverable SG1 - Project Selection Analysis	Individual	75	7.5%
Project Deliverable SG2 - Project Plan	Group	90	9.0%
Project Deliverable SG3 - Project Progress Report	Group	60	6.0%
Project Deliverable SG4 – Agile Project Plan	Individual	75	7.5%
WPR - Midterm	Individual	175	17.5%
Term End Exam	Individual	225	22.5%
Instructor Points – Checks on Learning/Reading	Individual	50	5.0%
Course Total:	1000	1000	

3. **Course Pedagogy.**

a. **Problem Sets (150 total points).** There will be two problem sets due over the semester. Problem Set 1 (75 points) will cover project initiation, cost estimation, and improving cost estimation. Problem Set 2 (75 points) will project scheduling, expediting a project, and resource loading and leveling.

b. **Quizzes (100 Points).** There will be two 50-point quizzes that will prepare students for the TEE.

c. **Course Project (300 Points).** Cadets will complete a semester-long course project comprising four stage gate deliverables: Project Selection Analysis, Project Plan, and Project Progress Report, and Project Agile Plan (worth 75, 90, 60, and 75 points, respectively). The second and third stage gates will be completed in small groups, while the first and fourth will be completed individually.

d. **Midterm Examination (175 Points)**. One 70-minute Written Partial Review (WPR) covering specified blocks of lessons as per the attached enclosure.

e. **Term End Examination (225 points).** One 3.5-hour Term End Examination (TEE) covers all course material. Points will be awarded based on the Cadet's ability to demonstrate their project management skills and knowledge acquired throughout the course.

f. **Instructor Points (50 points).** Instructor points will be awarded through in-class, pre-lesson quizzes, participation and conduct, peer evaluation feedback, and any additional assignments the instructors deem necessary to increase learning.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to the Current Academic Term (based on previous Course Assessments).**

a. **Course Objectives.** None

b. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

c. **Curriculum.** There were no significant changes to the course curriculum or structure during this term.

d. **Graded Requirements.** There were no changes to graded requirements from the previous semester.

e. **Other.** None.

2. **Proposed Changes for Next Academic Term.**

a. **Course Objectives.** No change.

b. **Textbook.** No change

c. **Curriculum.** Recommended changes for AT26-2 include slightly reducing the time effort of the TEE by adjusting the expediting a project question to two crash iterations rather than three. Consideration should be given to modifying the Agile Project Management Block to reduce redundancy between lessons. It is also recommended to integrate an Agile Lego exercise to reinforce concepts through practical application.

d. **Graded Requirements.** No change

e. **Other.** None

Day 1 Schedule

Enclosure 2: Course Syllabus

Block	LSN	Date	Lesson Name	Notes	Reading Sections
Project Initiation			**Project Initiation**		
	1	18-Aug	Definition of Project Management	Issue MS Project Template	1.1-1.3 (pp. 1-21) Optional: 1.5 (pp.26-28)
	2	20-Aug	Project Selection - I		2.1-2.2 (pp. 35-50)
	3	25-Aug	Project Selection - II	Course Textbook Check; SG 1 Launch	2.2-2.3 (pp. 50-60) *Stop at "8 Steps of the PPM Process"
	4	27-Aug	The PM, the PM Profession, and Stakeholder Mgmt		3.1-3.4 (pp. 83-110) 4.1-4.5 (pp. 119-136)
		29-Aug	Class Drop 1		
	5	3-Sep	Project Organization and Teams		5.1-5.8 (pp. 149-180) *Skip "Agile Team Roles" (pp. 174-175)
	6	5-Sep	PM Ethics and Quiz #1	Review and In-Class Quiz, LSN 1-6	Handouts
Project Planning			**Project Planning**		
	7	8-Sep	Project Planning	PS1 Launch	6.1 (pp. 193-212) 6.3 (pp. 221-224)
	8	10-Sep	Including Risk in Project Management	Project SG #1 Due by 2359 12 Sep; MS Project Template CHECKED START LSN 8	7.3-7.4 (pp. 261-273) *Stop at "General Simulation Analysis"
	9	15-Sep	Budgeting and Cost Estimation	SG2 Launch	7.1 (pp. 239-250) *Stop at "Budgeting with Agile"
	10	17-Sep	Improving Cost Estimates		7.2 (pp.251-261)
		19-Sep	Class Drop 2		
	11	22-Sep	Scheduling I - An Intro to PERT/CPM		8.1-8.2 (pp. 297-308) *Stop at "Calculating Activity Times"
	12	24-Sep	Scheduling II - Solving the Network		8.2 (pp. 308-318)
	13	29-Sep	Scheduling III - Uncertainty in a Project	Problem Set #1 Due by 2359 29 Sep; PS2 Launch	8.2 (pp. 318-325)
	14	1-Oct	**Midterm Review**		Review lessons 1-15
		3-Oct	Class Drop 3		
	15	6-Oct	Midterm	Midterm WPR, LSNs 1-15	
	16	8-Oct	MS Project Introduction Lab		MS Project step-by-step
	17	14-Oct	Expediting a Project		9.1 (pp. 356-364)
	18	16-Oct	Resource Loading and Leveling		9.2-9-4 (pp. 364-374)
	19	20-Oct	Allocating Scarce Resources & Multi-Project Scheduling in MS Project		9.5 (PP. 374-379)
	20	22-Oct	SG#2 Stakeholder Update/Baseline Schedule Brief	Problem Set #2 Due by 2359 22 Oct	
		24-Oct	Class Drop 4		
	21	27-Oct	MS Project Practice Lab		MS Project step-by-step
	22	29-Oct	Stage Gate #2 Working Session	Project SG #2 Due by 2359 29 Oct	
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	23	3-Nov	Monitoring a Project	SG3 Launch	10.1-10.2 (pp.397-409)
	24	5-Nov	Earned Value Management (EVM) I		10.3 (pp. 409-421)
	25	7-Nov	EVM II - LEGO Lab		
	26/27	13-Nov	Project Control & Auditing / EVM III -- Problem Solving Lab	Project SG #3 Due by 2359 13 Nov	12.1-12.5 (pp. 468-488)/ 11.1-11.4 (pp. 433-459)
	28	17-Nov	Project Termination		13.1-13.4 (pp. 499-517)
	29	19-Nov	Quiz #2	In-Class Quiz, LSNs 19-32	
Agile PM			**Agile Project Management**		
	30	24-Nov	Introduction to Agile	SG4 Launch	1.4 (pp.21-26), Agile Team Roles (pp. 174-175)
		26-Nov	Class Drop 5		
	31	2-Dec	Agile Planning and Scheduling		6.2 (pp. 213-221) 7.1 (pp. 250-251) 8.5 (pp. 339-341)
	32	4-Dec	Agile Project Execution		10.4 (pp.421-423)
	33	8-Dec	Agile Project Monitoring, Control, and Closure		11.3 (pp. 454-455), 13.2 (pp. 508); review p.423
	34	10-Dec	SG4 Working Session	Project SG #4 due by 2359 10 Dec	
	35	12-Dec	**Course Closeout & TEE Review**		Review LSNs 1-35
		15-Dec	Class Drop 6		
TEE					

Legend:
Graded Event
Notes / Special Event
Lab
Class Drop

Day 2 Schedule

Block	LSN	Date	Lesson Name	Notes	Reading Sections
Project Initiation			**Project Initiation**		
	1	19-Aug	Definition of Project Management	Issue MS Project Template	1.1-1.3 (pp. 1-21) Optional: 1.5 (pp.26-28)
	2	21-Aug	Project Selection - I		2.1-2.2 (pp. 35-50)
		22-Aug	Class Drop 1		
	3	26-Aug	Project Selection - II	Course Textbook Check; SG1 Launch	2.2-2.3 (pp. 50-60) *Stop at "8 Steps of the PPM Process"
	4	28-Aug	The PM, the PM Profession, and Stakeholder Mgmt		3.1-3.4 (pp. 83-110) 4.1-4.5 (pp. 119-136)
	5	2-Sep	Project Organization and Teams		5.1-5.8 (pp. 149-180) *Skip "Agile Team Roles" (pp. 174-175)
.	6	4-Sep	PM Ethics and Quiz #1	Review and In-Class Quiz, LSN 1-6	Handouts
Project Planning			**Project Planning**		
	7	9-Sep	Project Planning	PS1 Launch - 8 Sep	6.1 (pp. 193-212) 6.3 (pp. 221-224)
	8	11-Sep	Including Risk in Project Management	Project SG #1 Due by 2359 12 Sep; MS Project Template CHECKED START LSN 8	7.3-7.4 (pp. 261-273) *Stop at "General Simulation Analysis"
		12-Sep	Class Drop 2		
	9	16-Sep	Budgeting and Cost Estimation	SG2 Launch - 15 Sep	7.1 (pp. 239-250) *Stop at "Budgeting with Agile"
	10	18-Sep	Improving Cost Estimates		7.2 (pp.251-261)
	11	23-Sep	Scheduling I - An Intro to PERT/CPM		8.1-8.2 (pp. 297-308) *Stop at "Calculating Activity Times"
	12	25-Sep	Scheduling II - Solving the Network		8.2 (pp. 308-318)
		26-Sep	Class Drop 3		
	13	30-Sep	Scheduling III - Uncertainty in a Project	Problem Set #1 Due by 2359 29 Sep; PS2 Launch - 29 Sep	8.2 (pp. 318-325)
	14	2-Oct	**Midterm Review**		Review lessons 1-15
	15	7-Oct	Midterm	Midterm WPR, LSNs 1-15	
	16	9-Oct	MS Project Introduction Lab		MS Project step-by-step
		10-Oct	Class Drop 4		
	17	15-Oct	Expediting a Project		9.1 (pp. 356-364)
	18	17-Oct	Resource Loading and Leveling		9.2-9.4 (pp. 364-374)
	19	21-Oct	Allocating Scarce Resources & Multi-Project Scheduling in MS Project		9.5 (PP. 374-379)
	20	23-Oct	SG#2 Stakeholder Update/Baseline Schedule Brief	Problem Set #2 Due by 2359 22 Oct	
	21	28-Oct	MS Project Practice Lab		MS Project step-by-step
	22	30-Oct	Stage Gate #2 Working Session	Project SG #2 Due by 2359 29 Oct	
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	23	31-Oct	Monitoring a Project	SG3 Launch	10.1-10.2 (pp. 397-409)
	24	4-Nov	Earned Value Management (EVM) I		10.3 (pp. 409-421)
	25	6-Nov	EVM II - LEGO Lab		
	26	12-Nov	EVM III -- Problem Solving Lab		11.1-11.4 (pp. 433-459) *Skip "Scrum Events for Project
	27	14-Nov	Project Control & Auditing	Project SG #3 Due by 2359 13 Nov	12.1-12.5 (pp. 468-488)
	28	18-Nov	Project Termination		13.1-13.4 (pp. 499-517)
	29	20-Nov	Quiz #2	In-Class Quiz, LSNs 19-32	
Agile PM			**Agile Project Management**		
		21-Nov	Class Drop 5		
	30	25-Nov	Introduction to Agile	SG4 Launch - 24 Nov	1.4 (pp.21-26), Agile Team Roles (pp. 174-175)
	31	3-Dec	Agile Planning and Scheduling		6.2 (pp. 213-221) 7.1 (pp. 250-251) 8.5 (pp. 339-341)
	32	5-Dec	Agile Project Execution		10.4 (pp.421-423)
	33	9-Dec	Agile Project Monitoring, Control, and Closure		11.3 (pp. 454-455), 13.2 (pp. 508); review p.423
	34	11-Dec	SG4 Working Session / Course Closeout & TEE Review	Project SG #4 due by 2359 10 Dec	Review LSNs 1-35
TEE					

Legend:
Graded Event
Notes / Special Event
Lab
Class Drop

Enclosure 3: Course End Feedback

Feedback administered via Canvas, will be available for review later

TAB B AY2025 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-SE 17 May 2025

MEMORANDUM FOR COL Brandon Thompson, Program Director, Department of Systems Engineering, United States Military Academy

SUBJECT: AT25-2 Course Assessment – EM411: Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term (AT) 25-2.

2. POC for this action is the undersigned at daniel.finch@westpoint.edu.

DANIEL FINCH
Senior Instructor
EM411 Course Director

Encls
1. Course Assessment
2. Course Syllabus
3. Course End Feedback

DISTRIBUTION:

1 each EM411 Instructor, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. **Historic Course Offerings and Enrollments by Academic Year Term (AYT).**

a. **Offered.**

AYT 18		AYT 19		AYT 20		AYT 21		AYT 22		AYT 23		AYT 24		AYT 25	
18-1	18-2	19-1	19-1	20-2	18-1	23-1	23-2	22-1	22-2	23-1	23-2	24-1	24-2	25-1	25-2
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

b. **Enrollments.**

AYT	Enrollment
25-2	129
25-1	64
24-2	148
24-1	42
23-2	127
23-1	50
22-2	107
22-1	88
21-2	79
21-1	36
20-2	157
20-1	60
19-2	118
19-1	85
18-2	140
18-1	143

c. **Historical Term End Examination (TEE) Grade Averages.**

Academic Term	TEE Average
25-2	83.83%
25-1	81.21%
24-2	79.76%
24-1	79.73%
23-2	83.13%
23-1	82.97%
22-2	84.39%
22-1	82.58%
19-2	86.51%
19-1	87.93%
18-2	86.77%
18-1	86.80%
17-1	85.36%
16-2	83.64%

16-1	85.82%
15-1	85.75%

d. **Historical Course Final Grade Averages.**

Academic Term	EM411 Average
25-2 (Current)	87.72%
25-1	85.77%
24-2	86.53%
24-1	85.59%
23-2	86.40%
23-1	85.89%
22-2	87.60%
22-1	87.11%
21-2	87.47%
21-1	86.27%
20-2	88.51%
20-1	84.63%
19-2	86.51%
19-1	87.93%
19-1	87.93%
18-1	85.65%
18-2	86.59%
18-1	85.65%

Course Overview.

a. **Applicability.** Project Management (PM) is a vital skill for any leader serving in the military or civilian industry, and greatly enhances an organization's ability to successfully achieve its project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

b. **Scope.** Each cadet will learn the skills necessary to successfully initiate, plan, execute, monitor, and control a project. Topics include Project Selection, Project Manager Roles and Responsibilities, Organizational Structure, Project Planning, Budgeting and Scheduling, Resource Allocation, Monitoring and Controlling, Risk Assessment and Response Management, and Evaluation and Termination. A semester-long project will reinforce course concepts and provide cadets the opportunity to integrate PM software into the development of a project plan. Cadets will spend several lessons with project management software to ensure they are aware of technology that can be applied.

Course Strategy.

1. EM411 is divided into four major blocks that align with the lifecycle of a successful project.

i) Block 1: *Project Initiation* covers project management organization, roles and responsibilities, project selection methods, and stakeholder analysis and management.

ii) Block 2: *Project Planning* focuses on techniques used to define the project's scope and required work, create a project budget and schedule, improve cost and duration estimates, and explore the issues related to project scheduling, resourcing, and risk management.

iii) Block 3: *Project Execution* introduces project monitoring, controlling, auditing, and termination.

iv) Block 4: The *Agile Project Management* block will teach how to apply agile approaches, specifically Scrum, to projects.

2. A group project is assigned to reinforce and apply learned concepts and to bridge the gap between theory and practical application. Microsoft Project will be used to develop project management skills and increase Cadet productivity.

3. All lesson objectives are referenced with the Project Management Body of Knowledge (PMBOK).

d. **Endstate.** Each Cadet understands the application of project management and the complex interrelated tasks associated with completing projects on time, within budget, and to specification.

Section II – Course Assessment

Achievement of Course Objectives. The course continues to accomplish the six course objectives included in the table below. Emphasis remains the ability to successfully achieve project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

1. Identify project management practices used in current organizations.
2. Identify and describe the necessary qualities and skills required of successful project managers and their teams.
3. Successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.
4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the PMBOK.
5. Utilize Project Management software in planning and managing successful projects.
6. Exercise oral and written communication skills.

1. **Course Content.**

a. **Body of Knowledge.** The course achieved the initially stated desires and intent in a holistic analysis. The course objectives covered the disciplined-accepted body of knowledge (Project Management Institute or PMI) is relevant to an undergraduate course in project management.

b. **Effective Assessment.** The combination of problem sets, quizzes, WPRs, problem sets, and Stage Gates worked well for assessing student understanding of the material. Frequent checks on learning were insightful in allowing instructors to validate assumptions about students' ability to apply fundamental course concepts. As such, I highly recommend retaining regular checks on learning, quizzes, and problem sets, even if for minimal point values, to lead students to become familiar with the material and to validate instructor assumptions about student retention.

2. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

b. **Software.** Cadets use Microsoft Project 2021

c. **Graded Requirements.** The course was composed of 11 graded events totaling 1000 points. With the exclusion of instructor points, 800 of the course's 950 points (84%) were assessed using individual events.

Graded Event	***Type***	***Points***	***% of Course***
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Problem Set 2	Individual	75	7.5%
Quiz 1	Individual	50	5.0%
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Project Deliverable SG1 - Project Selection Analysis	Individual	75	7.5%
Project Deliverable SG2 - Project Plan	Group	90	9.0%
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Project Deliverable SG4 – Agile Project Plan	Individual	75	7.5%
WPR - Midterm	Individual	175	17.5%
Term End Exam	Individual	225	22.5%
Instructor Points – Checks on Learning/Reading	Individual	50	5.0%
Course Total:	1000	1000	

3. **Course Pedagogy.**

a. **Problem Sets (150 total points).** There will be two problem sets due over the semester. Problem Set 1 (75 points) will cover project initiation, cost estimation, and improving cost estimation. Problem Set 2 (75 points) will project scheduling, expediting a project, and resource loading and leveling.

b. **Quizzes (100 Points).** There will be two 50-point quizzes that will prepare students for the TEE.

c. **Course Project (300 Points).** Cadets will complete a semester-long course project comprising four stage gate deliverables: Project Selection Analysis, Project Plan, and Project Progress Report, and Project Agile Plan (worth 75, 90, 60, and 75 points, respectively). The second and third stage gates will be completed in small groups, while the first and fourth will be completed individually.

d. **Midterm Examination (175 Points)**. One 70-minute Written Partial Review (WPR) covering specified blocks of lessons as per the attached enclosure.

e. **Term End Examination (225 points).** One 3.5-hour Term End Examination (TEE) covers all course material. Points will be awarded based on the Cadet's ability to demonstrate their project management skills and knowledge acquired throughout the course.

f. **Instructor Points (50 points).** Instructor points will be awarded through in-class, pre-lesson quizzes, participation and conduct, peer evaluation feedback, and any additional assignments the instructors deem necessary to increase learning.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to the Current Academic Term (based on previous Course Assessments).**

a. **Course Objectives.** None

b. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

c. **Curriculum.** There were no significant changes to the course curriculum or structure during this term.

d. **Graded Requirements.** There were no changes to graded requirements from the previous semester.

e. **Other.** None.

2. **Proposed Changes for Next Academic Term.**

a. **Course Objectives.** No change.

b. **Textbook.** No change

c. **Curriculum.** Recommended changes for AT26-1 include slightly reducing the time effort of the TEE by adjusting the expediting a project question to two crash iterations rather than three. Consideration should be given to adding a data visualization lesson objective for Earned Value Management and Stage Gate 3 of the course project. The goal of the lesson objective is to develop the ability to transform raw data into accessible visual formats for clear communication and analysis. Additionally, it is recommended that the use of Gradescope be adopted for course grading. Gradescope allows for more efficient grading of course graded events, allowing more dedicated instructor time to cadet development.

d. **Graded Requirements.** No change

e. **Other.** None

Enclosure 2: Course Syllabus

Block	LSN	Date	Lesson Name	Notes	Reading Sections
Project Initiation			**Project Initiation**		
	1	7-Jan	Definition of Project Management	Issue MS Project Template	1.1-1.3 (pp. 1-21) Optional: 1.5 (pp.26-28)
		9-Jan	Class Drop 1		
	2	13-Jan	Project Selection - I		2.1-2.2 (pp. 35-50)
	3	15-Jan	Project Selection - II	Course Textbook Check; SG1 Launch	2.2-2.3 (pp. 50-60) *Stop at "8 Steps of the PPM Process"
	4	21-Jan	The PM, the PM Profession, and Stakeholder Mgmt		3.1-3.4 (pp. 83-110) 4.1-4.5 (pp. 119-136)
	5	23-Jan	Project Organization and Teams		5.1-5.8 (pp. 149-180) *Skip "Agile Team Roles" (pp. 174-175)
	6	27-Jan	PM Ethics and Quiz	Review and In-Class Quiz, LSN 1-6	Handouts
Project Planning			**Project Planning**		
	7	29-Jan	Project Planning	PS1 Launch	6.1 (pp. 193-212) 6.3 (pp. 221-224)
	8	31-Jan	Including Risk in Project Management	Project SG #1 Due by 2359 31 Jan; MS Project Template CHECKED START LSN 8	7.3-7.4 (pp. 261-273) *Stop at "General Simulation Analysis"
	9	3-Feb	Budgeting and Cost Estimation	SG2 Launch	7.1 (pp. 239-250) *Stop at "Budgeting with Agile"
	10	5-Feb	Improving Cost Estimates		7.2 (pp.251-261)
	11	10-Feb	Scheduling I - An Intro to PERT/CPM		8.1-8.2 (pp. 297-308) *Stop at "Calculating Activity Times"
	12	12-Feb	Scheduling II - Solving the Network		8.2 (pp. 308-318)
		14-Feb	Class Drop 2		
	13	18-Feb	Scheduling III - Uncertainty in a Project	Problem Set #1 Due by 2359 18 Feb; PS2 Launch	8.2 (pp. 318-325)
	14	20-Feb	Midterm Review		Review lessons 1-15
	15	24-Feb	Midterm	Midterm WPR, LSNs 1-15	
	16	26-Feb	MS Project Introduction Lab		MS Project step-by-step
		3-Mar	Class Drop 3		
	17	5-Mar	Expediting a Project		9.1 (pp. 356-364)
	18	7-Mar	Resource Loading and Leveling		9.2-9-4 (pp. 364-374)
	19	10-Mar	Allocating Scarce Resources & Multi-Project Scheduling in MS Project		9.5 (PP. 374-379)
	20	12-Mar	SG#2 Stakeholder Update/Baseline Schedule Brief	Problem Set #2 Due by 2359 12 Mar	
		14-Mar	Class Drop 4		
	21	24-Mar	MS Project Practice Lab		MS Project step-by-step
	22	26-Mar	Stage Gate #2 Working Session	Project SG #2 Due by 2359 26 Mar	
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	23	31-Mar	Monitoring a Project	SG3 Launch	10.1-10.2 (pp.397-409)
	24	2-Apr	Earned Value Management (EVM) I		10.3 (pp. 409-421)
	25	7-Apr	EVM II - LEGO Lab		
		9-Apr	Class Drop 5		
	26	11-Apr	EVM III -- Problem Solving Lab		11.1-11.4 (pp. 433-459) *Skip "Scrum Events for Project
	27	14-Apr	Project Control & Auditing	Project SG #3 Due by 2359 14 Apr	12.1-12.5 (pp. 468-488)
	28	16-Apr	Project Termination		13.1-13.4 (pp. 499-517)
	29	22-Apr	Quiz	In-Class Quiz, LSNs 19-32	
Agile PM			**Agile Project Management**		
	30	25-Apr	Introduction to Agile	SG4 Launch	1.4 (pp.21-26), Agile Team Roles (pp. 174-175)
	31	28-Apr	Agile Planning and Scheduling		6.2 (pp. 213-221) 7.1 (pp. 250-251) 8.5 (pp. 339-341)
	32	30-Apr	Agile Project Execution		10.4 (pp.421-423)
	33	5-May	Agile Project Monitoring, Control, and Closure | SG4 Working Session		11.3 (pp. 454-455), 13.2 (pp.508); review p.423
	34	7-May	Waterfall and Agile in the Defense Industry Guest Speaker	Project SG #4 due by 2359 7 May	
	35	9-May	Course Closeout & TEE Review		Review LSNs 1-39
TEE					

Legend:
Graded Event
Notes / Special Event
Lab
Class Drop

Enclosure 3: Course End Feedback

Feedback administered via Canvas, will be available for review later

TAB C AY2024 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-SE 17 May 2024

MEMORANDUM FOR COL James Schreiner, Program Director, Department of Systems Engineering, United States Military Academy

SUBJECT: AT24-2 Course Assessment – EM411: Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term (AT) 24-2.

2. POC for this action is the undersigned at sam.yoo@westpoint.edu.

SAM M. YOO
MAJ, AV/51A
EM411 Course Director

Encls
1. Course Assessment
2. Course Syllabus
3. Course End Feedback

DISTRIBUTION:

1 each EM411 Instructor, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. **Historic Course Offerings and Enrollments by Academic Year Term (AYT).**

a. **Offered.**

AYT 18		AYT 19		AYT 20		AYT 21		AYT 22		AYT 23		AYT 24	
18-1	18-2	19-1	19-1	20-2	18-1	23-1	23-2	22-1	22-2	23-1	23-2	24-1	24-2
X	X	X	X	X	X	X	X	X	X	X	X	X	X

b. **Enrollments.**

AYT	Enrollment
24-2	148
24-1	42
23-2	127
23-1	50
22-2	107
22-1	88
21-2	79
21-1	36
20-2	157
20-1	60
19-2	118
19-1	85
18-2	140
18-1	143

c. **Historical Term End Examination (TEE) Grade Averages.**

Academic Term	TEE Average
24-2	79.76%
24-1	79.73%
23-2	83.13%
23-1	82.97%
22-2	84.39%
22-1	82.58%
19-2	86.51%
19-1	87.93%
18-2	86.77%
18-1	86.80%
17-1	85.36%
16-2	83.64%
16-1	85.82%
15-1	85.75%

d. **Historical Course Final Grade Averages.**

Academic Term	EM411 Average
24-2 (Current)	86.53%
24-1	85.59%
23-2	86.40%
23-1	85.89%
22-2	87.60%
22-1	87.11%
21-2	87.47%
21-1	86.27%
20-2	88.51%
20-1	84.63%
19-2	86.51%
19-1	87.93%
19-1	87.93%
18-1	85.65%
18-2	86.59%
18-1	85.65%

Course Overview.

a. **Applicability.** Project Management (PM) is a vital skill for any leader serving in the military or civilian industry, and greatly enhances an organization's ability to successfully achieve its project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

b. **Scope.** Each cadet will learn the skills necessary to successfully initiate, plan, execute, monitor, and control a project. Topics include Project Selection, Project Manager Roles and Responsibilities, Organizational Structure, Project Planning, Budgeting and Scheduling, Resource Allocation, Monitoring and Controlling, Risk Assessment and Response Management, and Evaluation and Termination. A visit to Sikorsky, and a discussion panel with Project Managers will illustrate the real-world application of PM techniques and procedures and will provide context to the topics of this course. A semester-long project will reinforce course concepts and provide cadets the opportunity to integrate PM software into the development of a project plan. Cadets will spend several lessons with project management software to ensure they are aware of technology that can be applied.

Course Strategy.

1. EM411 is divided into four major blocks that align with the lifecycle of a successful project.

i) Block 1: *Project Initiation* covers project management organization, roles and responsibilities, project selection methods, and stakeholder analysis and management.

ii) Block 2: *Project Planning* focuses on techniques used to define the project's scope and required work, create a project budget and schedule, improve cost and duration estimates, and explore the issues related to project scheduling, resourcing, and risk management.

iii) Block 3: *Project Execution* introduces project monitoring, controlling, auditing, and termination.

iv) Block 4: The *Agile Project Management* block will teach how to apply agile approaches, specifically Scrum, to projects.

2. A group project is assigned to reinforce and apply learned concepts and to bridge the gap between theory and practical application. Microsoft Project will be used to develop project management skills and increase Cadet productivity.

3. All lesson objectives are referenced with the Project Management Body of Knowledge (PMBOK).

d. **Endstate.** Each Cadet understands the application of project management and the complex interrelated tasks associated with completing projects on time, within budget, and to specification.

Section II – Course Assessment

Achievement of Course Objectives. The course continues to accomplish the six course objectives included in the table below. Emphasis remains the ability to successfully achieve project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

1. Identify project management practices used in current organizations.
2. Identify and describe the necessary qualities and skills required of successful project managers and their teams.
3. Successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.
4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the PMBOK.
5. Utilize Project Management software in planning and managing successful projects.
6. Exercise oral and written communication skills.

1. **Course Content.**

a. **Body of Knowledge.** The course achieved the initially stated desires and intent in a holistic analysis. The course objectives covered the disciplined-accepted body of knowledge (Project Management Institute or PMI) is relevant to an undergraduate course in project management.

b. **Effective Assessment.** The combination of problem sets, lesson knowledge check quizzes (Canvas), WPRs, problem sets, and the Stage Gates worked well for assessing student understanding of the material. The frequent checks on learning were insightful in allowing instructors to validate assumptions about students' ability to apply fundamental course concepts. As such, I highly recommend retaining regular checks on learning, quizzes, and problem sets, even if for minimal point values, to lead students to become familiar with the material and to validate instructor assumptions about student retention.

2. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

b. **Software.** Cadets use Microsoft Project 2021

c. **Graded Requirements.** The course was composed of 11 graded events totaling 1000 points. With the exclusion of instructor points, 800 of the course's 950 points (84%) were assessed using individual events.

Graded Event	***Type***	***Points***	***% of Course***
Problem Set 1	Individual	80	8.0%
Problem Set 2	Individual	80	8.0%
Quiz	Individual	90	9.0%
Project Deliverable SG1 - Project Selection Analysis	Individual	75	7.5%
Project Deliverable SG2 - Project Plan	Group	90	9.0%
Project Deliverable SG3 - Project Progress Report	Group	60	6.0%
Project Deliverable SG4 – Agile Project Plan	Individual	75	7.5%
WPR - Midterm	Individual	150	15.0%
Term End Exam	Individual	200	20.0%
Lesson Knowledge Checks	Individual	60	6.0%
Instructor Points	Individual	40	4.0%
Course Total:		1000	1000

3. **Course Pedagogy.**

a. **Problem Sets (160 total points).** There will be two problem sets due over the semester. Problem Set 1 (80 points) will cover project initiation and planning. Problem Set 2 (80 points) will cover cost estimation, scheduling, loading, and leveling.

b. **Quizzes (80 Points).** There will be one 90-point quiz that will prepare students for the TEE.

c. **Course Project (300 Points).** Cadets will complete a semester-long course project comprising four stage gate deliverables: Project Selection Analysis, Project Plan, and Project Progress Report, and Project Agile Plan (worth 75, 90, 60, and 75 points, respectively). The second and third stage gates will be completed in small groups, while the first and fourth will be completed individually.

d. **Midterm Examination (150 Points)**. One 70-minute Written Partial Review (WPR) covering specified blocks of lessons as per the attached enclosure.

e. **Term End Examination (200 points).** One 3.5-hour Term End Examination (TEE) covers all course material. Points will be awarded based on the Cadet's ability to demonstrate their project management skills and knowledge acquired throughout the course.

f. **Lesson Knowledge Checks (60 points).** Cadets will take multiple knowledge checks throughout the semester, including in-class short quizzes and practical exercise assignments.

g. **Instructor Points (40 points).** Instructor points will be awarded through in-class, pre-lesson quizzes, participation and conduct, peer evaluation feedback, and any additional assignments the instructors deem necessary to increase learning.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to the Current Academic Term (based on previous Course Assessments).**

a. **Course Objectives.** None

b. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

c. **Curriculum.** The major change this term involved revising the Agile block of instruction. This involved transitioning the Agile course from the use of Jira to simpler tools using Microsoft products to focus on the delivery of concepts versus learning a new software tool. This change left all previous core material tied to the traditional PM concepts in place.

d. **Graded Requirements.** The number of lesson knowledge checks were doubled from the previous semester (30 to 60 points), and fully administered online via Canvas.

Additionally, the WPR and Quiz were both administered with Canvas. Points were removed from the TEE (230 to 200 points) so that more points were awarded evenly throughout the course. SG4 was revised as an Agile assignment heavily using Microsoft-based products such as Excel and PowerPoints instead of Jira.

e. **Other.** None.

2. **Proposed Changes for Next Academic Term.**

a. **Course Objectives.** No change.

b. **Textbook.** No change

c. **Curriculum.** With the major revision of the Agile Block of PM this term, the continued refinement of all Agile lessons, practical exercises, and SG4 is recommended. The core Agile material is built but can be further improved with additional real-world examples and PEs to help Cadets better relate to the material. The major adjustment was removing the Jira-based software aspects and focusing on simpler Microsoft-based tools to illustrate Agile/Scrum. It is also recommended to incorporate an Agile Case study as a primer after the Intro to Agile lesson. Videos were also incorporated into this semester's curriculum, which can continue to be refined as suitable material (particularly with real-life examples can be found). Lastly, it should be explicitly clarified that any note cards/ sheets used for graded events must be handwritten in ink (not digitally reproduced, or printed in any form) to avoid Cadet abuse of note sharing.

d. **Graded Requirements.** Overall, pre-lesson preparation for classes was improved from the previous semester. Incorporating additional knowledge checks helped increase the engagement and preparation of students.

e. **Other.** None

Enclosure 2: Course Syllabus

Block	LSN	Date	Lesson Name	Notes	Reading Sections
Project Initiation			**Project Initiation**		
	1	8-Jan	Definition of Project Management		1.1-1.3 (pp. 1-21) Optional: 1.5 (pp.26-28)
	2	10-Jan	Project Initiation & Selection - I		2.1-2.2 (pp. 35-50)
	3	16-Jan	Project Initiation & Selection - II	**Course Textbook Check**	2.2-2.3 (pp. 50-60) *Stop at "8 Steps of the PPM Process"
	4	18-Jan	PM and Profession & PM Ethics		3.1-3.4 (pp. 83-110) PMBOK Handouts
	5	22-Jan	Stakeholder Management	**Issue MS Project Template**	4.1-4.5 (pp. 119-136)
	6	24-Jan	Project Organization and Teams		5.1-5.8 (pp. 149-180) *Skip "Agile Team Roles" (pp. 174-175)
Project Planning			**Project Planning**		
	7	26-Jan	**Class Drop**		
	8	29-Jan	Project Planning	**MS Project Template CHECKED START LSN 8**	6.1 (pp. 193-212) 6.3 (pp. 221-224)
	9	31-Jan	Including Risk in Project Management	**Project SG #1 Due by 2300 04 FEB**	7.3-7.4 (pp. 261-273) *Stop at "General Simulation Analysis"
	10	5-Feb	Budgeting and Cost Estimation		7.1 (pp. 239-250) *Stop at "Budgeting with Agile"
	11	7-Feb	Improving Cost Estimates		7.2 (pp.251-261)
	12	9-Feb	**Class Drop**		
	13	12-Feb	Scheduling I - An Intro to PERT/CPM		8.1-8.2 (pp. 297-308) *Stop at "Calculating Activity Times"
	14	14-Feb	Scheduling II - Solving the Network	**Problem Set #1 Due by 2300 14 FEB**	8.2 (pp. 308-318)
	15	20-Feb	Scheduling III - Uncertainty in a Project		8.2 (pp. 318-325)
	16	22-Feb	**Midterm Review**		Review lessons 1-15
	17	26-Feb	**Midterm**	**Midterm WPR, LSNs 1-15**	
	18	28-Feb	**MS Project Introduction Lab**		MS Project step-by-step
	19	1-Mar	**Class Drop**		
	20	4-Mar	Expediting a Project		9.1 (pp. 356-364)
	21	6-Mar	Resource Loading and Leveling		9.2-9-4 (pp. 364-374)
	22	11-Mar	Allocating Scarce Resources & Multi-Project Scheduling in MS Project		9.5 (PP. 374-379)
	23	13-Mar	**MS Project Practice Lab**		MS Project step-by-step
	24	15-Mar	**Stage Gate #2 Working Session**	**Problem Set #2 Due by 2300 14 MAR**	
	25	18-Mar	Monitoring a Project		10.1-10.2 (pp.397-409)
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	26	20-Mar	**Class Drop**	**Project SG #2 Due by 2300 20 MAR**	
			SPRING BREAK 23MAR-1APR		
	27	2-Apr	Earned Value Management (EVM) I		10.3 (pp. 409-421)
	28	4-Apr	**EVM II - LEGO Lab**		
	29	8-Apr	**EVM III -- Problem Solving Lab**		11.1-11.4 (pp. 433-459) *Skip "Scrum Events for Project Control" (pp. 454-455)
	30	10-Apr	Project Control & Auditing		12.1-12.5 (pp. 468-488)
	31	12-Apr	**Class Drop**	**Project SG #3 Due by 2300 11 APR**	
	32	15-Apr	Project Termination		13.1-13.4 (pp. 499-517)
	33	17-Apr	**Quiz**	**In-Class Quiz, LSNs 19-32**	
	34	22-Apr	**Project Site Visit (CEAC) -All Hours**		
Agile PM			**Agile Project Management**		
	35	24-Apr	Introduction to Agile		1.4 (pp.21-26), Agile Team Roles (pp. 174-175)
	36	29-Apr	Agile Planning and Scheduling		6.2 (pp. 213-221) 7.1 (pp. 250-251) 8.5 (pp. 339-341)
	37	1-May	Agile Project Execution		10.4 (pp.421-423)
	38	6-May	Agile Project Monitoring, Control, and Closure | SG4 Working Session	**Project SG #4 due by 2300 09 MAY**	11.3 (pp. 454-455), 13.2 (pp.508); review p.423
	39	8-May	**Course Closeout & TEE Review**		Review LSNs 1-39
	40	10-May	**Class Drop**		
			TEE		

Enclosure 3: Course End Feedback

Feedback administered via Canvas, will be available for review later

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D AY2023 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-SE 20 May 2023

MEMORANDUM FOR COL James Schreiner, Program Director, Department of Systems Engineering, United States Military Academy

SUBJECT: AT23-2 Course Assessment – EM411: Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term (AT) 23-2.

2. POC for this action is the undersigned at matt.young@westpoint.edu.

MATT A. YOUNG
MAJ, SC
EM411 Course Director

Encls
1. Course Assessment
2. Course Instructional Memorandum and Syllabus
3. Course End Feedback

DISTRIBUTION:

1 each EM411 instructors, Program Director, and Department Head

Enclosure 1: Course Assessment

Section I – Course Information and History

1. **Historic Course Offerings and Enrollments by Academic Year Term (AYT).**

a. **Offered.**

AYT 18		AYT 19		AYT 20		AYT 21		AYT 22		AYT 23	
18-1	18-2	19-1	19-1	20-2	18-1	21-2	21-1	22-1	22-2	23-1	23-2
X	X	X	X	X	X	X	X	X	X	X	X

b. **Enrollments.**

AYT	Enrollment
23-2	127
23-1	50
22-2	107
22-1	88
21-2	79
21-1	36
20-2	157
20-1	60
19-2	118
19-1	85
18-2	140
18-1	143

c. **Historical Term End Examination (TEE) Grade Averages.**

Academic Term	TEE Average
23-2	83.13%
23-1	82.97%
22-2	84.39%
22-1	82.58%
19-2	86.51%
19-1	87.93%
18-2	86.77%
18-1	86.80%
17-1	85.36%
16-2	83.64%
16-1	85.82%
15-1	85.75%

d. **Historical Course Final Grade Averages.**

Academic Term	EM411 Average
23-2 (Current)	86.40%
23-1	85.89%
22-2	87.60%
22-1	87.11%
21-2	87.47%
21-1	86.27%
20-2	88.51%
20-1	84.63%
19-2	86.51%
19-1	87.93%
19-1	87.93%
18-1	85.65%
18-2	86.59%
18-1	85.65%

Course Overview.

a. **Applicability.** Project Management (PM) is a vital skill for any leader serving in the military or civilian industry, and greatly enhances an organization's ability to successfully achieve its project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

b. **Scope.** Each cadet will learn the skills necessary to successfully initiate, plan, execute, monitor, and control a project. Topics include Project Selection, Project Manager Roles and Responsibilities, Organizational Structure, Project Planning, Budgeting and Scheduling, Resource Allocation, Monitoring and Controlling, Risk Assessment and Response Management, and Evaluation and Termination. A visit to Sikorsky, and a discussion panel with Project Managers will illustrate the real-world application of PM techniques and procedures and will provide context to the topics of this course. A semester-long project will reinforce course concepts and provide cadets the opportunity to integrate PM software into the development of a project plan. Cadets will spend several lessons in a computer lab environment to ensure they are aware of technology that can be applied.

Course Strategy.

1. EM411 is divided into three major blocks that align with the lifecycle of a successful project.

i) Block 1: Project Initiation covers project management organization, roles and responsibilities, project selection methods, and stakeholder analysis and management.

ii) Block 2: Project Planning focuses on techniques used to define the project's scope and required work, create a project budget and schedule, improve cost and duration estimates, and explore the issues related to project scheduling, resourcing, and risk management.

iii) Block 3: Project Execution introduces project monitoring, controlling, auditing, and termination.

2. A group project is assigned to reinforce and apply learned concepts and to bridge the gap between theory and practical application. Microsoft Project will be used to develop project management skills and increase cadet productivity.

3. All lesson objectives are referenced with the Project Management Body of Knowledge (PMBOK).

d. **Endstate.** Each Cadet understands the application of project management and the complex interrelated tasks associated with completing projects on time, within budget, and to specification.

Section II – Course Assessment

Achievement of Course Objectives. The course continues to accomplish the six course objectives included in the table below. Emphasis remains the ability to successfully achieve project deliverables within given cost, time, and resource constraints. Within the Systems Engineering discipline, Project Management is nested with the Solution Implementation Phase of the Systems Decision Process (SDP).

1. Identify project management practices used in current organizations.
2. Identify and describe the necessary qualities and skills required of successful project managers and their teams.
3. Successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.
4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the PMBOK.
5. Utilize Project Management software in planning and managing successful projects.
6. Exercise oral and written communications skills.

1. **Course Content.**

a. **Body of Knowledge.** In a holistic analysis, the course achieved the initially stated desires and intent. The course objectives covered the disciplined-accepted body of knowledge (Project Management Institute or PMI) are relevant to an undergraduate course in project management.

b. **Effective Assessment.** The combination of problem sets, checks on learning (Kahoot/MS Forms) quizzes, WPRs, problem sets, and the Stage Gates worked well for assessing student understanding of material. The frequent checks on learning were insightful in that they allowed instructors to validate assumptions about students' ability to apply fundamental course concepts. As such, I highly recommend retaining intermittent checks on learning, quizzes, and problem sets, even if for minimal point values, to force students to become familiar with material and to validate instructor assumptions about student retention.

2. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

b. **Software.** Cadets use Microsoft Project 2021

c. **Graded Requirements.** The course was composed of 13 graded events totaling 1000 points. With the exclusion of instructor points, 790 of the course's 950 points (83%) were assessed using individual events.

Graded Event	*Type*	*Points*	*% of Course*
Problem Set 1	Individual	50	5.0%
Problem Set 2	Individual	50	5.0%
Problem Set 3	Individual	50	5.0%
Quiz 1	Individual	60	6.0%
Quiz 2	Individual	60	6.0%
Project Deliverable SG1 - Project Selection Analysis	Individual	60	6.0%
Project Deliverable SG2 - Project Charter	Group	80	8.0%
Project Deliverable SG3 - Project Plan	Group	80	8.0%
Project Deliverable SG4 - Project Progress Report	Individual	80	8.0%
WPR 1 - Midterm	Individual	120	12.0%
Term End Exam	Individual	220	22.0%
Lesson Knowledge Checks	Individual	40	4.0%
Instructor Points	Individual	50	5.0%
Course Total:		1000	100

3. **Course Pedagogy.**

a. **Problem Sets (150 total points).** There will be three problem sets due over the course of the semester. Problem Set 1 (50 points) will cover project initiation and project planning. Problem Set 2 (50 points) will cover project planning and cost estimation. Problem Set 3 (50 points) will cover scheduling, loading, and leveling.

b. **Quizzes (120 Points).** There will be two 60-point quizzes that will prepare the students for Midterm and the TEE.

c. **Course Project (300 Points).** Cadets will complete a semester-long course project, consisting of four stage gate deliverables: Project Selection Analysis, Project Charter, Project Plan, and Project Progress Report (worth 60, 80, 80, and 80 points, respectively). The second and third stage gates will be completed in small groups, while the first and fourth stage gate will be completed individually.

d. **Midterm Examination (120 Points)**. One 70-minute Written Partial Review (WPR) covering specified blocks of lessons as per the attached enclosure.

e. **Term End Examination (220 points).** One 3.5-hour Term End Examination (TEE) covering all material covered in the course. Points will be awarded based on the cadet's ability to demonstrate their project management skills and knowledge acquired throughout the course.

f. **Lesson Knowledge Checks (40 points).** Cadets will take multiple knowledge checks throughout the semester.

g. **Instructor Points (50 points).** Instructor points will be awarded through In-class participation and conduct, practical exercises, and additional assignments that the instructors deem necessary to increase learning.

6. **Administrative Guidance.**

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Assessment of Changes to Current Academic Term (based on previous Course Assessments).**

a. **Course Objectives.** None

b. **Textbook.** *Project Management a Managerial Approach,* 11th Edition, by Jack R. Meredith and Scott M. Shafer.

c. **Curriculum.** Minor changes were made to the order of lessons in Block 2, Project Planning based upon AT23-1's graded event performance. Lesson 12 (Including Risk in Project Management) was moved to Lesson 08 to introduce risk earlier in the project planning phase. All other lessons were shifted accordingly. This change improved the flow of the concepts in Block 2, Project Planning and increased WPR 1 Risk Management, Topic IV from 58.92% in AT23-1 to 83.46% in AT23-2.

d. **Graded Requirements.** The PM Group Case Study was eliminated and replaced a Stage Gate 4 working session. This change was made to reduce group graded events below 20%, as directed by the PD. Additionally, we turned Stage Gate #1 (NPV/DCF) from a group graded event to an individual graded event to reduce group graded events below 20%. This change also enabled Cadets who struggle with NPV/DCF to learn individually. As a result, we saw the NPV/DCF portion of the TEE increase from an 70.3% average in AT23-1 to an 81.2% in AT23-2.

e. **Other.** None.

2. **Proposed Changes for Next Academic Term.**

a. **Course Objectives.** No change

b. **Textbook.** No change

c. **Curriculum.** An Agile Project Management approach is applied to each section of the textbook and is prevalent within the Project Management Institute. However, there is one class built in the syllabus to cover the topic at the end of the term. We recommend building a more robust Agile Project Management section into the course to identify and utilize standards implemented by the PMI through the Project Management Body of Knowledge (PMBOK). Additionally, Cadet Capstone teams are using an Agile approach to solve their complex problem set. Performance on the Agile PM section on the TEE was the lowest, which we believe is due to its newness at the very end of the course. Additionally, as the last subject area, it was not assessed through any other means prior to the TEE. Placing it at strategic transitions between Project Selection, Project Planning, Project Execution, Project Monitoring/Controlling, and Project Termination would allow it to be assessed. We recommend replacing 4-5 classes with a more focused Agile PM emphasis.

d. **Graded Requirements.** We recommend reducing the traditional Thayer Hall Stage Gate Project from four stage gates to three. After exercising the traditional PM approach themed Stage Gates, the final Stage Gate will embrace an Agile PM approach. Additionally, we should investigate methods to require Cadets to engage the material outside of the classroom. We recommend breaking up the problem sets into homework assignments due weekly or biweekly.

e. **Other.** None.

Block	LSN	Date	Lesson Name	Notes	Reading Sections
			Project Initiation		
Project Initiation	1	11-Jan	Definition of Project Management		1.1-1.3 (pp. 1-21) Optional: 1.5 (pp.26-28)
	2	13-Jan	Project Initiation & Selection - I	Issue MS Project Template	2.1-2.2 (pp. 35-50)
	3	18-Jan	Project Initiation & Selection - II		2.2-2.3 (pp. 50-60) *Stop at "8 Steps of the PPM Process"
	4	20-Jan	PM and Profession & PM Ethics		3.1-3.4 (pp. 83-110) PMBOK Handouts
	5	23-Jan	Class Drop		
	6	26-Jan	Stakeholder Management		4.1-4.5 (pp. 119-136)
	7	30-Jan	Project Organization and Teams	MS Project Template Due NLT 2300	5.1-5.8 (pp. 149-180) *Skip "Agile Team Roles" (pp. 174-175)
			Project Planning		
Project Planning	8	1-Feb	Project Planning		6.1 (pp. 193-212) 6.3 (pp. 221-224)
	9	3-Feb	Including Risk in Project Management		7.3-7.4 (pp. 261-273) *Stop at "General Simulation Analysis"
	10	6-Feb	Budgeting and Cost Estimation		7.1 (pp. 239-250) *Stop at "Budgeting with Agile"
	11	9-Feb	Improving Cost Estimates		7.2 (pp.251-261)
	12	13-Feb	Quiz 1	Quiz 1, LSNs 1-11	
	13	16-Feb	Scheduling I - An Intro to PERT/CPM	Problem Set #1 Due by 2300	8.1-8.2 (pp. 297-308) *Stop at "Calculating Activity Times"
	14	22-Feb	Scheduling II - Solving the Network		8.2 (pp. 308-318)
	15	24-Feb	Scheduling III - Uncertainty in a Project		8.2 (pp. 318-325)
	16	27-Feb	Midterm Review	Project Stage Gate #1 Due NLT 2300	Review lessons 1-15
	17	1-Mar	Midterm	Midterm WPR, LSNs 1-16	
	18	3-Mar	Class Drop		
			Spring Break		
	19	13-Mar	MS Project Introduction Lab		MS Project step-by-step
	20	15-Mar	Expediting a Project	Problem Set #2 Due by 2300	9.1 (pp. 356-364)
	21	17-Mar	Resource Loading and Leveling		9.2-9-4 (pp. 364-374)
	22	20-Mar	Allocating Scarce Resources & Multi-Project Scheduling in MS Project	Project Stage Gate #2 Due by 2300	9.5 (PP. 374-379)
	23	23-Mar	MS Project Practice Lab		MS Project step-by-step
	24	27-Mar	Project Stage Gate #3 Working Session		
			Project Execution, Monitor and Control, Termination		
Project Execution / Monitor and Control / Termination	25	29-Mar	Monitoring a Project		10.1-10.2 (pp.397-409)
	26	31-Mar	Earned Value Management (EVM) I		10.3 (pp. 409-421)
	27	3-Apr	EVM II - LEGO Lab (In DSE Conference Room)		
	28	6-Apr	EVM III -- Problem Solving Lab	Problem Set #3 Due by 2300	11.1-11.4 (pp. 433-459) *Skip "Scrum Events for Project Control" (pp. 454-455)
	29	10-Apr	Class Drop		
	30	12-Apr	Project Control & Auditing	Project Stage Gate #3 (MSP) Due by 2300	12.1-12.5 (pp. 468-488)
	31	14-Apr	Project Termination		13.1-13.4 (pp. 499-517)
	32	17-Apr	Quiz 2	Quiz 2, LSNs 19-31	
	33	20-Apr	Project Site Visit (Sikorsky) - I & J hours		
	34	24-Apr	Class Drop		
	35	26-Apr	Project Site Visit (Sikorsky) - G & H hours		
	36	1-May	Agile Planning		Agile Team Roles (pp. 174-175) 6.2 (pp. 213-221) 10.4 (pp.421-423)
	37	5-May	Project Stage Gate #4 Working Session	Project Stage Gate #4 Due by 2300	
	38	8-May	Class Drop		
	39	10-May	PM in Action (Guest Speaker) - Deans Hour - Kendrick Auditorium		
	40	11-May	Course Closeout & TEE Review		Review lessons 1-38
			TEE		
TEE	TEE	TBD	Term End Exam		

Legend:
Graded Event
Notes / Special Event
Lab
Class Drop

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB E AY2022 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE 17 December 2021

MEMORANDUM FOR COL Paul Evangelista, Engineering Management Program Director, Department of Systems Engineering

SUBJECT: AT22-1 Course Assessment – EM411, Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term 2022-1.

2. EM411 Project Management (PM) teaches cadets the skills required to lead an organization in achieving their performance goals with respect to cost, schedule, and performance. This course is nested with the Project Management Institute's PM Body of Knowledge (PMBOK). Course topics include project selection, project manager roles and responsibilities, Agile approach, organizational structure, project planning, budgeting, scheduling, resource allocation, monitoring and controlling, evaluation and termination, and risk assessment and ethics in the profession. Case studies and a course project based on the Cadet Barracks Construction Project at USMA illustrate real world PM techniques and procedures in practice and provide context to the conceptual aspects of this course. The course project and a problem set provide students with the opportunities to integrate PM software (Microsoft Project - MSP) into a Project Action Plan.

3. I recommend five primary changes for improvement to the course:

a. Continue to include the course long PM project based on Barracks Renovation or CEAC construction. If permissible, include a trip section to walk through the barracks construction or CEAC construction and provide an opportunity for cadets to engage with the construction project manager.

b. Sustain breaking the course project into smaller graded events. Project Stage Gate 1 was split into two deliverables. This allowed cadets to work on the project earlier in the course and aligned with the lessons when course material was taught. If possible, consider splitting Stage Gate 2 and 3.

c. Reinstated the Term End Examination. The TEE was removed in AY21 as a result of COVID policy requirements. WPR2 was removed and the TEE was reinstated. TEE worth 200 points.

d. Coordinate for a trip section to Sikorsky in Connecticut in the Spring Semester. Contact Mr. Bryan Dougherty, Program Manager at Sikorsky at bryan.dougherty1@gmail.com.

e. Sustain the guest lecture PM in Action. The PM in Action lecture was well received by cadets this semester and it reinforced course concepts.

4. POC for this action is the undersigned at brandon.thompson@westpoint.edu.

BRANDON S. THOMPSON
LTC, MI/FA49
EM411 Course Director

Encls
1. Course Assessment
2. Lesson Objectives / Syllabus
3. EM411 Course Objective Assessment
4. EM Program Student Outcomes Crosswalk
5. Course End Feedback
6. Cadet Comments

Enclosure 1: Course Assessment – EM411 Project Management

Section I – Course Description

1. **Redbook Description.**

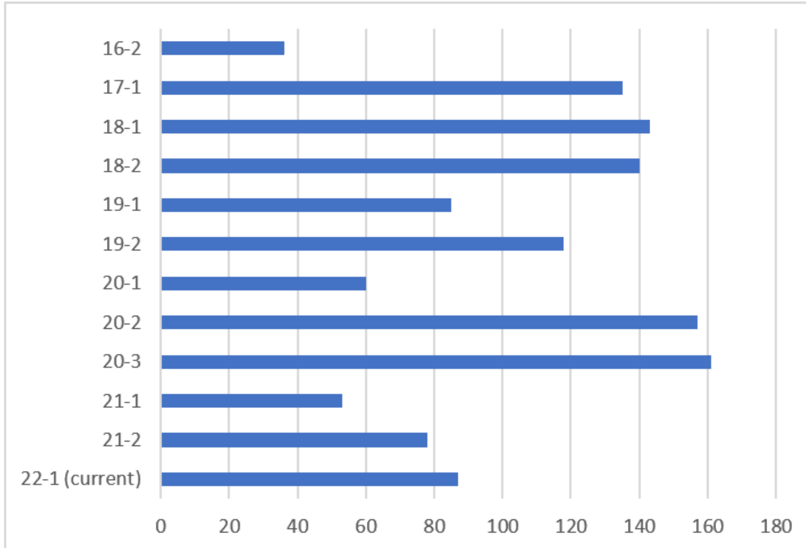
This course develops skills required to lead an organization to the achievement of their objectives through the proper application of the management of planning, implementing, and controlling the organization activities, personnel, and resources. The course focuses on the Implementation phase of the Systems Decision Process (SDP). Topics include project selection, roles and responsibilities of the project manager, planning the project, budgeting the project, scheduling the project, allocating resources to the project, monitoring and controlling the project, evaluating and terminating the project, risk assessment and management, organizational structure and human resources. Case studies illustrate problems and how to solve them. Course assignments are designed to help students learn and apply project management techniques taught in the course. The class design project will provide students with the opportunity to integrate project management software, Microsoft Project, into the preparation of an Engineering Management Project Plan.

a. **Lessons.** 35 @ 75 min (2.00 Att/wk) LABS: 0 @ 0 min
b. **Special Requirements.** One (1) half-day Trip Section
c. **Offered.**

AY16		**AY17**		**AY18**		**AY19**		**AY20**			**AY21**		**AY22**	
16-1	16-2	17-1	17-2	20-1	20-1	19-1	19-2	20-1	20-2	20-3	21-2	21-2	22-1	22-2
X	X	X		X	X	X	X	X	X	X	X	X	X	X

2. **Enrollment.**

Academic Term	EM411 Enrollment
16-2	36
17-1	135
18-1	143
18-2	140
19-1	85
19-2	118
20-1	60
20-2	157
16-1	161
21-1	53
21-2	78
22-1 (current)	87



3. **Current Course Objectives.**

a. Identify project management practices used in current organizations.

b. Identify and describe the necessary qualities and skills required of successful project managers.

c. Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.

d. Identify and utilize project management standards implemented by the Project Management Institute through the Project Management Body of Knowledge.

e. Utilize Microsoft Project 2019© as an effective tool in managing successful projects.

f. Utilize case studies to bridge the gap between project management theory and practice.

g. Exercise oral and written communications skills.

4. **Textbooks.**

a. *Project Management a Managerial Approach,* 10th Edition, by Jack R. Meredith, Scott M. Shafer, and Samuel J. Mantel Jr.

5. **Course Strategy.** This course is divided into three major blocks that generally align with the lifecycle of a project. The first block, **Project Initiation** consists of five lessons that deal with project selection, the project management profession, conflict and negotiation techniques, and project teams. The second block, **Project Planning**, consists of eighteen lessons and focuses on initial project planning, techniques to improve cost and duration estimates, creating a project budget and schedule, exploring the issues related to project scheduling and resourcing, Agile, and risk management. It covers topics such as PERT, CPM, project uncertainty, simulation, resource loading and leveling and allocating scarce resources. It also includes a site visit to see project management in the real world and introduces the

course project. This block culminates with four lessons to familiarize cadets with Microsoft Project. The third block, **Project Execution,** consists of twelve lessons and introduces project monitoring, earned value analysis, project control, ethics, auditing, and terminating a project. One lesson is dedicated to Project Management in Action and typically includes a guest lecturer to provide cadets with a real-world example of project management in industry. This block culminates with group project submissions and a case study presentation. During each block, practical exercises are used to illustrate relevant and applicable project management issues and bridge the gap between theory and practical application. Microsoft Project is used during the second half of the course to develop project management skills and increase cadet productivity. A detailed lesson syllabus is included in Enclosure 3.

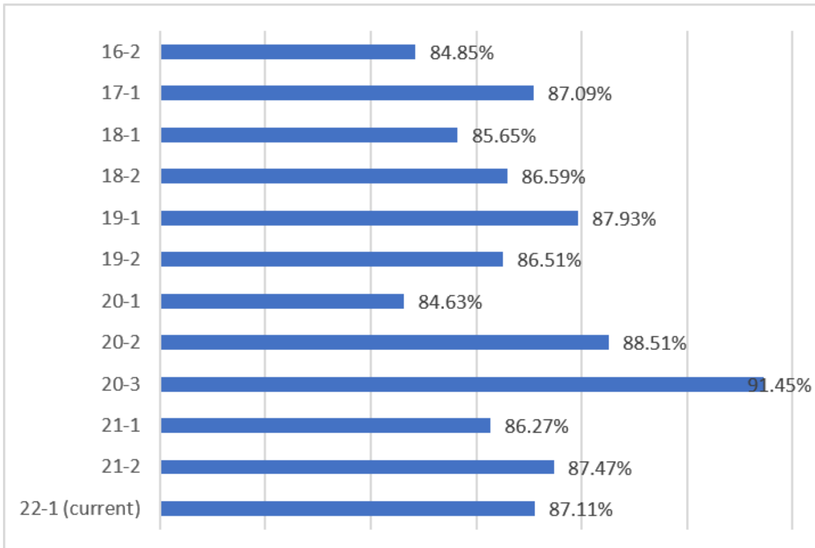
6. **Graded Requirements.**

Graded Event	***Points***	***Lesson***	***% of Course***
Project Deliverable PG1a	50	9	5.0%
Quiz 1	50	10	5.0%
Problem Set 1	35	13	3.5%
Problem Set 2	40	17	4.0%
Midterm	100	18	10.0%
Project Deliverable PG1b	100	20	10.0%
Problem Set 3	40	23	4.0%
Problem Set 4	35	25	3.5%
Project Deliverable PG2	100	29	10.0%
Quiz 2	50	32	5.0%
Case Study Presentation	50	38	5.0%
Project Deliverable PG3	100	40	10.0%
Term End Exam	200	-	20.0%
Instructor Points	50	-	5.0%
Course Total:	1000		

Section II – Course Assessment

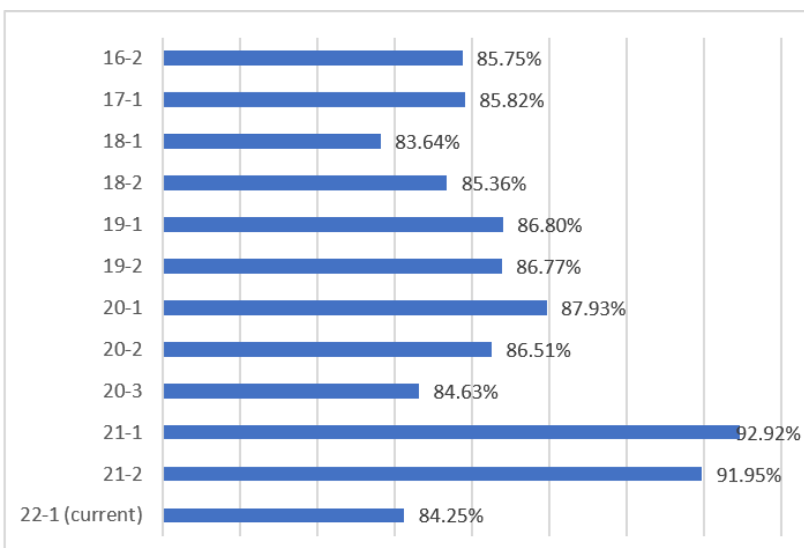
1. **Achievement of Course Objectives.**
   a. Course Objectives/Syllabus. Enclosure 2.
   b. Course Objective Assessment Table. Enclosure 3.
   c. Results of Course End Feedback (web-based). Enclosure 5.
   d. Results of Cadet Free-Form Comments. Enclosure 6.
   e. Historical Course Final Grade Averages.

Academic Term	EM411 Average
16-2	84.85%
17-1	87.09%
18-1	85.65%
18-2	86.59%
19-1	87.93%
19-2	86.51%
20-1	84.63%
20-2	88.51%
20-3	91.45%
21-1	86.27%
21-2	87.47%
22-1 (current)	87.11%



   f. Historical Term End Examination Grade Averages.

Academic Term	EM411 TEE Average
16-2	83.64%
17-1	85.36%
18-1	86.80%
18-2	86.77%
19-1	87.93%
19-2	86.51%
20-1	84.63%
20-2	92.92%
20-3	91.95%
21-1	84.25%
21-2	85.76%
22-1 (current)	84.25%



2. **Course Objectives.**

a. **Body of Knowledge.** This course develops skills required to lead an organization to the achievement of their objectives through the proper application of the management of planning, implementing, and controlling the organization activities, personnel, and resources. The course focuses on the Implementation phase of the Systems Decision Process (SDP). Topics include project selection, roles and responsibilities of the project manager, planning the project, budgeting the project, scheduling the project, allocating resources to the project, monitoring and controlling the project, evaluating and terminating the project, risk assessment and management, organizational structure and human resources. Case studies illustrate problems and how to solve them. Course assignments are designed to help students learn and apply project management techniques taught in the course. The class design project will provide students with the opportunity to integrate project management software, Microsoft Project, into the preparation of an Engineering Management Project Plan.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provide an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 10th Edition, by Jack R. Meredith, Scott M. Shafer, and Samuel J. Mantel Jr. sufficiently covers the material outlined in the course objectives. The text is relatively easy to read, though at times cumbersome due to the length of sections. The Project Management Institute's PMBOK is used as a complementary source for project management knowledge. The Ethics lesson is derived from the PMBOK Code of Ethics.

b. **Software.** Microsoft Project 2019 is used throughout the second half of the course. Cadets are taught how to manage a project schedule, budget and resources using MS Project through in-class labs, practical exercises, and during the course project.

c. **Graded Requirements.** The graded requirements in EM411 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. The use of text material, computer-based practical exercises, hands-on labs, and case study analysis has proven to be sufficient for assessing course knowledge for engineering, management, and economics majors alike. There are, however, some recommended changes in section III.

4. **Course Pedagogy.**

a. **Course Project.** The course is allotted 3.5 credit hours due to the significant course project. The basic objectives for the course project required the cadets to go through the entire project management process from initiation and selection through monitoring and control. The project is based on the numerous upgrades to facilities currently underway at USMA to provide realism to the project.

b. **Case Study.** We maintained the 50-point graded Case Study from the previous semester. The intent was for students to research case studies from the Project Management

Institute and present a 10-minute presentation highlighting the project management tools that were applied or should have been applied in the case study. Due to COVID19 restrictions including a shortened schedule, the case study presentation assignment was modified to require cadet groups to record a 10-minute video presentation. All six groups' presentations were then played in one lesson. Recommend sustaining the recorded presentations until classes return to 40 lessons and cadets are allowed in the classroom.

c. **Instructor Points.** Cadet performance was measured based on participation in class and preparation for class and awarded points accordingly. Recommend establishing a grading rubric for instructor points for consistency between sections.

d. **Term End Exam.** A comprehensive term end exam was administered covering course content from all lessons. The TEE was the same exam that was administered in previous years prior to COVID.

5. **EM Program Student Outcomes Crosswalk.** Enclosure 5.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Changes to Current Academic Term.**
   a. **Course Objectives.** None.
   b. **Textbook.** None
   c. **Curriculum.** The course was taught in person with all students present in class.
   d. **Graded Requirements.** Recommend sustaining the four problem set model and add lesson quizzes where needed to account for the Instructor grade. Both quizzes were conducted via Blackboard. Blackboard quizzes require careful consideration due to increased time requirements and creative grading. Multiple choice questions are the preferred option for Blackboard tests but do not provide a chance for partial credit. The TEE was reinstated as a 200 point assignment given during TEE week.
   e. **Other.** None.

2. **Proposed Change for Next Academic Term.**
   a. **Course Objectives.** None.
   b. **Textbook.** None.
   c. **Curriculum.** See five recommended changes above.
   d. **Graded Requirements.** None.
   e. **Other.** None.

Enclosure 2: EM411 Lesson Objectives/Syllabus

Lesson 1: Definition of Project Management
1. Define a project, its six characteristics and differentiate it from a function (K)
2. Explain the three direct goals of every project (K)
3. Explain the project life cycle and the two prevalent shapes it can take (K)

Lesson 2: Project Initiation and Project Selection
1. Apply non-numeric project selection models (sacred cow, operating necessity, competitive necessity, product line extension and comparative benefit, sustainability) (S)
2. Apply numeric project selection models (payback period, discounted cash flow, weighted scoring model.) (S)
3. Explain Project Portfolio Management (PPM) (K)

Lesson 3: The Project Manager and the Profession
1. Identify the key differences between a project manager and a functional manager (K)
2. List and describe the responsibilities, demands, and skills and attributes of an effective project manager (K)
3. List and explain desirable team characteristics (K)

Lesson 4: Conflict and the Art of Negotiation
1. Analyze project team problems and conflicts. Determine sources and categories of conflict within the project life cycle (S)
2. Explain the process of principled negotiation to include its three requirements and the four main points of the method (K)

Lesson 5: Project Organization and Teams
1. Identify the three primary types of project organizations with advantages and disadvantages of each type (K)
2. Explain the factors to be considered when staffing and managing project teams (K)

Lesson 6: Project Planning I
1. Explain the purpose of the initial project coordination and the contents of the Project Charter (K)
2. Develop a Work Breakdown Structure (WBS) for a project (S)
3. Develop a Linear Responsibility Chart for a project (S)

Lesson 7: Project Planning II – Agile Project Management
1. Explain the purpose of Agile Project Management (K)
2. Explain the differences between the Waterfall and Agile Project Management methodologies (K)

Lesson 8: Quiz 1/Project Planning Review
Execute concepts from Lessons 1-5 in a graded environment (S)

Lesson 9: Budgeting and Cost Estimation
1. Apply top-down and bottom-up budgeting strategies to include direct/indirect costs and fully costing a project (S)
2. Differentiate category-oriented versus program-oriented budgeting (S)

Lesson 10: Improving Cost Estimates
1. Recognize sources of budget uncertainty (K)
2. Implement methods to improve cost estimates, including using Learning Curves (S)

Lesson 11: Including Risk in Project Management
1. Analyze project risk to include: Risk Identification, Risk Analysis, and Response to Risk and create a Risk Assessment given a scenario (S)

Lesson 12: Scheduling I – An Introduction to PERT and CPM
1. Create CPM networks and conduct a forward and backward pass to determine the critical path in a network and determine slack (S)
2. Explain the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) (K)

Lesson 13: DROP

Lesson 14: Scheduling II – Solving the Network
Calculate probabilistic activity times (S)

Lesson 15: Scheduling III – Uncertainty in a Project
1. Analyze the probability of completing a project on time (S)
2. Analyze the duration of a project, given known risk (S)

Lesson 16: Problem Solving Lab – Scheduling
Review all skills for project scheduling

Lesson 17: Midterm Exam Review

Lesson 18: Midterm Exam
Execute concepts from Lessons 1-14 in a graded environment

Lesson 19: Class Drop

Lesson 20: Expediting a Project (Resource Allocation)
Expedite a project by "crashing" tasks and use of fast-tracking (S)

Lesson 21: Resource Loading and Leveling
Evaluate resource usage and conduct resource loading and leveling (S)

Lesson 22: Introduction to Microsoft Project
Use computer software packages (MS Project) to enhance a Project Managers ability to monitor, assess, and control a project (S)

Lesson 23: Allocating Scarce Resources and Multi-Project Scheduling in MSP
Apply scarce resource allocation using priority rules (to include across multiple projects) (S)

Lesson 24: Microsoft Project Practice Lab
Use computer software packages (MS Project) to enhance a Project Manager's ability to plan, monitor, assess, and control a project (S)

Lesson 25: Project Stage Gate #2 Working Session
Implement all MS Project skills for the course project's second deliverable

Lesson 26: Monitoring a Project
Explain the plan-monitor-control cycle and considerations for designing a monitoring system (K)

Lesson 27: Earned Value Management I
1. Define earned value terminology and explain the EVM process (K)
2. Apply earned value analysis to various projects (S)

Lesson 28: Earned Value Management II
1. Define earned value terminology and explain the EVM process (K)
2. Apply earned value analysis to various projects (S)

Lesson 29: Problem Solving Lab – Earned Value Management
Review all skills for earned value management

Lesson 30: DROP

Lesson 31: Project Control and Auditing
1. Explain the control cycle (K)
2. Describe scope creep and change control (K)
3. Explain the audit process to include the purpose, depth, and timing of an audit (K)

Lesson 32: Quiz 2/Project Termination
1. Execute concepts from Lessons 19-31 in a graded environment (S)
2. Explain the varieties of project termination (K)
3. Analyze the factors to consider in determining when to terminate a project (S)

Lesson 33: Ethics in the Profession
1. Describe the Project Management Code of Ethics and the Ethical Decision Making Framework (B)
2. Explain the differences between PM CoE and the ethical standards of an Army Officer (B)

Lesson 34: Final Exam Review

Lesson 35: DROP

Lesson 36: Final Exam
Execute concepts from Lessons 16-35 in a graded environment

Lesson 37: DROP

Lesson 38: PM Case Study Presentations
Leverage skills from lessons 1-37 to analyze and present a Case Study (S)

Lesson 39: Course Closeout
Course review and closeout

Lesson 40: DROP

EM411 Syllabus

Phase	Lesson	Date	Lesson Name	Graded Event	Reading Sections
Project Initiation			**Project Initiation**		
	1	16-Aug	Definition of Project Management		1.1-1.4 (pp. 1-23)
	2	18-Aug	Project Initiation		2.1-2.3 (pp. 28-55)
	3	20-Aug	PM and Profession		3.1-3.4 (pp. 77-105)
	4	24-Aug	Conflict and the Art of Negotiation		4.1-4.6 (pp. 118-137)
	5	26-Aug	Project Organization and Teams	**MS Project Template due - 10 Instructor Points**	5.1-5.8 (pp. 145-176)
Project Planning			**Project Planning**		
	6	30-Aug	Project Planning I		6.1, 6.3 (pp. 187-208, 214-217)
	7	1-Sep	Project Planning II - Agile Project Management		6.2 (pp. 208-214)
	8	3-Sep	**Class Drop**		
	9	8-Sep	PM Meeting for Stage Gate 1a	**PM Meeting for Stage Gate 1a Due NLT 1610 8 SEP**	
	10	10-Sep	**Quiz 1**	**Quiz 1**	
	11	13-Sep	Budgeting and Cost Estimation		7.1 (pp. 233-246)
	12	16-Sep	Improving Cost Estimates		7.2 (pp.246-256)
	13	20-Sep	Including Risk in Project Management		7.3-7.4 (pp. 256-279)
	14	24-Sep	Scheduling I - An Intro to PERT/CPM	**Problem Set #1 Due by 1610 24 SEP**	
	15	29-Sep	Scheduling II - Solving the Network		Review 8.1-8.2 (pp. 291-312)
	16	1-Oct	Scheduling III - Uncertainty in a Project		8.2 (pp. 312-319)
	17	4-Oct	**Midterm Review**	**Problem Set #2 Due by 1610 4 OCT**	Review lessons 1-15
	18	7-Oct	**Midterm**	**Midterm WPR**	Review lessons 1-15
	19	13-Oct	**Class Drop**		
	20	15-Oct	Expediting a Project	**Project Stage Gate #1 Due by 1610 15 OCT**	9.1 (pp. 348-356)
	21	19-Oct	Resource Loading and Leveling		9.2-9-4 (pp. 356-366)
	22	21-Oct	Introduction to MS Project	**Problem Set #3 Due by 1610 22 OCT**	MS Project step-by-step
	23	25-Oct	Allocating Scarce Resources and Multi-Project Scheduling in MS Project		
	24	27-Oct	**MS Project Practice Lab**		MS Project step-by-step
	25	29-Oct	Project Stage Gate #2 Working Session		
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	26	1-Nov	Monitoring a Project		10.1-10.2 (pp.389-398)
	27	4-Nov	Earned Value Management I (Lab)	**Problem Set #4 Due by 1610 4 NOV**	
	28	8-Nov	Earned Value Management II		10.3 (pp. 398-411)
	29	10-Nov	**Problem Solving Lab - Earned Value Management**		11.1-11.3 (pp.421-442)
	30	12-Nov	Project Control & Auditing	**Project Stage Gate #2 (MSP) Due by 1610 12 NOV**	12.1-12.5 (pp. 459-476)
	31	15-Nov	Project Termination and Ethics in the Profession		13.1-13.4 (pp. 488-505); PMBOK
	32	19-Nov	**Quiz 2**	**Quiz 2**	
	33	22-Nov	**PM in Action (Dean's Hour Brief)**		
	34	24-Nov	**Class Drop**		
	35	29-Nov	Project Working Session		
	36	1-Dec	**Class Drop**		
	37	3-Dec	**Project Site Visit - Bradley Barracks**	**PM Case Study Due by 1610 4 DEC**	
	38	6-Dec	PM Case Study Presentations		
	39	9-Dec	**Course Closeout & TEE Review**		Review lessons 18-38
	40	10-Dec	**Class Drop**	**Project Stage Gate #3 Due by 1610 10 DEC**	PMBoK Handout

Enclosure 3: EM411 Course Objective Assessment

	EM411 Course Objectives	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	**Identify project management practices used in current organizations.**	**3.33**	**4.00**	Quiz 1 WPR #1 Case Study WPR #2
2.	**Identify and describe the necessary qualities and skills required of successful project managers.**	**3.67**	**4.00**	Quiz 1 Project SG #1 WPR #1 Project SG #2 Case Study Project SG #3
3.	**Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.**	**3.67**	**3.50**	Quiz 1 Problem Set #1 Project SG #1 Problem Set #2 WPR #1 Problem Set #3 Problem Set #4 Project SG #2 Quiz 2 WPR #2 Project SG #3
4.	**Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the Project Management Body of Knowledge (PMBOK).**	**3.67**	**3.25**	Quiz 1 Problem Set #1 Project SG #1 WPR #1 Project SG #2 Project SG #3
5.	**Utilize Microsoft Project 2019© as an effective tool in managing successful projects.**	**3.67**	**3.50**	Problem Set #4 Project SG #2 Project SG #3
6.	**Utilize case studies to bridge the gap between project management theory and practice.**	**3.33**	**3.50**	Project SG #1 Project SG #2 Case Study WPR #2
7.	**Exercise oral and written communications skills.**	**3.67**	**3.50**	Project SG #1 Project SG #2 Project SG #3 Class Participation
	Overall Assessment		**3.67**	All Graded Events

NOTES:
1) Graded Events included: Quizzes 1-2, Problem Sets 1-4, WPRs 1-2, Project Stage Gates 1-3, Case Study, Participation
2) Student Assessment is based on the End-of-Course Survey
3) Course Director Assessment is:
   -- *Based on course average for "All graded events"
   -- All others based on individual questions on graded events to include TEE question results

Scale: 1 = Unsatisfactory, 2 = Developing, 3 = Satisfactory, 4 = Exemplary

Enclosure 4: EM Program Student Outcome Crosswalk.

EM Program Student Outcomes \ EM411 Course Objectives	1. Identify project management practices used in current organizations.	2. Identify and describe the necessary qualities and skills required of successful project managers.	3. Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.	4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the Project Management Body of Knowledge (PMBOK).	5. Utilize Microsoft Project 2019© as an effective tool in managing successful projects.	6. Utilize case studies to bridge the gap between project management theory and practice.	7. Exercise oral and written communications skills.
1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	X		X	X			
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors				X		X	
3. an ability to communicate effectively with a range of audiences						X	X
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts				X		X	
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives						X	X
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions			X	X	X		
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies	X	X		X			

Enclosure 5: Course End Feedback

EM411 AT21-2 Course End Evaluation

		5	4	3	2	1			
					EM411 - I1-10				
Academy				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q1	My fellow students contributed to my learning in this course.	12 33.33%	20 55.56%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q2	My motivation to learn and to continue learning has increased because of this course.	10 27.78%	22 61.11%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q3	My personal schedule allows me enough time to reflect on the material I have learned in class.	13 36.11%	19 52.78%	4 11.11%	0 0.00%	0 0.00%	36	4.3	4
Q4	My personal schedule allows me enough time to adequately prepare for my optimum academic performance.	12 33.33%	19 52.78%	5 13.89%	0 0.00%	0 0.00%	36	4.2	4

					EM411 - I1-10				
Academy Remote				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q5	Graded events (WPR's, Projects, Papers, TEE's, etc.) in this course were effective and fair in the remote environment.	18 50.00%	18 50.00%	0 0.00%	0 0.00%	0 0.00%	36	4.5	5
Q6	I took more responsibility for my own learning in this course as a result of switching to a remote learning environment.	13 36.11%	15 41.67%	7 19.44%	1 2.78%	0 0.00%	36	4.1	4
Q7	I was able to maintain my attention in this course during remote class sessions to the same extent as in-person classes.	9 25.00%	15 41.67%	7 19.44%	4 11.11%	1 2.78%	36	3.8	4
Q8	I had to work harder in this course to achieve a similar amount of learning because of the transition to a remote environment.	3 8.33%	10 27.78%	9 25.00%	13 36.11%	1 2.78%	36	3.0	2
Q9	I had to spend more time in this course to do the work required in the remote environment.	4 11.11%	7 19.44%	11 30.56%	13 36.11%	1 2.78%	36	3.0	2
Q10	Participating in remote learning made me want to continue to learn on my own at USMA and beyond.	6 17.14%	18 51.43%	10 28.57%	0 0.00%	1 2.86%	35	3.8	4

					EM411 - I1-10				
Cognitive Domain				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q11	In this course, my critical thinking ability increased.	12 34.29%	21 60.00%	2 5.71%	0 0.00%	0 0.00%	35	4.3	4
Q12	The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a2:1 ratio of out-of-class time versus in-class time.	10 27.78%	26 72.22%	0 0.00%	0 0.00%	0 0.00%	36	4.3	4
Q13	The classroom environment (e.g. desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn.	10 27.78%	20 55.56%	6 16.67%	0 0.00%	0 0.00%	36	4.1	4
Q14	In this course, my ability to think creatively and take intellectual risks increased.	9 25.00%	22 61.11%	4 11.11%	1 2.78%	0 0.00%	36	4.1	4

					EM411 - I1-10				
Systems Engineering				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q15	This course helped me learn to design, manage or reengineer systems or processes.	13 36.11%	20 55.56%	3 8.33%	0 0.00%	0 0.00%	36	4.3	4
Q16	This course taught me to communicate effectively both orally and in writing.	7 19.44%	28 77.78%	1 2.78%	0 0.00%	0 0.00%	36	4.2	4
Q17	This course improved my ability to solve real-world problems through quantitative techniques.	11 30.56%	24 66.67%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q18	This course provided me with practical, problem-solving experiences applicable to my future as an Army officer.	12 33.33%	23 63.89%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q19	Course exercises and designs were realistic experiences that enabled open ended analysis, design, or implementation of systems.	14 38.89%	20 55.56%	2 5.56%	0 0.00%	0 0.00%	36	4.3	4
Q20	This course has motivated me to pursue deeper learning in related systems engineering or engineering management topics.	11 30.56%	20 55.56%	5 13.89%	0 0.00%	0 0.00%	36	4.2	4
Q21	This course improved my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	9 25.00%	22 61.11%	5 13.89%	0 0.00%	0 0.00%	36	4.1	4
Q22	This course improved my ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	7 19.44%	23 63.89%	6 16.67%	0 0.00%	0 0.00%	36	4.0	4
Q23	This course improved my ability to communicate effectively with a range of audiences.	6 16.67%	25 69.44%	5 13.89%	0 0.00%	0 0.00%	36	4.0	4
Q24	This course improved my ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal context	11 30.56%	25 69.44%	0 0.00%	0 0.00%	0 0.00%	36	4.3	4
Q25	This course improved my ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	13 36.11%	22 61.11%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q26	This course improved my ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	8 22.22%	23 63.89%	5 13.89%	0 0.00%	0 0.00%	36	4.1	4
Q27	This course improved my ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	10 27.78%	23 63.89%	3 8.33%	0 0.00%	0 0.00%	36	4.2	4

EM411 - I1-10

Systems and Decision Sciences		Responses (%)						Course	
		SA	A	N	D	SD	N	Mean	Med.
Q28	Recognize moral issues and make ethical decisions as a leader of character within each stage of the system lifecycle.	11 30.56%	21 58.33%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q29	Apply innovative, multidisciplinary systems thinking to identify a problem, specify the needs of multiple stakeholders.	11 30.56%	24 66.67%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q30	Design and conduct systems experiments, including collecting, analyzing and interpreting data.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q31	Define the problem, design solutions, make decisions, and implement the chosen engineering solution within a broad global and societal context.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q32	Perform systems decision-making that considers qualitative and quantitative aspects from multidisciplinary and non-traditional engineering domains.	8 22.22%	25 69.44%	3 8.33%	0 0.00%	0 0.00%	36	4.1	4
Q33	Accurately, clearly, concisely, and persuasively report findings, conclusions, and recommendations to the decision-makers and stakeholders in a multicultural context.	11 30.56%	23 63.89%	2 5.56%	0 0.00%	0 0.00%	36	4.3	4
Q34	Lead interdisciplinary teams to implement effective and efficient solutions within non-traditional engineering domains.	10 27.78%	24 66.67%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q35	Demonstrate the skills and interest for intellectual growth and learning for a career of professional excellence and service to the nation as an officer in the United States Army.	7 19.44%	25 69.44%	4 11.11%	0 0.00%	0 0.00%	36	4.1	4
Q36	Design or re-engineer a system or process in order to develop innovative alternatives that meet the needs of the client within realistic environmental constraints.	8 22.22%	24 66.67%	4 11.11%	0 0.00%	0 0.00%	36	4.1	4
Q37	Define and measure system performance to validate solutions.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4

Academy - I		Responses (%)						Individual	
		SA	A	N	D	SD	N	Mean	Med.
Q38	This instructor encouraged students to be responsible for their own learning.	20 55.56%	15 41.67%	1 2.78%	0 0.00%	0 0.00%	36	4.5	5
Q39	My Instructor made effective use of an online platform for instruction and collaboration.	25 69.44%	11 30.56%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5
Q40	This instructor used effective techniques for learning, both in class and for out-of-class assignments.	26 72.22%	10 27.78%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5
Q41	My instructor cared about my learning in this course.	27 75.00%	9 25.00%	0 0.00%	0 0.00%	0 0.00%	36	4.8	5
Q42	My instructor demonstrated respect for cadets as individuals.	30 83.33%	6 16.67%	0 0.00%	0 0.00%	0 0.00%	36	4.8	5

Academy Remote - I		Responses (%)						Individual	
		SA	A	N	D	SD	N	Mean	Med.
Q43	My instructor used an effective blend of techniques during remote classes.	25 69.44%	11 30.56%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5

Cognitive Domain - I		Responses (%)						Individual	
		SA	A	N	D	SD	N	Mean	Med.
Q44	This instructor stimulated my thinking.	22 61.11%	14 38.89%	0 0.00%	0 0.00%	0 0.00%	36	4.6	5
Q45	This instructor encouraged students to think creatively and take intellectual risks.	15 50.00%	14 46.67%	1 3.33%	0 0.00%	0 0.00%	30	4.5	5

TAB F AY2021 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-SE 14 May 2021

MEMORANDUM FOR COL Paul Evangelista, Engineering Management Program Director, Department of Systems Engineering

SUBJECT: AT21-2 Course Assessment – EM411, Project Management

1. The attached report summarizes the EM411 assessment and curriculum change actions for Academic Term 2021-2.

2. EM411 Project Management (PM) teaches cadets the skills required to lead an organization in achieving their performance goals with respect to cost, schedule, and performance. This course is nested with the Project Management Institute's PM Body of Knowledge (PMBOK). Course topics include project selection, project manager roles and responsibilities, Agile approach, organizational structure, project planning, budgeting, scheduling, resource allocation, monitoring and controlling, evaluation and termination, and risk assessment and ethics in the profession. Case studies and a course project based on the Cadet Barracks Construction Project at USMA illustrate real world PM techniques and procedures in practice and provide context to the conceptual aspects of this course. The course project and a problem set provide students with the opportunities to integrate PM software (Microsoft Project - MSP) into a Project Action Plan.

3. I recommend five primary changes for improvement to the course:

a. Continue to include the course long PM project based on Barracks Renovation or CEAC construction. If permissible, include a trip section to walk through the barracks construction or CEAC construction and provide an opportunity for cadets to engage with the construction project manager.

b. Sustain breaking the course project into smaller graded events. Project Stage Gate 1 was split into two deliverables. This allowed cadets to work on the project earlier in the course and aligned with the lessons when course material was taught. If possible, consider splitting Stage Gate 2 and 3.

c. Recommend reinstating the Term End Examination. Remove the Final Exam (WPR2) from the schedule. Move the due date for the Project Stage Gate 3 forward in the course. Reduce Stage Gate 2 and Stage Gate 3 to 75 points each and make the TEE worth 200 points (150 points from removing WPR2).

d. Coordinate for a trip section to Sikorsky in Connecticut in the Spring Semester. Contact Mr. Bryan Dougherty, Program Manager at Sikorsky at bryan.dougherty1@gmail.com.

e. Sustain the guest lecture PM in Action. The PM in Action lecture was well received by cadets this semester and it reinforced course concepts.

4. POC for this action is the undersigned at brandon.thompson@westpoint.edu.

BRANDON S. THOMPSON
LTC, MI/FA49
EM411 Course Director

Encls
1. Course Assessment
2. Lesson Objectives / Syllabus
3. EM411 Course Objective Assessment
4. EM Program Student Outcomes Crosswalk
5. Course End Feedback
6. Cadet Comments

Enclosure 1: Course Assessment – EM411 Project Management

Section I – Course Description

1. **Redbook Description.**

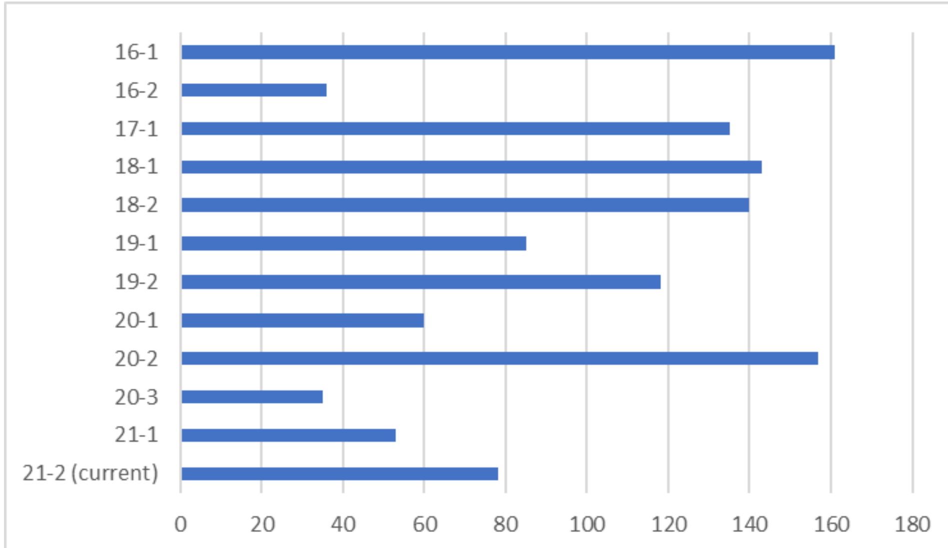
This course develops skills required to lead an organization to the achievement of their objectives through the proper application of the management of planning, implementing, and controlling the organization activities, personnel, and resources. The course focuses on the Implementation phase of the Systems Decision Process (SDP). Topics include project selection, roles and responsibilities of the project manager, planning the project, budgeting the project, scheduling the project, allocating resources to the project, monitoring and controlling the project, evaluating and terminating the project, risk assessment and management, organizational structure and human resources. Case studies illustrate problems and how to solve them. Course assignments are designed to help students learn and apply project management techniques taught in the course. The class design project will provide students with the opportunity to integrate project management software, Microsoft Project, into the preparation of an Engineering Management Project Plan.

a. **Lessons.** 35 @ 75 min (2.00 Att/wk) LABS: 0 @ 0 min
b. **Special Requirements.** One (1) half-day Trip Section
c. **Offered.**

AY16		AY17		AY18		AY19		AY20			AY21		AY22	
16-1	16-2	17-1	17-2	20-1	20-1	19-1	19-2	20-1	20-2	20-3	21-2	21-2	22-1	22-2
X	X	X		X	X	X	X	X	X	X	X	X	X	X

2. **Enrollment.**

Academic Term	EM411 Enrollment
16-1	161
16-2	36
17-1	135
18-1	143
18-2	140
19-1	85
19-2	118
20-1	60
20-2	157
16-1	161
21-1	53
21-2 (current)	78



3. **Current Course Objectives.**

a. Identify project management practices used in current organizations.

b. Identify and describe the necessary qualities and skills required of successful project managers.

c. Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.

d. Identify and utilize project management standards implemented by the Project Management Institute through the Project Management Body of Knowledge.

e. Utilize Microsoft Project 2019© as an effective tool in managing successful projects.

f. Utilize case studies to bridge the gap between project management theory and practice.

g. Exercise oral and written communications skills.

4. **Textbooks.**

a. *Project Management a Managerial Approach,* 10th Edition, by Jack R. Meredith, Scott M. Shafer, and Samuel J. Mantel Jr.

5. **Course Strategy.** This course is divided into three major blocks that generally align with the lifecycle of a project. The first block, **Project Initiation** consists of five lessons that deal with project selection, the project management profession, conflict and negotiation techniques, and project teams. The second block, **Project Planning**, consists of eighteen lessons and focuses on initial project planning, techniques to improve cost and duration estimates, creating a project budget and schedule, exploring the issues related to project scheduling and resourcing, Agile, and risk management. It covers topics such as PERT, CPM, project uncertainty, simulation, resource loading and leveling and allocating scarce resources. It also includes a site visit to see project management in the real world and introduces the

course project. This block culminates with four lessons to familiarize cadets with Microsoft Project. The third block, **Project Execution,** consists of twelve lessons and introduces project monitoring, earned value analysis, project control, ethics, auditing, and terminating a project. One lesson is dedicated to Project Management in Action and typically includes a guest lecturer to provide cadets with a real-world example of project management in industry. This block culminates with group project submissions and a case study presentation. During each block, practical exercises are used to illustrate relevant and applicable project management issues and bridge the gap between theory and practical application. Microsoft Project is used during the second half of the course to develop project management skills and increase cadet productivity. This course traditionally includes a Term End Examination (TEE); however, due to the COVID19 pandemic, AT21-1 classes were taught remotely for the second half of the semester and the TEE was modified into a Final Examination (WPR2) administered during class. A detailed lesson syllabus is included in Enclosure 3.

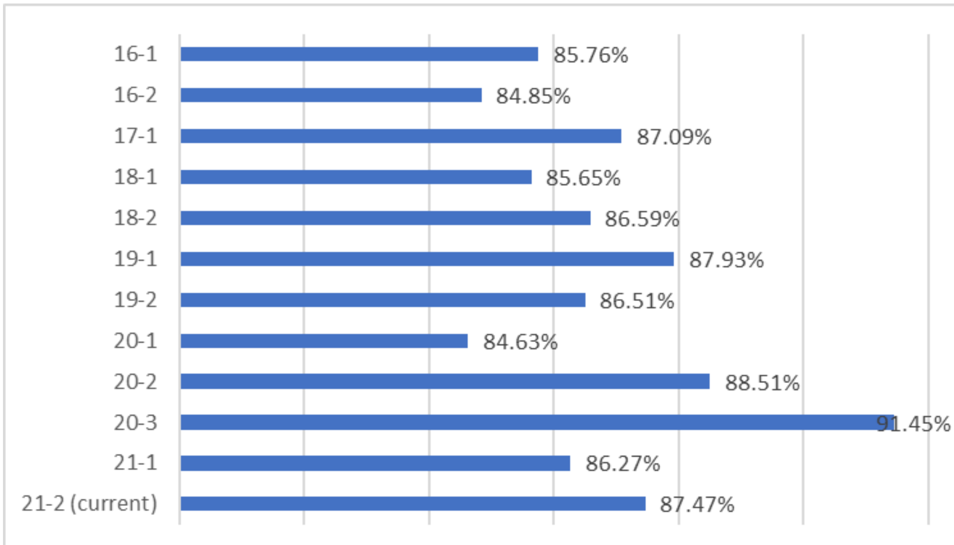
6. Graded Requirements.

AT21-2 Graded Events				
Graded Event	***Points***	***Lesson***	***Date***	***% of Course***
Project Deliverable PG1a	50	8	5-Feb	5.0%
Quiz 1	50	9	8-Feb	5.0%
Problem Set 1	35	12	17-Feb	3.5%
Problem Set 2	40	16	1-Mar	4.0%
Midterm	150	18	5-Mar	15.0%
Project Deliverable PG1b	100	19	11-Mar	10.0%
Problem Set 3	40	22	19-Mar	4.0%
Problem Set 4	35	25	29-Mar	3.5%
Project Deliverable PG2	100	28	8-Apr	10.0%
Quiz 2	50	30	14-Apr	5.0%
Final Examination	150	35	3-May	15.0%
Case Study Presentation	50	37	7-May	5.0%
Project Deliverable PG3	100	38	11-May	10.0%
Instructor Points	50	-	11-May	5.0%
Course Total:	1000			

Section II – Course Assessment

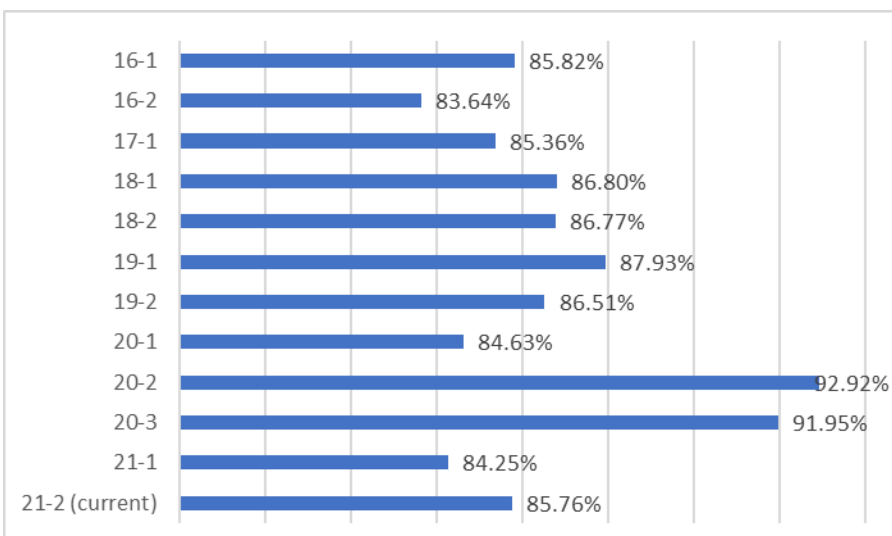
1. **Achievement of Course Objectives.**
    a. Course Objectives/Syllabus. Enclosure 2.
    b. Course Objective Assessment Table. Enclosure 3.
    c. Results of Course End Feedback (web-based). Enclosure 5.
    d. Results of Cadet Free-Form Comments. Enclosure 6.
    e. Historical Course Final Grade Averages.

Academic Term	EM411 Average
16-1	85.76%
16-2	84.85%
17-1	87.09%
18-1	85.65%
18-2	86.59%
19-1	87.93%
19-2	86.51%
20-1	84.63%
20-2	88.51%
20-3	91.45%
21-1	86.27%
21-2 (current)	87.47%



    f. Historical Term End Examination Grade Averages.

Academic Term	EM411 TEE Average
16-1	85.82%
16-2	83.64%
17-1	85.36%
18-1	86.80%
18-2	86.77%
19-1	87.93%
19-2	86.51%
20-1	84.63%
20-2	92.92%
20-3	91.95%
21-1	84.25%
21-2 (current)	85.76%



2. **Course Objectives.**

a. **Body of Knowledge.** This course develops skills required to lead an organization to the achievement of their objectives through the proper application of the management of planning, implementing, and controlling the organization activities, personnel, and resources. The course focuses on the Implementation phase of the Systems Decision Process (SDP). Topics include project selection, roles and responsibilities of the project manager, planning the project, budgeting the project, scheduling the project, allocating resources to the project, monitoring and controlling the project, evaluating and terminating the project, risk assessment and management, organizational structure and human resources. Case studies illustrate problems and how to solve them. Course assignments are designed to help students learn and apply project management techniques taught in the course. The class design project will provide students with the opportunity to integrate project management software, Microsoft Project, into the preparation of an Engineering Management Project Plan.

b. **Effective Assessment.** The end of course survey and the department's major-graded-event content approval process provide an adequate assessment tool for an impartial and effective assessment of the course objectives.

3. **Course Processes.**

a. **Textbook.** *Project Management a Managerial Approach,* 10th Edition, by Jack R. Meredith, Scott M. Shafer, and Samuel J. Mantel Jr. sufficiently covers the material outlined in the course objectives. The text is relatively easy to read, though at times cumbersome due to the length of sections. The Project Management Institute's PMBOK is used as a complementary source for project management knowledge. The Ethics lesson is derived from the PMBOK Code of Ethics.

b. **Software.** Microsoft Project 2019 is used throughout the second half of the course. Cadets are taught how to manage a project schedule, budget and resources using MS Project through in-class labs, practical exercises, and during the course project.

c. **Graded Requirements.** The graded requirements in EM411 are appropriate for gauging student comprehension, extending fundamental ideas, and demonstrating the real-world relevance of the material. The use of text material, computer-based practical exercises, hands-on labs, and case study analysis has proven to be sufficient for assessing course knowledge for engineering, management, and economics majors alike. There are, however, some recommended changes in section III.

4. **Course Pedagogy.**

a. **Course Project.** The course is allotted 3.5 credit hours due to the significant course project. The basic objectives for the course project required the cadets to go through the entire project management process from initiation and selection through monitoring and control. The project is based on the numerous upgrades to facilities currently underway at USMA to provide realism to the project.

b. **Case Study.** We maintained the 50-point graded Case Study from the previous semester. The intent was for students to research case studies from the Project Management

Institute and present a 10-minute presentation highlighting the project management tools that were applied or should have been applied in the case study. Due to COVID19 restrictions including a shortened schedule, the case study presentation assignment was modified to require cadet groups to record a 10-minute video presentation. All six groups' presentations were then played in one lesson. Recommend sustaining the recorded presentations until classes return to 40 lessons and cadets are allowed in the classroom.

c. **Instructor Points.** Cadet performance was measured based on participation in class and preparation for class and awarded points accordingly. Recommend establishing a grading rubric for instructor points for consistency between sections.

d. **Term End Exam.** In accordance with department guidance issued, the TEE was replaced with a smaller Final exam issued on lesson 37. The exam covered lesson material from the second half of the course.

5. **EM Program Student Outcomes Crosswalk.** Enclosure 5.

Section III – Assessment of Recent Changes and Proposed Future Changes.

1. **Changes to Current Academic Term.**
   a. **Course Objectives.** None.
   b. **Textbook.** None
   c. **Curriculum.** The course was taught in a hybrid manner with some students present in the classroom and some remote via Blackboard Connect. The last three weeks of class were all conduct remote due to increases in COVID cases. Class lectures were recorded and posted prior to class. Cadets were required to view the lesson videos before class and come to class prepared to work on practical exercise. The Project Management in Action lessons were canceled due to the shortened semester.
   d. **Graded Requirements.** Recommend sustaining the four problem set model and add lesson quizzes where needed to account for the Instructor grade. Both quizzes and WPR2 were conducted via Blackboard. Blackboard quizzes require careful consideration due to increased time requirements and creative grading. Multiple choice questions are the preferred option for Blackboard tests but do not provide a chance for partial credit. The TEE was modified into a 150-point Final Exam (WPR2) conducted during class time via Blackboard.
   e. **Other.** None.

2. **Proposed Change for Next Academic Term.**
   a. **Course Objectives.** None.
   b. **Textbook.** None.
   c. **Curriculum.** See five recommended changes above.
   d. **Graded Requirements.** Recommend reinstating the TEE and removing the Final Exam. Move Quiz 2 to a lesson that allows you to test the major course concepts for the second half of the semester.
   e. **Other.** None.

Enclosure 2: EM411 Lesson Objectives/Syllabus

Lesson 1: Definition of Project Management
1. Define a project, its six characteristics and differentiate it from a function (K)
2. Explain the three direct goals of every project (K)
3. Explain the project life cycle and the two prevalent shapes it can take (K)

Lesson 2: Project Initiation and Project Selection
1. Apply non-numeric project selection models (sacred cow, operating necessity, competitive necessity, product line extension and comparative benefit, sustainability) (S)
2. Apply numeric project selection models (payback period, discounted cash flow, weighted scoring model.) (S)
3. Explain Project Portfolio Management (PPM) (K)

Lesson 3: The Project Manager and the Profession
1. Identify the key differences between a project manager and a functional manager (K)
2. List and describe the responsibilities, demands, and skills and attributes of an effective project manager (K)
3. List and explain desirable team characteristics (K)

Lesson 4: Conflict and the Art of Negotiation
1. Analyze project team problems and conflicts. Determine sources and categories of conflict within the project life cycle (S)
2. Explain the process of principled negotiation to include its three requirements and the four main points of the method (K)

Lesson 5: Project Organization and Teams
1. Identify the three primary types of project organizations with advantages and disadvantages of each type (K)
2. Explain the factors to be considered when staffing and managing project teams (K)

Lesson 6: Project Planning I
1. Explain the purpose of the initial project coordination and the contents of the Project Charter (K)
2. Develop a Work Breakdown Structure (WBS) for a project (S)
3. Develop a Linear Responsibility Chart for a project (S)

Lesson 7: Project Planning II – Agile Project Management
1. Explain the purpose of Agile Project Management (K)
2. Explain the differences between the Waterfall and Agile Project Management methodologies (K)

Lesson 8: Quiz 1/Project Planning Review
Execute concepts from Lessons 1-5 in a graded environment (S)

Lesson 9: Budgeting and Cost Estimation
1. Apply top-down and bottom-up budgeting strategies to include direct/indirect costs and fully costing a project (S)
2. Differentiate category-oriented versus program-oriented budgeting (S)

Lesson 10: Improving Cost Estimates
1. Recognize sources of budget uncertainty (K)
2. Implement methods to improve cost estimates, including using Learning Curves (S)

Lesson 11: Including Risk in Project Management
1. Analyze project risk to include: Risk Identification, Risk Analysis, and Response to Risk and create a Risk Assessment given a scenario (S)

Lesson 12: Scheduling I – An Introduction to PERT and CPM
1. Create CPM networks and conduct a forward and backward pass to determine the critical path in a network and determine slack (S)
2. Explain the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) (K)

Lesson 13: DROP

Lesson 14: Scheduling II – Solving the Network
Calculate probabilistic activity times (S)

Lesson 15: Scheduling III – Uncertainty in a Project
1. Analyze the probability of completing a project on time (S)
2. Analyze the duration of a project, given known risk (S)

Lesson 16: Problem Solving Lab – Scheduling
Review all skills for project scheduling

Lesson 17: Midterm Exam Review

Lesson 18: Midterm Exam
Execute concepts from Lessons 1-14 in a graded environment

Lesson 19: Class Drop

Lesson 20: Expediting a Project (Resource Allocation)
Expedite a project by "crashing" tasks and use of fast-tracking (S)

Lesson 21: Resource Loading and Leveling
Evaluate resource usage and conduct resource loading and leveling (S)

Lesson 22: Introduction to Microsoft Project
Use computer software packages (MS Project) to enhance a Project Managers ability to monitor, assess, and control a project (S)

Lesson 23: Allocating Scarce Resources and Multi-Project Scheduling in MSP
Apply scarce resource allocation using priority rules (to include across multiple projects) (S)

Lesson 24: Microsoft Project Practice Lab
Use computer software packages (MS Project) to enhance a Project Manager's ability to plan, monitor, assess, and control a project (S)

Lesson 25: Project Stage Gate #2 Working Session
Implement all MS Project skills for the course project's second deliverable

Lesson 26: Monitoring a Project
Explain the plan-monitor-control cycle and considerations for designing a monitoring system (K)

Lesson 27: Earned Value Management I
1. Define earned value terminology and explain the EVM process (K)
2. Apply earned value analysis to various projects (S)

Lesson 28: Earned Value Management II
1. Define earned value terminology and explain the EVM process (K)
2. Apply earned value analysis to various projects (S)

Lesson 29: Problem Solving Lab – Earned Value Management
Review all skills for earned value management

Lesson 30: DROP

Lesson 31: Project Control and Auditing
1. Explain the control cycle (K)
2. Describe scope creep and change control (K)
3. Explain the audit process to include the purpose, depth, and timing of an audit (K)

Lesson 32: Quiz 2/Project Termination
1. Execute concepts from Lessons 19-31 in a graded environment (S)
2. Explain the varieties of project termination (K)
3. Analyze the factors to consider in determining when to terminate a project (S)

Lesson 33: Ethics in the Profession
1. Describe the Project Management Code of Ethics and the Ethical Decision Making Framework (B)
2. Explain the differences between PM CoE and the ethical standards of an Army Officer (B)

Lesson 34: Final Exam Review

Lesson 35: DROP

Lesson 36: Final Exam
Execute concepts from Lessons 16-35 in a graded environment

Lesson 37: DROP

Lesson 38: PM Case Study Presentations
Leverage skills from lessons 1-37 to analyze and present a Case Study (S)

EM411 Syllabus

Phase	Lesson	Date	Lesson Name	Graded Event	Reading Sections
Project Initiation			**Project Initiation**		
	1	19-Jan	Definition of Project Management		1.1-1.4 (pp. 1-23)
	2	20-Jan	Project Initiation		2.1-2.3 (pp. 28-55)
	3	22-Jan	PM and Profession		3.1-3.4 (pp. 77-105)
	4	26-Jan	Conflict and the Art of Negotiation		4.1-4.6 (pp. 118-137)
	5	27-Jan	Project Organization and Teams	**MS Project Template due - 10 Instructor Points**	5.1-5.8 (pp. 145-176)
Project Planning			**Project Planning**		
	6	29-Jan	Project Planning I		6.1, 6.3 (pp. 187-208, 214-217)
	7	2-Feb	Project Planning II - Agile Project Management		6.2 (pp. 208-214)
	8	4-Feb	**Class Drop**	**PM Meeting for Stage Gate 1a Due NLT 1630 5 FEB**	
	9	8-Feb	Project Planning Review / **Quiz 1**	**Quiz 1**	
	10	10-Feb	Budgeting and Cost Estimation		7.1 (pp. 233-246)
	11	12-Feb	Improving Cost Estimates		7.2 (pp.246-256)
	12	17-Feb	Including Risk in Project Management	**Problem Set #1 Due**	7.3-7.4 (pp. 256-279)
	13	19-Feb	Scheduling I - An Intro to PERT/CPM		
	14	23-Feb	Scheduling II - Solving the Network		Review 8.1-8.2 (pp. 291-312)
	15	25-Feb	Scheduling III - Uncertainty in a Project		8.2 (pp. 312-319)
	16	1-Mar	**Midterm Review**	**Problem Set #2 Due**	Review lessons 1-15
	17	3-Mar	**Class Drop**		
	18	5-Mar	**Midterm**	**Midterm WPR**	Review lessons 1-15
	19	11-Mar	Expediting a Project	**Project Stage Gate #1 Due by 1630 11 MAR**	9.1 (pp. 348-356)
	20	15-Mar	Resource Loading and Leveling		9.2-9-4 (pp. 356-366)
	21	17-Mar	**Class Drop**		
	22	19-Mar	Introduction to MS Project	**Problem Set #3 Due**	MS Project step-by-step
	23	23-Mar	Allocating Scarce Resources and Multi-Project Scheduling in MS Project		
	24	25-Mar	**MS Project Practice Lab**		MS Project step-by-step
	25	29-Mar	Project Stage Gate #2 Working Session	**Problem Set #4 Due**	
Project Execution / Monitor and Control / Termination			**Project Execution, Monitor and Control, Termination**		
	26	1-Apr	Monitoring a Project		10.1-10.2 (pp.389-398)
	27	6-Apr	Earned Value Management I (Lab)		
	28	8-Apr	Earned Value Management II	**Project Stage Gate #2 (MSP) Due by 1630 8 APR**	10.3 (pp. 398-411)
	29	12-Apr	**Problem Solving Lab - Earned Value Management**		11.1-11.3 (pp.421-442)
	30	14-Apr	**Quiz 2**	**Quiz 2**	
	31	16-Apr	Project Control & Auditing		12.1-12.5 (pp. 459-476)
	32	20-Apr	Project Termination and Ethics in the Profession		13.1-13.4 (pp. 488-505); PMBOK
	33	23-Apr	**Class Drop**		
	34	28-Apr	**Final Exam Review**		Review lessons 18-34
	35	3-May	**Final Examination**	**Final WPR**	Review lessons 18-34
	36	5-May	**PM in Action (Dean's Hour Brief)**		
	37	7-May	**Class Drop**	**PM Case Study Due by 1630 9 MAY**	
	38	11-May	PM Case Study Presentations	**Project Stage Gate #3 Due by 1630 11 MAY**	PMBoK Handout
	39				
	40				

Enclosure 3: EM411 Course Objective Assessment

	EM411 Course Objectives	Direct Assessment	Course Director's Assessment	Applicable Graded Events
1.	**Identify project management practices used in current organizations.**	**3.33**	**4.00**	Quiz 1 WPR #1 Case Study WPR #2
2.	**Identify and describe the necessary qualities and skills required of successful project managers.**	**3.33**	**4.00**	Quiz 1 Project SG #1 WPR #1 Project SG #2 Case Study Project SG #3
3.	**Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.**	**3.33**	**3.50**	Quiz 1 Problem Set #1 Project SG #1 Problem Set #2 WPR #1 Problem Set #3 Problem Set #4 Project SG #2 Quiz 2 WPR #2 Project SG #3
4.	**Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the Project Management Body of Knowledge (PMBOK).**	**3.33**	**3.00**	Quiz 1 Problem Set #1 Project SG #1 WPR #1 Project SG #2 Project SG #3
5.	**Utilize Microsoft Project 2019© as an effective tool in managing successful projects.**	**3.00**	**3.00**	Problem Set #4 Project SG #2 Project SG #3
6.	**Utilize case studies to bridge the gap between project management theory and practice.**	**3.33**	**3.50**	Project SG #1 Project SG #2 Case Study WPR #2
7.	**Exercise oral and written communications skills.**	**3.33**	**3.50**	Project SG #1 Project SG #2 Project SG #3 Class Participation
	Overall Assessment		**3.50**	All Graded Events

NOTES:
1) Graded Events included: Quizzes 1-2, Problem Sets 1-4, WPRs 1-2, Project Stage Gates 1-3, Case Study, Participation
2) Student Assessment is based on the End-of-Course Survey
3) Course Director Assessment is:
  -- *Based on course average for “All graded events”
  -- All others based on individual questions on graded events to include TEE question results

Scale: 1 = Unsatisfactory, 2 = Developing, 3 = Satisfactory, 4 = Exemplary

Enclosure 4: EM Program Student Outcome Crosswalk.

EM Program Student Outcomes \ EM411 Course Objectives	1. Identify project management practices used in current organizations.	2. Identify and describe the necessary qualities and skills required of successful project managers.	3. Be able to successfully plan, schedule, resource, monitor, control, terminate, and audit a wide array of projects.	4. Identify and utilize project management standards implemented by the Project Management Institute (PMI) through the Project Management Body of Knowledge (PMBOK).	5. Utilize Microsoft Project 2019© as an effective tool in managing successful projects.	6. Utilize case studies to bridge the gap between project management theory and practice.	7. Exercise oral and written communications skills.
1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	X		X	X			
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors				X		X	
3. an ability to communicate effectively with a range of audiences						X	X
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts				X		X	
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives						X	X
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions			X	X	X		
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies	X	X		X			

Enclosure 5: Course End Feedback

EM411 AT21-2 Course End Evaluation

		5	4	3	2	1			
					EM411 - I1-10				
Academy				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q1	My fellow students contributed to my learning in this course.	12 33.33%	20 55.56%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q2	My motivation to learn and to continue learning has increased because of this course.	10 27.78%	22 61.11%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q3	My personal schedule allows me enough time to reflect on the material I have learned in class.	13 36.11%	19 52.78%	4 11.11%	0 0.00%	0 0.00%	36	4.3	4
Q4	My personal schedule allows me enough time to adequately prepare for my optimum academic performance.	12 33.33%	19 52.78%	5 13.89%	0 0.00%	0 0.00%	36	4.2	4

					EM411 - I1-10				
Academy Remote				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q5	Graded events (WPR's, Projects, Papers, TEE's, etc.) in this course were effective and fair in the remote environment.	18 50.00%	18 50.00%	0 0.00%	0 0.00%	0 0.00%	36	4.5	5
Q6	I took more responsibility for my own learning in this course as a result of switching to a remote learning environment.	13 36.11%	15 41.67%	7 19.44%	1 2.78%	0 0.00%	36	4.1	4
Q7	I was able to maintain my attention in this course during remote class sessions to the same extent as in-person classes.	9 25.00%	15 41.67%	7 19.44%	4 11.11%	1 2.78%	36	3.8	4
Q8	I had to work harder in this course to achieve a similar amount of learning because of the transition to a remote environment.	3 8.33%	10 27.78%	9 25.00%	13 36.11%	1 2.78%	36	3.0	2
Q9	I had to spend more time in this course to do the work required in the remote environment.	4 11.11%	7 19.44%	11 30.56%	13 36.11%	1 2.78%	36	3.0	2
Q10	Participating in remote learning made me want to continue to learn on my own at USMA and beyond.	6 17.14%	18 51.43%	10 28.57%	0 0.00%	1 2.86%	35	3.8	4

					EM411 - I1-10				
Cognitive Domain				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q11	In this course, my critical thinking ability increased.	12 34.29%	21 60.00%	2 5.71%	0 0.00%	0 0.00%	35	4.3	4
Q12	The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a2:1 ratio of out-of-class time versus in-class time.	10 27.78%	26 72.22%	0 0.00%	0 0.00%	0 0.00%	36	4.3	4
Q13	The classroom environment (e.g. desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn.	10 27.78%	20 55.56%	6 16.67%	0 0.00%	0 0.00%	36	4.1	4
Q14	In this course, my ability to think creatively and take intellectual risks increased.	9 25.00%	22 61.11%	4 11.11%	1 2.78%	0 0.00%	36	4.1	4

					EM411 - I1-10				
Systems Engineering				Responses (%)				Course	
		SA	A	N	D	SD	N	Mean	Med.
Q15	This course helped me learn to design, manage or reengineer systems or processes.	13 36.11%	20 55.56%	3 8.33%	0 0.00%	0 0.00%	36	4.3	4
Q16	This course taught me to communicate effectively both orally and in writing.	7 19.44%	28 77.78%	1 2.78%	0 0.00%	0 0.00%	36	4.2	4
Q17	This course improved my ability to solve real-world problems through quantitative techniques.	11 30.56%	24 66.67%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q18	This course provided me with practical, problem-solving experiences applicable to my future as an Army officer.	12 33.33%	23 63.89%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q19	Course exercises and designs were realistic experiences that enabled open ended analysis, design, or implementation of systems.	14 38.89%	20 55.56%	2 5.56%	0 0.00%	0 0.00%	36	4.3	4
Q20	This course has motivated me to pursue deeper learning in related systems engineering or engineering management topics.	11 30.56%	20 55.56%	5 13.89%	0 0.00%	0 0.00%	36	4.2	4
Q21	This course improved my ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	9 25.00%	22 61.11%	5 13.89%	0 0.00%	0 0.00%	36	4.1	4
Q22	This course improved my ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	7 19.44%	23 63.89%	6 16.67%	0 0.00%	0 0.00%	36	4.0	4
Q23	This course improved my ability to communicate effectively with a range of audiences.	6 16.67%	25 69.44%	5 13.89%	0 0.00%	0 0.00%	36	4.0	4
Q24	This course improved my ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal context	11 30.56%	25 69.44%	0 0.00%	0 0.00%	0 0.00%	36	4.3	4
Q25	This course improved my ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	13 36.11%	22 61.11%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q26	This course improved my ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	8 22.22%	23 63.89%	5 13.89%	0 0.00%	0 0.00%	36	4.1	4
Q27	This course improved my ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	10 27.78%	23 63.89%	3 8.33%	0 0.00%	0 0.00%	36	4.2	4

EM411 - I1-10

Systems and Decision Sciences		Responses (%) SA	A	N	D	SD	N	Course Mean	Med.
Q28	Recognize moral issues and make ethical decisions as a leader of character within each stage of the system lifecycle.	11 30.56%	21 58.33%	4 11.11%	0 0.00%	0 0.00%	36	4.2	4
Q29	Apply innovative, multidisciplinary systems thinking to identify a problem, specify the needs of multiple stakeholders.	11 30.56%	24 66.67%	1 2.78%	0 0.00%	0 0.00%	36	4.3	4
Q30	Design and conduct systems experiments, including collecting, analyzing and interpreting data.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q31	Define the problem, design solutions, make decisions, and implement the chosen engineering solution within a broad global and societal context.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q32	Perform systems decision-making that considers qualitative and quantitative aspects from multidisciplinary and non-traditional engineering domains.	8 22.22%	25 69.44%	3 8.33%	0 0.00%	0 0.00%	36	4.1	4
Q33	Accurately, clearly, concisely, and persuasively report findings, conclusions, and recommendations to the decision-makers and stakeholders in a multicultural context.	11 30.56%	23 63.89%	2 5.56%	0 0.00%	0 0.00%	36	4.3	4
Q34	Lead interdisciplinary teams to implement effective and efficient solutions within non-traditional engineering domains.	10 27.78%	24 66.67%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4
Q35	Demonstrate the skills and interest for intellectual growth and learning for a career of professional excellence and service to the nation as an officer in the United States Army.	7 19.44%	25 69.44%	4 11.11%	0 0.00%	0 0.00%	36	4.1	4
Q36	Design or re-engineer a system or process in order to develop innovative alternatives that meet the needs of the client within realistic environmental constraints.	8 22.22%	24 66.67%	4 11.11%	0 0.00%	0 0.00%	36	4.1	4
Q37	Define and measure system performance to validate solutions.	9 25.00%	25 69.44%	2 5.56%	0 0.00%	0 0.00%	36	4.2	4

Academy - I		Responses (%) SA	A	N	D	SD	N	Individual Mean	Med.
Q38	This instructor encouraged students to be responsible for their own learning.	20 55.56%	15 41.67%	1 2.78%	0 0.00%	0 0.00%	36	4.5	5
Q39	My Instructor made effective use of an online platform for instruction and collaboration.	25 69.44%	11 30.56%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5
Q40	This instructor used effective techniques for learning, both in class and for out-of-class assignments.	26 72.22%	10 27.78%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5
Q41	My instructor cared about my learning in this course.	27 75.00%	9 25.00%	0 0.00%	0 0.00%	0 0.00%	36	4.8	5
Q42	My instructor demonstrated respect for cadets as individuals.	30 83.33%	6 16.67%	0 0.00%	0 0.00%	0 0.00%	36	4.8	5

Academy Remote - I		Responses (%) SA	A	N	D	SD	N	Individual Mean	Med.
Q43	My instructor used an effective blend of techniques during remote classes.	25 69.44%	11 30.56%	0 0.00%	0 0.00%	0 0.00%	36	4.7	5

Cognitive Domain - I		Responses (%) SA	A	N	D	SD	N	Individual Mean	Med.
Q44	This instructor stimulated my thinking.	22 61.11%	14 38.89%	0 0.00%	0 0.00%	0 0.00%	36	4.6	5
Q45	This instructor encouraged students to think creatively and take intellectual risks.	15 50.00%	14 46.67%	1 3.33%	0 0.00%	0 0.00%	30	4.5	5

SECTION II

EXAMINATIONS

TAB G WPR/Midterm

Instructor Solution

Name: ____________________________

Section: ____________________________

Time Complete: ____________________________

EM411 – Project Management
AT26-1 Midterm Examination (Version A)

1. This is a 75-minute Written Partial Review worth 175 points. The exam consists of six topic areas and 15 pages, including this cover page. Ensure that your exam is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: This is a closed book exam. You may utilize a nongraphic calculator, one double-sided 5" x 8" handwritten notecard, and the formula reference provided. **Ensure your name is written on the notecard and turned in with the Exam**.

3. Feel free to disassemble the exam. A stapler is available to reattach the pages when you are finished. When reassembling your exam, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. EXTRA PAPER: Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name, and section.

6. GRADING SUMMARY:

	Section	*Possible*	*Earned*
I.	Project Selection and Initiation	25	
II.	Project Organization, Teams, & Stakeholders	25	
III.	Project Planning	20	
IV.	Risk in Project Management	20	
V.	Budgeting & Cost Estimation	35	
VI.	Scheduling	50	
	Total:	175	

This Midterm Examination will be released from academic security on Friday, 17 Oct 2025. Prior to that time, I will not discuss any aspects of this WPR with anyone except an EM411 instructor. By signing my name below, I acknowledge this requirement.

I. Project Initiation and Selection (25 pts)

Questions 1-10: Multiple Choice (2pts each): Select the BEST answer to each question from the options provided.

1. In non-numeric project selection, the "Competitive Necessity" approach is typically used when:

a) A company must launch a project to stay relevant in the market
b) An executive insists on a project
c) A project must comply with safety laws
d) The payback period is short

2. Weighted scoring models are most often used to overcome the following disadvantage of profitability selection models.

a) The inability to account for other qualitative or strategic factors
b) The inability to account for project results beyond the payback period
c) The inability to account for time value of money
d) The inability to account for cash flow

3. There are three characteristics that **all** projects share. Which of the following is one of those characteristics?

a) The PM sometimes makes trade-offs
b) The PM must consider time, schedule, and budget
c) The project is one-time occurrence
d) The outcome of the project may change over time

4. When using Net Present Value (NPV)/Discounted Cash Flow (DCF), ______________________.

a) a rate of return or hurdle rate is required
b) the time value of money is not accounted for
c) risk is measured in the time it takes to recoup the initial investment
d) multiple criteria are considered in the evaluation

5. When faced with conflicting expectations among key stakeholders relating to a project's goals, project managers must make appropriate ____________ in order to balance competing demands related to these objectives.

a) weights
b) guidelines
c) trade-offs
d) tasks

6. The three direct goals of every project are _____________, ______________, and _______________.

a) quality, scope, time
b) scope, time, cost
c) cost, time, budget
d) schedule, cost, time

7. Which of the following is true of the payback period model:

a) It considers cash flow once the initial investment is recovered
b) A shorter payback period is more attractive
c) It accounts for annual changes in cash flows
d) It is adequate for calculating risk for the duration of the project

8. True or False, all businesses which maintain a project portfolio employ a Project Management Office to oversee project delivery.

a) True
b) False

9. A project selection method that subjectively ranks projects in order to select the "best" project for the company to undertake.

a) sustainability
b) product line extension
c) comparative benefit
d) operating necessity

10. A company must initiate a project to meet a change in government regulations. This project selection method is best classified as:

a) sustainability
b) product line extension
c) comparative benefit
d) operating necessity

11. (5 pts) Your company is considering a 4-year engineering project that requires an **immediate investment of $255,000**. You will only see cash flows in **years 2 and 4 – which you estimate to be $125,000 for each year**. Your company expects an **3% rate of return** on any investment, and you estimate 4**% annual inflation** for the next 4 years. **Using only DCF/NPV, should the company invest in this project?** Justify your answer. Show all work.

Year	0	1	2	3	4
Cash Flow	($255,000.00)	$0.00	$125,000.00	$0.00	$125,000.00

NPV = -255,000 + 125,000 / $(1 + .03 + .04)^2$ + 125,000 / $(1 + .03 + .04)^4$
= ($50,458.26)

No, at the end of the 4-years, the NPV is negative. In terms of profitability, this project is a non-profitable endeavor.

II. Project Organization, Teams, and Stakeholders (25 pts)

12. (4 pts) What two items best describe a **Projectized Organization (select two):**

a) Focus on standalone projects
b) Project Managers retain significant authority
c) Maximum flexibility in the use of staff
d) Team members are mentored by Functional Managers

13. (7 pts) Match the following **advantages** to the correct Project Organization.

Organization Description	
A	*There is less anxiety about what happens post - project*
B	*High level of commitment and motivation*
A	*Specialists in the division can be grouped to share knowledge and experience*
B	*The lines of communication are shortened*
A	*Unity of command exists*
B	*The ability to make swift decisions is greatly enhanced*
B	*Supports a holistic approach to the project*

Organization	
A	*Functional Organization*
B	*Projectized Organization*

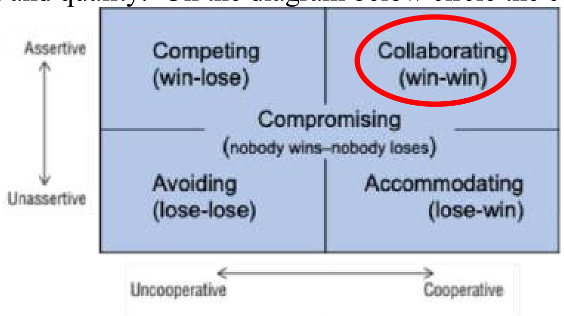
14. (2 pts) Desired traits for an effective Project Manager include all of the following EXCEPT:

a) Uses a Systems approach
b) Possesses high technical skills and analysis mindset
c) Able to act as a facilitator and generalist
d) The ability to get the job done

15. (3 pts) True or False, traditional categories of conflict include (1) Interpersonal conflict, (2) Differing goals and expectations, and (3) Uncertainty regarding decision authority.

a) True
b) False

16. (3 pts) A contractor wants to use cheaper material to save money, but the project team insists on higher-grade material for long term durability. Both sides meet, analyze lifecycle costs, and agree on a mid-tier option that balances cost and quality. On the diagram below circle the conflict resolution strategy used.



17. (6 pts) A project lifecycle includes Initiation, Execution, and Closure phases. Match the following types of conflict with the Project Stage in which they are most commonly found.

	Type of Conflict
C	Scheduling Deadlines and Budget
A	Technical Objectives and Resource Commitment
B	Functional Manager Commitment and Scheduling

	Project Stage
A	Project Initiation
B	Project Execution
C	Project Closure

III. Project Planning (20 pts)

18. (3 pts) A Project Launch Meeting______________ allows the PM to discuss the project in sufficient detail with key stakeholders, in order to develop a general understanding of the technical scope, areas of responsibility, tentative schedules and budgets, and needed risk management group.

19. (3 pts) The __Project Charter__________________________is a high-level document that helps define the scope and gives the PM authority to direct activities, request resources and spend funds.

20. (4 pts) A Work Breakdown Structure (WBS) is the main tool for managing project's scope and is typically broken down to the _____work package___________________level, which can be managed by one individual.

21. (8 pts) Match the following descriptions of Project Charter Elements to the correct name from the list of options. ***(NOTE: You will only use 4 of the 6 names listed.)***

Project Charter Element Descriptions	
B	*A more detailed statement of the projects goals, their priorities, what constitutes success, and measurable goals*
C	*High-level description of the project and its requirements, including managerial and technical approaches to the work.*
A	*A short summary of the project's overall scope and objectives.*
G	*This section covers potential problems as well as potential opportunities that could affect the project.*

Elements	
A	*Purpose*
B	*Objectives*
C	*Overview*
D	*Schedules*
E	*Resources*
F	*Stakeholders*
G	*Risk Management Plan*
H	*Evaluation Methods*

22. (2 pts) When developing a Linear Responsibility Matrix, a person who is ultimately answerable for the correct and thorough completion of a deliverable is given which role?

a) Responsible (R)
b) Accountable (A)
c) Consulted (C)
d) Informed (I)

IV. Risk in Project Management (20 pts)

23. During Project Planning, you utilize a quantitative risk analysis technique to identify two major risks to the project.

Part a. (9 points) Complete the Risk Management Annex below, by filling in each blank in the Table.

Risk #	**Risk**	**S**	**L**	**D**	**RPN**	**Risk Response Type**	**Risk Response Action**
1	Custom Repair Parts are defective, which could cause a major mechanical failure	6	4	7	**168**	**Mitigate**	Develop and implement an inspection and testing procedure for all mechanical parts delivered. This will improve our ability to detect a flaw prior to installation.
2	FAA Travel Restrictions delay key team members from traveling to competition	8	3	4	**96**		

Part b. (4 points) There are three major types of risk. Identify which type(s) of risk have been identified in the table above.

Risk #1 = ____Preventable_________   Risk #2 = ___External___________

Part c. (3 points) Initial Risk Assessments are focused mainly on externalities, while risk associated with schedule, budget, and resource allocation is addressed during the project planning phase.

a. True

b. False

Part d. (4 points) If you were the PM, and you were **considering two approaches** of risk response to either remove the risk from the project to another stakeholder or do nothing, what **two responses** are you referring to? (select 2)

a. Avoid
b. Mitigate
c. Transfer
d. Accept

V. Budgeting and Cost Estimation (35 pts)

24. After completing an initial project plan, you begin the process of budgeting and cost estimation. You received the following information from your team members:

Resources	***Number required***	***Cost per unit***	***Total Cost***
Floor panels	10	$11,000 each	$110,000.00
Wall panels	12	$7,000 each	$84,000.00
Window panels	8	$3,000 each	$24,000.00
Roofing panels	8	$5,000 each	$40,000.00
Truss	4	$15,500 each	$62,000.00
Door	6	$2,500 each	$15,000.00
TOTAL MATERIALS COST			***$335,000.00***
Project Manager	300 hrs	$450/day	*$16,875.00*
Foreman	200 hrs	$250/day	*$6,250.00*
Document Control Mgr	100 hrs	$100/day	*$1,875.00*
TOTAL INTERNAL LABOR COST			***$25,000.00***
Inspector (Contractor)	48 hrs	$150/day	*$900.00*
Floor Sub-Contract	250 hrs	$24/hour	*$6,000.00*
Vertical Sub-Contract	200 hrs	$48/hour	*$9,600.00*
Roof Sub-Contract	150 hrs	$52/hour	*$7,800.00*
TOTAL CONTRACTED LABOR COST			***$24,300.00***

Organization Charges:
- Overhead Charges at 10% of internal labor costs,
- General Administrative costs at 5% of internal and contracted labor costs.
- Your organization includes a 15% contingency on all direct and indirect costs.

a) (10 pts) Given the project plan and organizational charges above, provide the Project Manager's **cost baseline or "fully costed" project budget** (SHOW YOUR WORK).

Overhead	0.10	
G&A	0.05	
Contingency	0.15	
Materials	$ 335,000.00	
Internal Labor	$ 25,000.00	
Contracted Labor	$ 24,300.00	
Direct Costs		$384,300.00
Overhead	$2,500.00	
G&A	$2,465.00	
Indirect		$4965.00
Direct + Indirect		$389,365.00
Contingency		$58,404.75
Fully Costed		$447,769.75

b) (2 pts) The budget table above (Question 26, Page 9) is an example of ___________________.

a. Activity-Oriented Budget
b. Top-Down Budget
c. Category-Oriented Budget
d. Parametric Budget

25. (2 pts) It costs the contractor $6 million for its 26-person crew to complete a 4-month project. The Budget Office estimates that it will cost $4.75 million for the crew to complete a 2.5-month project. What **budget technique** is the Budget Office using to complete this estimate? What type of **budgeting strategy** is this?

Budget Technique: Analogous, Parametric, Scaling, or Rule of Thumb/Expert Opinion **(circle one)**

Budget Strategy: Top-Down, Bottom-Up, Iterative **(circle one)**

26. (3 pts) There are 5 common sources of Budget Uncertainty. Select 3 of them from the list below.

1. Technological Shock
2. Change Management process
3. Poor Estimating Techniques
4. Personnel Turnover
5. Project Charter details

27. (10 pts) In order to better estimate task completion times for elements in the WBS, you consider the impact of learning curves on the placement of floor panels. Your team calculates that the **learning rate of 90 percent** will be **fully realized after the 10th attempt**. They calculate the **10th attempt will take 6 hours to complete**. If this is true, how long would it take to place the **1st floor panel**?

$T_n = T_1 * n^r$; Hrs to complete the task on the 1st attempt. n = 10, $T_{10} = 6, lr = 0.88$

$6 = T_1 * 10^r$

$6 = T_1 * 10^{-0.1520}$ $\quad r = \frac{\ln(lr)}{\ln(2)} \quad r = \frac{\ln(.90)}{\ln(2)} \quad r = -0.1520$

$T_1 = 6 / 0.7046$

$T_1 = 8.5\ hours$

28. (8 pts) During project execution, you learn that the team actually only required 5 **hours to complete the 10th attempt, a new fully realized time.** If it took **80.3 hours** to complete the first 10 attempts, how long will it take to complete 30 floor panels.

$T_{1-10} = 80.3\ hours$

$T_{realized} = 5\ hours$

$T_{1-10} + T_{11-30} = Total\ Time$

80.3 hours + (5 * 20) = Total Time = 180.3 hours.

VI. Scheduling (50 pts)

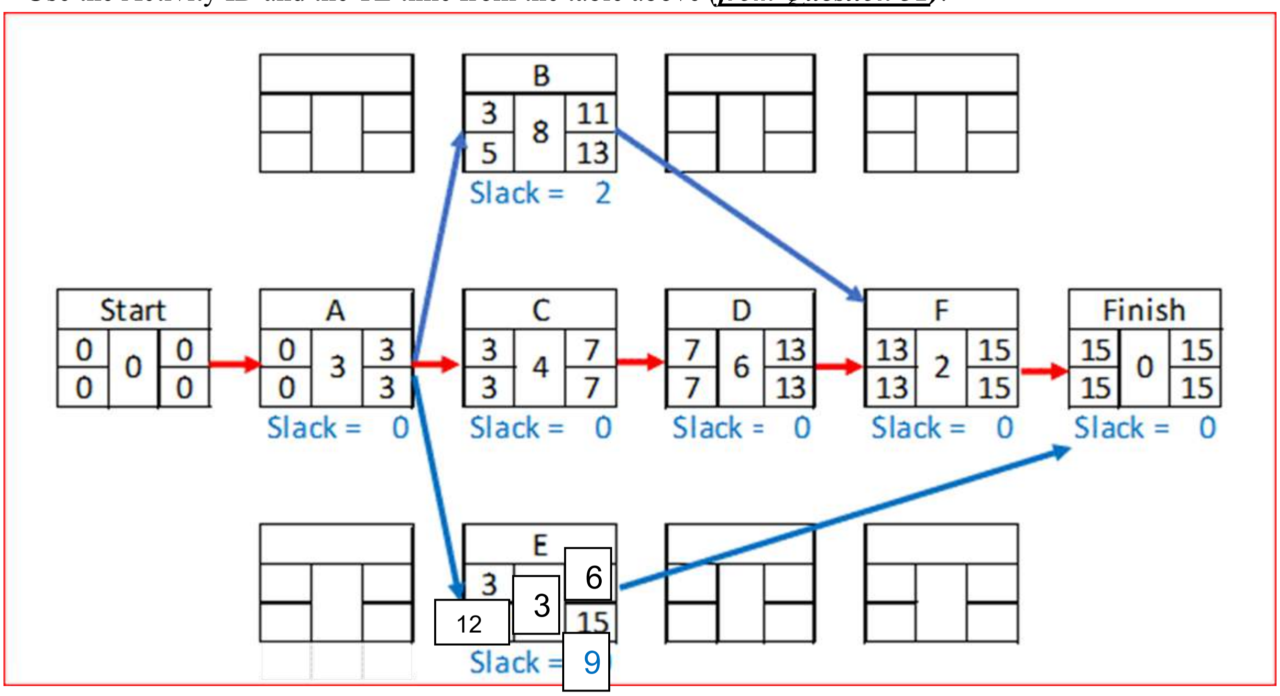
29. (2 pt) ________________ allows an acceleration of a successor activity but risks the over allocation of scarce resources.

    a. Uncertainty
    b. Lag time
    c. Slack or Float
    d. Lead time

30. (8 pts) In the table below, ***complete*** the TEs and variances for activities B and E. Note that some answers have already been provided to you. ***Round TEs as you've been taught in class. Round Variance to the nearest hundredths place.***

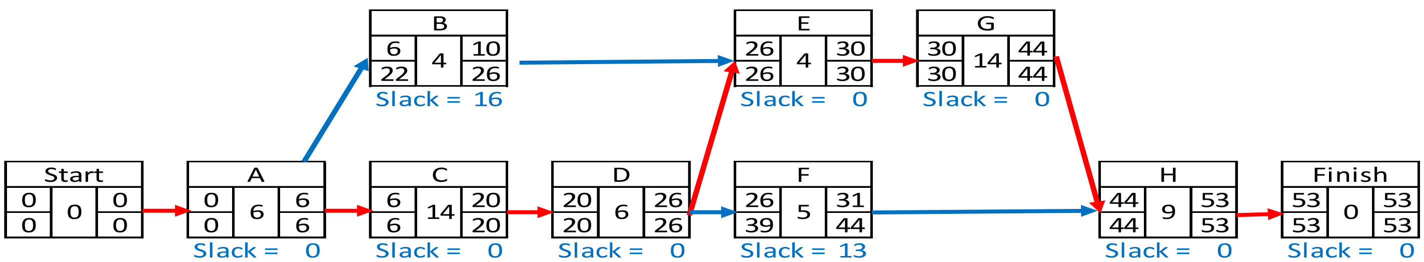
Activity ID	*Activity Name*	*Duration (weeks)*					
		Pred.	*Optimistic*	*Most Likely*	*Pessimistic*	*TE*	*Variance*
A	Fence-off Project Site	--	2	3	4	3.0	0.11
B	Replace Roofing Panels	A	5	7	10	8.0	0.69
C	Gut Interior	A	3	3	5	4.0	0.11
D	Install New Interior	C	4	5	8	6.0	0.44
E	Power Wash Exterior	A	2	2	4	**3.0**	**0.11**
F	Remove Site Fencing	B, D	2	2	2	2.0	0.00

31. (18 pts) In the space provided draw a <u>legible</u> network diagram, per ***DSE Standard,*** for this project. Use the Activity ID and the TE time from the table above ***(<u>from Question 31</u>)***.



32. After further analysis, several additional tasks were added to the project, and task duration estimates were improved.

Activity ID	*Duration (Days)*		
	Pred.	***TE***	***Variance***
A	--	6.0	0.44
B	B	4.0	0.11
C	A	14.0	1.78
D	C	6.0	0.44
E	B, D	4.0	0.25
F	D	5.0	0.00
G	E	14.0	0.25
H	F, G	9.0	1.36



Answer the questions below, **given the updated table and network diagram shown above**.

a. (5 pts) What is the project variance (σ^2)?

0.44 + 1.78 + 0.44 + 0.25 + 0.25 + 1.36 = 4.52

b. (5 pts) What is the probability of completing the project in **53 days**?

Answer: 50%

c. (5 pts) What is the probability of completing the project within **48 days**?

$$Z = \frac{(D-\mu)}{\sqrt{\sigma_\mu{}^2}} = \frac{(48-53)}{\sqrt{4.52}} = \frac{-5}{2.13} = -2.3474$$

P(Z< -2.3474) = 1 – P(Z< - 2.3474) = 1 – 0.9906 = .0094 or .94%

d. (5 pts) How many days will you need to complete the project if the CEO asks for a ***91%* assurance** that you'll meet your estimated date?

$D = Z * \sigma + \mu$

$D = (1.345 * 2.126) + 53$

$D =$ 55.85 Round up to 56 days

33. (2 pt) ________________ delays a successor activity and may result in an under-allocation of resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

Outstanding Student Work

Name: Eric Dailey

Section: G1

Time Complete: ______

EM411 – Project Management
AT26-1 Midterm Examination (Version A)

1. This is a 75-minute Written Partial Review worth 175 points. The exam consists of six topic areas and 13 pages, including this cover page. Ensure that your exam is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: This is a closed book exam. You may utilize a nongraphic calculator, one double-sided 5" x 8" handwritten notecard, and the formula reference provided. **Ensure your name is written on the notecard and turned in with the Exam.**

3. Feel free to disassemble the exam. A stapler is available to reattach the pages when you are finished. When reassembling your exam, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. EXTRA PAPER: Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name, and section.

6. GRADING SUMMARY:

	Section	*Possible*	*Earned*
I.	Project Selection and Initiation	25	
II.	Project Organization, Teams, & Stakeholders	25	
III.	Project Planning	20	
IV.	Risk in Project Management	20	
V.	Budgeting & Cost Estimation	35	
VI.	Scheduling	50	
	Total:	175	

This Midterm Examination will be released from academic security on Friday, 17 Oct 2025. Prior to that time, I will not discuss any aspects of this WPR with anyone except an EM411 instructor. By signing my name below, I acknowledge this requirement.

Signature: Eric Dailey

Name Eric Dailey

I. Project Initiation and Selection (25 pts)

Questions 1-10: Multiple Choice (2pts each): Select the BEST answer to each question from the options provided.

1. In non-numeric project selection, the "Competitive Necessity" approach is typically used when:

a) A company must launch a project to stay relevant in the market
b) An executive insists on a project
c) A project must comply with safety laws
d) The payback period is short

2. Weighted scoring models are most often used to overcome the following disadvantage of profitability selection models.

a) The inability to account for other qualitative or strategic factors
b) The inability to account for project results beyond the payback period
c) The inability to account for time value of money
d) The inability to account for cash flow

3. There are three characteristics that **all** projects share. Which of the following is one of those characteristics?

a) The PM sometimes makes trade-offs
b) The PM must consider time, schedule, and budget
c) The project is one-time occurrence
d) The outcome of the project may change over time

4. When using Net Present Value (NPV)/Discounted Cash Flow (DCF), ____________________.

a) a rate of return or hurdle rate is required
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c) risk is measured in the time it takes to recoup the initial investment
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5. When faced with conflicting expectations among key stakeholders relating to a project's goals, project managers must make appropriate ___________ in order to balance competing demands related to these objectives.

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b) guidelines
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6. The three direct goals of every project are ___________, _____________, and, _____________.

a) quality, scope, time
b) scope, time, cost
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Name Eric Dailey

7. Which of the following is true of the payback period model:

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8. True or False, all businesses which maintain a project portfolio employ a Project Management Office to oversee project delivery.

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b) False

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11. (5 pts) Your company is considering a 4-year engineering project that requires an **immediate investment of $255,000.** You will only see cash flows in year 2 and in year 4 – **which you estimate to be $125,000 for each year**. Your company expects an **3% rate of return** on any investment, and you estimate **4% annual inflation** for the next 4 years. **Using only DCF/NPV, should the company invest in this project?** Justify your answer. Show all work.

Year	0	1	2	3	4
Cash Flow	$-255,000	0	$125,000	0	$125,000

$$-255{,}000 + \left(\frac{0}{1.07} + \frac{125{,}000}{1.07^2} + \frac{0}{1.07^3} + \frac{125{,}000}{1.07^4}\right)$$

$$= \$-50{,}455.26$$

No because the project will not lead to financial gains.

Name Eric Dailey

II. Project Organization, Teams, and Stakeholders (25 pts)

12. (4 pts) What two items best describe a **Projectized Organization <u>(select two)</u>:**

a) Focus on standalone projects
b) Project Managers retain significant authority
c) Maximum flexibility in the use of staff
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B	*High level of commitment and motivation*
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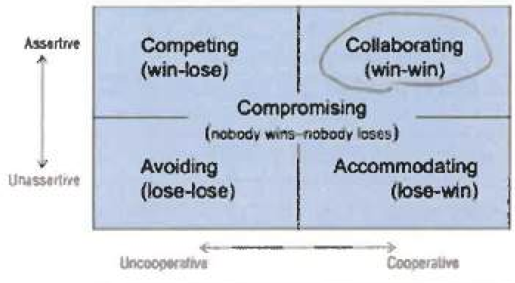
Name Eric Dailey

16. (3 pts) A contractor wants to use cheaper material to save money, but the project team insists on higher-grade material for long term durability. Both sides meet, analyze lifecycle costs, and agree on a mid-tier option that balances cost and quality. On the diagram below circle the conflict resolution strategy used.



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	Type of Conflict
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A	Technical Objectives and Resource Commitment
B	Functional Manager Commitment and Scheduling

	Project Stage
A	Project Initiation
B	Project Execution
C	Project Closure

III. Project Planning (20 pts)

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19. (3 pts) The Project Charter is a high-level document that helps define the scope and gives the PM authority to direct activities, request resources and spend funds.

20. (4 pts) A Work Breakdown Structure (WBS) is the main tool for managing project's scope and is typically broken down to the Work Package level, which can be managed by one individual.

Name Eric Dailey

21. (8 pts) Match the following descriptions of Project Charter Elements to the correct name from the list of options. ***(NOTE: You will only use 4 of the 8 elements listed.)***

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H	*Evaluation Methods*

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Name Eric Dailey

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a. (9 pts) Complete the Risk Management Annex below, by filling in each blank in the Table.

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b. (4 pts) There are three major types of risk. Identify which type(s) of risk have been identified in the table above.

Risk #1 = Preventable   Risk #2 = External

c. (3 pts) True or False, Initial Risk Assessments are focused mainly on externalities, while risk associated with schedule, budget, and resource allocation is addressed during the project planning phase.

a. True

b. False

d. (4 pts) If you were the PM, and you were **considering two approaches** of risk response to either remove the risk from the project to another stakeholder or do nothing, what **two responses** are you referring to? (select 2)

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b) Mitigate
c) Transfer
d) Accept

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V. Budgeting and Cost Estimation (35 pts)

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Organization Charges:

- Overhead Charges at 10% of internal labor costs,
- General Administrative costs at 5% of internal and contracted labor costs.
- Your organization includes a 15% contingency on all direct and indirect costs.

a) (10 pts) Given the project plan and organizational charges above, provide the Project Manager's **cost baseline or "fully costed" project budget** (SHOW YOUR WORK).

Baseline = Subtotal + Contingency

Subtotal = Direct + Indirect

Direct = $384,300

Indirect = 0.1(25,000) + 0.05(25,000 + 24,300) = $4,965

Contingency = 0.15(384,300 + 4,965) = $58,389.75

Baseline = $384,300 + $4,965 + $58,389.75

Answer: = $447,654.75

b) (2 pts) The budget table above (Question 24, Page 9) is an example of ________________.

a. Activity-Oriented Budget
b. Top-Down Budget
c. Category-Oriented Budget
d. Parametric Budget

25. (2 pts) It costs the contractor $6 million for its 26-person crew to complete a 4-month project. The Budget Office estimates that it will cost $4.75 million for the crew to complete a 2.5-month project. What **cost estimating technique** is the Budget Office using to complete this estimate? What type of **budgeting strategy** is this?

Cost Estimating Tech: Analogous, Parametric, Scaling, Rule of Thumb/Expert Opinion **(circle one)**

Budget Strategy: Top-Down, Bottom-Up, Iterative **(circle one)**

26. (3 pts) There are 5 common sources of Budget Uncertainty. Select 3 of them from the list below.

1. Technological Shock
2. Change Management process
3. Poor Estimating Techniques
4. Personnel Turnover
5. Project Charter details

Blank Page

27. (10 pts) In order to better estimate task completion times for elements in the WBS, you consider the impact of learning curves on the placement of floor panels. Your team calculates that the **learning rate of 90 percent** will be **fully realized after the 10th attempt**. They calculate the **10th attempt will take 6 hours to complete**. If this is true, how long would it take to place the **1st floor panel**?

$lr = 0.9$ $T_{10} = 6$ T_1 ? $r = \frac{\ln(lr)}{\ln(2)} = -0.152$

$T_n = T_1 n^r$

$6 = T_1 \cdot 10^{-0.152}$

$T_1 = 8.51$ hours

ANS

Answer:

28. (8 pts) During project execution, you learn that the team actually only required 5 **hours to complete the 10th attempt, a new fully realized time.** If it took **80.3 hours** to complete the first 10 attempts, how long will it take to complete 30 floor panels.

80.3 hours + 5(20) = 180.3 hours

ANS

Answer:

VI. Scheduling (50 pts)

29. (2 pts) ________________ allows an acceleration of a successor activity but risks the over allocation of scarce resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

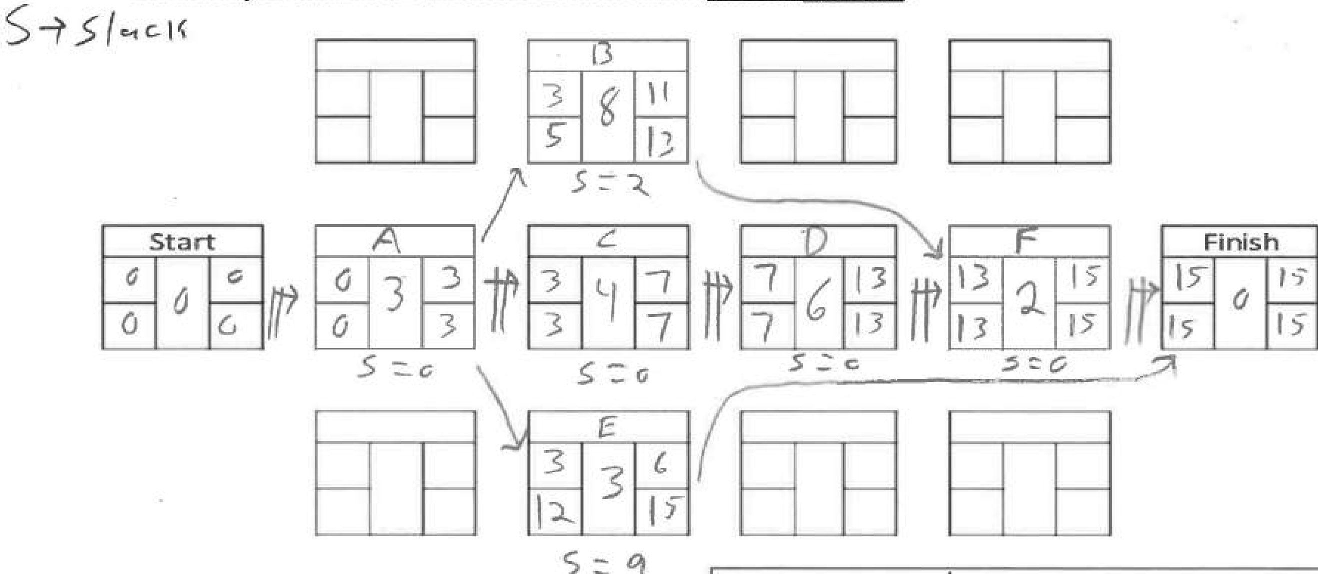
$\sigma^2 = \left(\frac{b-a}{6}\right)^2$

30. (8 pts) In the table below, ***complete*** the TEs and variances for activities B and E. Note that some answers have already been provided to you. ***Round TEs as you've been taught in class. Round Variance to the nearest hundredths place.***

Activity ID	*Activity Name*	*Duration (weeks)*					
		Pred.	*Optimistic*	*Most Likely*	*Pessimistic*	*TE*	*Variance*
A	Fence-off Project Site	--	2	3	4	3.0	0.11
B	Replace Roofing Panels	A	5	7	10	8	0.69
C	Gut Interior	A	3	3	5	4.0	0.11
D	Install New Interior	C	4	5	8	6.0	0.44
E	Power Wash Exterior	A	2	2	4	3	0.11
F	Remove Site Fencing	B, D	2	2	2	2.0	0.00

31. (18 pts) In the space provided *draw a legible network diagram per* ***DSE Standard****, for this project.* Use the Activity ID and the TE time from the table above ***(from Question 31)***.

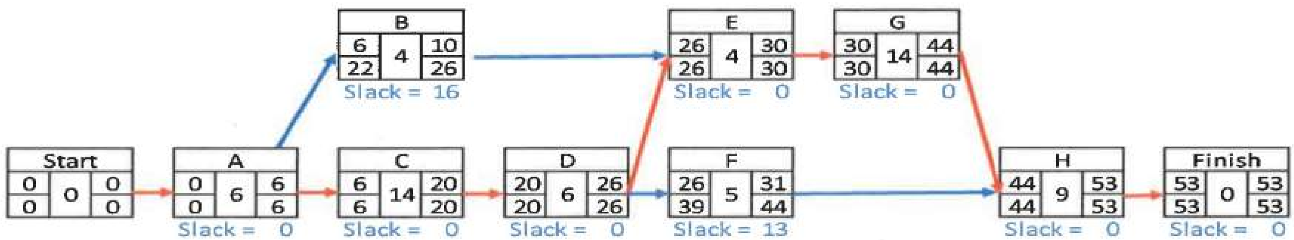
S → Slack



Project Duration: 15 weeks
Critical Path: A, C, D, F

32. After further analysis, several additional tasks were added to the project, and task duration estimates were improved.

Activity ID	***Duration (Days)***		
	Pred.	***TE***	***Variance***
A	--	6.0	0.44
B	B	4.0	0.11
C	A	14.0	1.78
D	C	6.0	0.44
E	B, D	4.0	0.25
F	D	5.0	0.00
G	E	14.0	0.25
H	F, G	9.0	1.36



CP: A, C, D, E, G, H

Answer the questions below, **given the updated table and network diagram shown above.**

a. (5 pts) What is the project variance (σ^2)?

$\sigma^2 \rightarrow$ Sum Variance of activities on CP
$= 0.44 + 1.78 + 0.44 + 0.25 + 0.25 + 1.36 = 4.52$
ANS

b. (5 pts) What is the probability of completing the project in **53 days**?

$D \rightarrow 53 \quad \mu \rightarrow 53$

$$Z = \frac{0}{\sqrt{4.52}} = 0$$

Probability = 0.5
ANS

Answer: 0.5

c. (5 pts) What is the probability of completing the project within **48 days**?

$$z = \frac{48 - 53}{\sqrt{4.52}} = -2.35$$

$$z = 2.35 \rightarrow 0.9906$$

$$1 - 0.9906 = \underline{\underline{0.0094}} \text{ ANS}$$

Answer: 0.0094

d. (5 pts) How many days will you need to complete the project if the CEO asks for a ***91%* assurance** that you'll meet your estimated date?

$$z \rightarrow 1.34$$

$$1.34 = \frac{D - 53}{\sqrt{4.52}}$$

$$D = 1.34(\sqrt{4.52}) + 53 = 55.84$$

$\underline{\underline{56 \text{ days}}}$ ANS

Answer: 56 days

33. (2 pts)_____________ delays a successor activity and may result in an under-allocation of resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

Blank Page

EM411 WPR/MidTerm

● Graded

Student

Eric Dailey

Total Points

173 / 175 pts

Question 1

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

Question 2

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 3

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Click here to replace this description.

Question 4

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 5

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 6

(no title) — **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 7

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 8

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 9

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 10

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 11

(no title) **5** / 5 pts

✔ **– 0 pts** Correct

– 2 pts Math error. Formula is correct.

– 5 pts Incorrect formula, results, and reasoning.

– 1 pt Incorrect or missing investment recommendation.

– 1 pt Solve the NPV.

– 4 pts Incorrect NPV but recommendation follows.

Question 12

(no title) **4** / 4 pts

✔ **– 0 pts** Correct

– 2 pts One Correct

– 4 pts None Correct

Question 13

(no title) 6 / 7 pts

– **0 pts** Correct

✔ – **1 pt** One Incorrect. Correct Answers: A, B, A, B, A, B, B

– **2 pts** Two Incorrect. Correct Answers: A, B, A, B, A, B, B

– **3 pts** Three Incorrect. Correct Answers: A, B, A, B, A, B, B

– **4 pts** Four Incorrect. Correct Answers: A, B, A, B, A, B, B

– **5 pts** Five Incorrect. Correct Answers: A, B, A, B, A, B, B

– **6 pts** Six Incorrect. Correct Answers: A, B, A, B, A, B, B

– **7 pts** Seven Incorrect. Correct Answers: A, B, A, B, A, B, B

Question 14

(no title) 2 / 2 pts

✔ – **0 pts** Correct

– **2 pts** Incorrect. The correct answer is: b) Possesses high technical skills and analysis mindset.

Question 15

(no title) 3 / 3 pts

✔ – **0 pts** Correct

– **3 pts** Incorrect

Question 16

(no title) 3 / 3 pts

✔ – **0 pts** Correct

– **3 pts** Incorrect

Question 17

(no title) 6 / 6 pts

✔ – **0 pts** Correct

– **2 pts** One incorrect. Correct answers are C, A, B

– **4 pts** Two incorrect. Correct answers are C, A, B

– **6 pts** All incorrect. Correct answers are C, A, B

Question 18

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 1.5 pts Partially correct

– 3 pts Incorrect

Question 19

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 3 pts Incorrect

Question 20

(no title) **4** / 4 pts

✔ **– 0 pts** Correct

– 2 pts Partially Correct

– 4 pts Incorrect

Question 21

(no title) **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts 1 Incorrect

– 4 pts 2 Incorrect

– 6 pts 3 Incorrect

– 8 pts 4 Incorrect

Question 22

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 23

(no title) **20** / 20 pts

23.1 **Part a** **9** / 9 pts

- ✔ **− 0 pts** Correct
- **− 9 pts** All incorrect
- **− 6 pts** RPNs incorrect
- **− 3 pts** Risk type incorrect

23.2 **Part b** **4** / 4 pts

- ✔ **− 0 pts** Correct
- **− 4 pts** None correct
- **− 2 pts** One correct

23.3 **Part c** **3** / 3 pts

- ✔ **− 0 pts** Correct
- **− 3 pts** Incorrect

23.4 **Part d** **4** / 4 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** One correct
- **− 4 pts** None correct

Question 24

(no title) **11** / 12 pts

24.1 **Part a** **9** / 10 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** Incorrect Indirect Costs
- ✔ **− 1 pt** Math error
- **− 2 pts** Incorrect contingency
- **− 1 pt** Incorrect direct cost

24.2 **Part b** **2** / 2 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** Incorrect

Question 25

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 1 pt 1 Correct

– 2 pts Nothing Correct

Question 26

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 1 pt 2 Correct

Question 27

(no title) **10** / 10 pts

✔ **– 0 pts** Correct

– 2 pts Math errors

– 3 pts Incorrect variables

– 3 pts Incorrect equation

– 3 pts "r" not calculated

Question 28

(no title) **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts Wrong equation/approach

– 1 pt Wrong # of iterations

– 1 pt Math error

Question 29

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 30

(no title) **8** / 8 pts

✔ **− 0 pts** Correct

− 2 pts One mistake

− 2 pts Rounding errors

Question 31

(no title) **18** / 18 pts

✔ **− 0 pts** Correct

− 8 pts Multiple errors

− 2 pts Slack not ID'd

− 2 pts Incorrect backward/backward pass

− 2 pts Incorrect relationships

Question 32

(no title) **20** / 20 pts

32.1 **Part a** **5** / 5 pts

✔ **− 0 pts** Correct

− 5 pts Incorrect

− 1 pt Partial credit

32.2 **Part b** **5** / 5 pts

✔ **− 0 pts** Correct

− 5 pts Incorrect

32.3 **Part c** **5** / 5 pts

✔ **− 0 pts** Correct

− 2 pts Partial credit for work shown

− 1 pt Math or Z table error

− 5 pts Blank

32.4 **Part d** **5** / 5 pts

✔ **− 0 pts** Correct

− 1 pt Partial credit - equation/z value/math errors

− 5 pts Blank or no meaningful work

− 2 pts Partial credit - wrong approach

Question 33

(no title) **2** / 2 pts

✔ **− 0 pts** Correct

− 2 pts Incorrect

Fair-to-Good Student Work

Name: Amir Ali

Section: H1

Time Complete: ____________

EM411 – Project Management
AT26-1 Midterm Examination (Version A)

1. This is a 75-minute Written Partial Review worth 175 points. The exam consists of six topic areas and 13 pages, including this cover page. Ensure that your exam is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: This is a closed book exam. You may utilize a nongraphic calculator, one double-sided 5" x 8" handwritten notecard, and the formula reference provided. **Ensure your name is written on the notecard and turned in with the Exam**.

3. Feel free to disassemble the exam. A stapler is available to reattach the pages when you are finished. When reassembling your exam, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. EXTRA PAPER: Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name, and section.

6. GRADING SUMMARY:

	Section	*Possible*	*Earned*
I.	Project Selection and Initiation	25	
II.	Project Organization, Teams, & Stakeholders	25	
III.	Project Planning	20	
IV.	Risk in Project Management	20	
V.	Budgeting & Cost Estimation	35	
VI.	Scheduling	50	
	Total:	175	

This Midterm Examination will be released from academic security on Friday, 17 Oct 2025. Prior to that time, I will not discuss any aspects of this WPR with anyone except an EM411 instructor. By signing my name below, I acknowledge this requirement.

Signature: [signature]

Name Amir Ali

I. Project Initiation and Selection (25 pts)

Questions 1-10: Multiple Choice (2pts each): Select the BEST answer to each question from the options provided.

1. In non-numeric project selection, the "Competitive Necessity" approach is typically used when:

a) A company must launch a project to stay relevant in the market
b) An executive insists on a project
c) A project must comply with safety laws
d) The payback period is short

2. Weighted scoring models are most often used to overcome the following disadvantage of profitability selection models.

a) The inability to account for other qualitative or strategic factors
b) The inability to account for project results beyond the payback period
c) The inability to account for time value of money
d) The inability to account for cash flow

3. There are three characteristics that **all** projects share. Which of the following is one of those characteristics?

a) The PM sometimes makes trade-offs
b) The PM must consider time, schedule, and budget
c) The project is one-time occurrence
d) The outcome of the project may change over time

4. When using Net Present Value (NPV)/Discounted Cash Flow (DCF), ____________________.

a) a rate of return or hurdle rate is required
b) the time value of money is not accounted for
c) risk is measured in the time it takes to recoup the initial investment
d) multiple criteria are considered in the evaluation

5. When faced with conflicting expectations among key stakeholders relating to a project's goals, project managers must make appropriate ___________ in order to balance competing demands related to these objectives.

a) weights
b) guidelines
c) trade-offs
d) tasks

6. The three direct goals of every project are ____________, _____________, and, _____________.

a) quality, scope, time
b) scope, time, cost
c) cost, time, budget
d) schedule, cost, time

Name Amir Ali

7. Which of the following is true of the payback period model:

   a) It considers cash flow once the initial investment is recovered
   b) A shorter payback period is more attractive
   c) It accounts for annual changes in cash flows
   d) It is adequate for calculating risk for the duration of the project

8. True or False, all businesses which maintain a project portfolio employ a Project Management Office to oversee project delivery.

   a) True
   b) False

9. A project selection method that subjectively ranks projects in order to select the "best" project for the company to undertake.

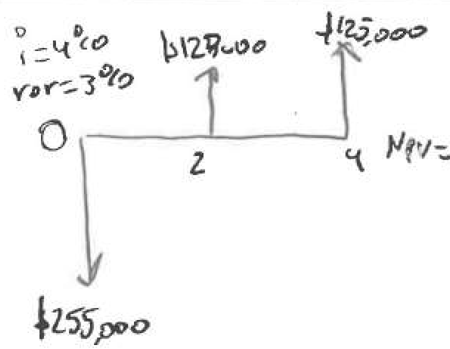
   a) sustainability
   b) product line extension
   c) comparative benefit
   d) operating necessity

10. A company must initiate a project to meet a change in government regulations. This project selection method is best classified as:

   a) sustainability
   b) product line extension
   c) comparative benefit
   d) operating necessity

11. (5 pts) Your company is considering a 4-year engineering project that requires an **immediate investment of $255,000**. You will only see cash flows in **year 2 and in year 4 – which you estimate to be $125,000 for each year**. Your company expects an **3% rate of return** on any investment, and you estimate **4% annual inflation** for the next 4 years. **Using only DCF/NPV, should the company invest in this project?** Justify your answer. Show all work.

Year	0	1	2	3	4
Cash Flow	255,000	0	125,000	0	125,00



$$NPV = -255,000 + \frac{125000}{(1.07)^2} + \frac{125000}{(1.07)^4} = -50458.26$$

NPV = $ 50458.26

Name Amir Ali

II. Project Organization, Teams, and Stakeholders (25 pts)

12. (4 pts) What two items best describe a **Projectized Organization** **(select two):**

a) Focus on standalone projects
b) Project Managers retain significant authority
c) Maximum flexibility in the use of staff
d) Team members are mentored by Functional Managers

13. (7 pts) Match the following **advantages** to the correct Project Organization.

	Organization Description
B	*There is less anxiety about what happens post - project*
A	*High level of commitment and motivation*
A	*Specialists in the division can be grouped to share knowledge and experience*
B	*The lines of communication are shortened*
A	*Unity of command exists*
B	*The ability to make swift decisions is greatly enhanced*
A	*Supports a holistic approach to the project*

Organization	
A	*Functional Organization*
B	*Projectized Organization*

14. (2 pts) Desired traits for an effective Project Manager include all of the following EXCEPT:

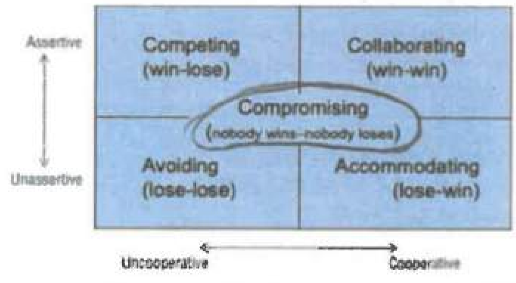
a) Uses a Systems approach
b) Possesses high technical skills and analysis mindset
c) Able to act as a facilitator and generalist
d) The ability to get the job done

15. (3 pts) True or False, traditional categories of conflict include (1) Interpersonal conflict, (2) Differing goals and expectations, and (3) Uncertainty regarding decision authority.

a) True
b) False

Name Amir A

16. (3 pts) A contractor wants to use cheaper material to save money, but the project team insists on higher-grade material for long term durability. Both sides meet, analyze lifecycle costs, and agree on a mid-tier option that balances cost and quality. On the diagram below circle the conflict resolution strategy used.



17. (6 pts) A project lifecycle includes Initiation, Execution, and Closure phases. Match the following types of conflict with the Project Stage in which they are most commonly found.

	Type of Conflict
C	Scheduling Deadlines and Budget
A	Technical Objectives and Resource Commitment
B	Functional Manager Commitment and Scheduling

	Project Stage
A	Project Initiation
B	Project Execution
C	Project Closure

III. Project Planning (20 pts)

18. (3 pts) A ~~Proj~~ launch allows the PM to discuss the project in sufficient detail with key stakeholders, in order to develop a general understanding of the technical scope, areas of responsibility, tentative schedules and budgets, and needed risk management group.

19. (3 pts) The Project Charter is a high-level document that helps define the scope and gives the PM authority to direct activities, request resources and spend funds.

20. (4 pts) A Work Breakdown Structure (WBS) is the main tool for managing project's scope and is typically broken down to the Work Package level, which can be managed by one individual.

Name Amir Ali

21. (8 pts) Match the following descriptions of Project Charter Elements to the correct name from the list of options. ***(NOTE: You will only use 4 of the 8 elements listed.)***

Project Charter Element Descriptions	
B	*A more detailed statement of the projects goals, their priorities, what constitutes success, and measurable goals*
C	*High-level description of the project and its requirements, including managerial and technical approaches to the work.*
A	*A short summary of the project's overall scope and objectives.*
G	*This section covers potential problems as well as potential opportunities that could affect the project.*

Elements	
A	*Purpose*
B	*Objectives*
C	*Overview*
D	*Schedules*
E	*Resources*
F	*Stakeholders*
G	*Risk Management Plan*
H	*Evaluation Methods*

22. (2 pts) When developing a Linear Responsibility Matrix, a person who is ultimately answerable for the correct and thorough completion of a deliverable is given which role?

a) Responsible (R)
b) Accountable (A)
c) Consulted (C)
d) Informed (I)

Name Amir Ali

IV. Risk in Project Management (20 pts)

23. During Project Planning, you utilize a quantitative risk analysis technique to identify two major risks to the project.

a. (9 pts) Complete the Risk Management Annex below, by filling in each blank in the Table.

Risk #	**Risk**	**S**	**L**	**D**	**RPN**	**Risk Response Type**	**Risk Response Action**
1	Custom Repair Parts are defective, which could cause a major mechanical failure	6	4	7	168	mitigate	Develop and implement an inspection and testing procedure for all mechanical parts delivered. This will improve our ability to detect a flaw prior to installation.
2	FAA Travel Restrictions delay key team members from traveling to competition	8	3	4	96		

b. (4 pts) There are three major types of risk. Identify which type(s) of risk have been identified in the table above.

Risk #1 = preventable   Risk #2 = external

c. (3 pts) True or False, Initial Risk Assessments are focused mainly on externalities, while risk associated with schedule, budget, and resource allocation is addressed during the project planning phase.

a. True

b. False

d. (4 pts) If you were the PM, and you were **considering two approaches** of risk response to either remove the risk from the project to another stakeholder or do nothing, what **two responses** are you referring to? (select 2)

a) Avoid
b) Mitigate
c) Transfer
d) Accept

Name Amir Ali

V. Budgeting and Cost Estimation (35 pts)

24. After completing an initial project plan, you begin the process of budgeting and cost estimation. You received the following information from your team members:

Resources	*Number required*	*Cost per unit*	*Total Cost*
Floor panels	10	$11,000 each	$110,000.00
Wall panels	12	$7,000 each	$84,000.00
Window panels	8	$3,000 each	$24,000.00
Roofing panels	8	$5,000 each	$40,000.00
Truss	4	$15,500 each	$62,000.00
Door	6	$2,500 each	$15,000.00
TOTAL MATERIALS COST			***$335,000.00***
Project Manager	300 hrs	$450/day	*$16,875.00*
Foreman	200 hrs	$250/day	*$6,250.00*
Document Control Mgr	100 hrs	$100/day	*$1,875.00*
TOTAL INTERNAL LABOR COST			***$25,000.00***
Inspector (Contractor)	48 hrs	$150/day	*$900.00*
Floor Sub-Contract	250 hrs	$24/hour	*$6,000.00*
Vertical Sub-Contract	200 hrs	$48/hour	*$9,600.00*
Roof Sub-Contract	150 hrs	$52/hour	*$7,800.00*
TOTAL CONTRACTED LABOR COST			***$24,300.00***

Organization Charges:
- Overhead Charges at 10% of internal labor costs,
- General Administrative costs at 5% of internal and contracted labor costs.
- Your organization includes a 15% contingency on all direct and indirect costs.

a) (10 pts) Given the project plan and organizational charges above, provide the Project Manager's **cost baseline or "fully costed" project budget** (SHOW YOUR WORK).

Overhead .10(25000) = 2500

GA = .05(25000 + 24,300) = 2465

indirect = 2500 + 2465 = 4965

+ direct = 335,000 + 25000 + 24,300 = 384300

Cot.coycey = .15(384,300 + 4965) = 58389.75

Full Cost = 58389.75 + 384300 + 4965 =

Answer: $447654.75

b) (2 pts) The budget table above (Question 24, Page 9) is an example of ________________.

a. Activity-Oriented Budget
b. Top-Down Budget
c. Category-Oriented Budget
d. Parametric Budget

25. (2 pts) It costs the contractor $6 million for its 26-person crew to complete a 4-month project. The Budget Office estimates that it will cost $4.75 million for the crew to complete a 2.5-month project. What **cost estimating technique** is the Budget Office using to complete this estimate? What type of **budgeting strategy** is this?

Cost Estimating Tech: Analogous, Parametric, Scaling, Rule of Thumb/Expert Opinion **(circle one)**

Budget Strategy: Top-Down, Bottom-Up, Iterative **(circle one)**

26. (3 pts) There are 5 common sources of Budget Uncertainty. Select 3 of them from the list below.

1. Technological Shock
2. Change Management process
3. Poor Estimating Techniques
4. Personnel Turnover
5. Project Charter details

Blank Page

27. (10 pts) In order to better estimate task completion times for elements in the WBS, you consider the impact of learning curves on the placement of floor panels. Your team calculates that the **learning rate of 90 percent** will be **fully realized after the 10th attempt**. They calculate the 10th **attempt will take 6 hours to complete**. If this is true, how long would it take to place the **1st floor panel**?

$T_{10} = 6$ $\quad lr = .9 \quad n = 10$

$$r = \frac{\ln(.9)}{\ln(2)} = -0.152$$

$$6 = T_1 (10)^{-0.152}$$

Answer: $\frac{6}{0.70469} = \frac{T_1 (0.70469)}{0.70469} = 8.51$ Hours

28. (8 pts) During project execution, you learn that the team actually only required 5 **hours to complete the 10th attempt, a new fully realized time.** If it took **80.3 hours** to complete the first 10 attempts, how long will it take to complete 30 floor panels.

$5(20) =$

$100 + 80.3 = 180.3$

Answer: 180.3 hours

VI. Scheduling (50 pts)

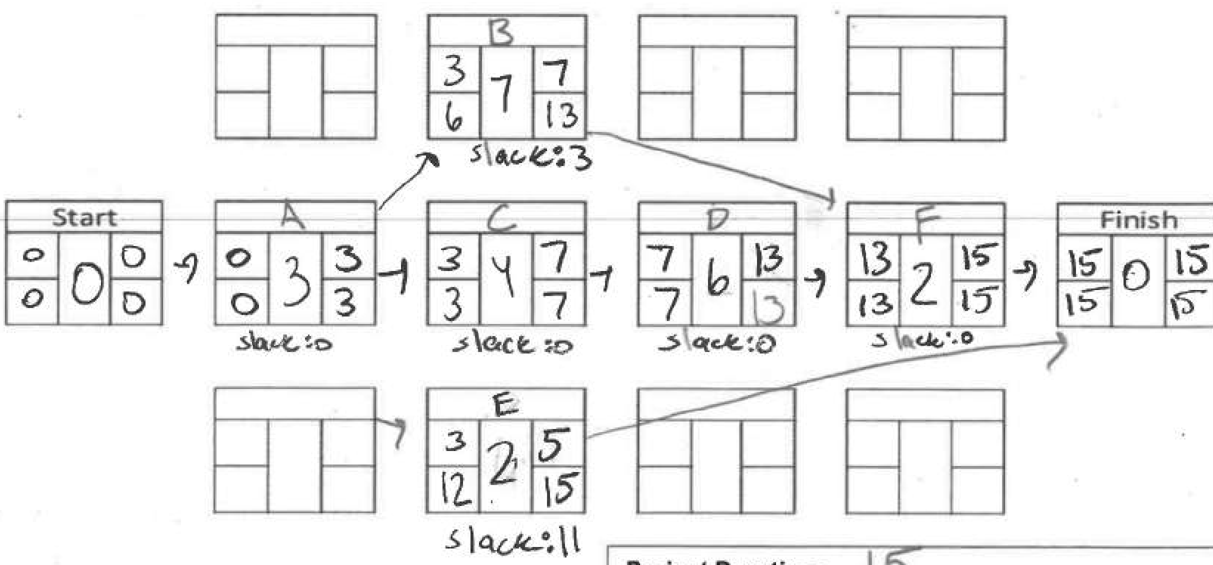
29. (2 pts) ________________ allows an acceleration of a successor activity but risks the over allocation of scarce resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

30. (8 pts) In the table below, ***complete*** the TEs and variances for activities B and E. Note that some answers have already been provided to you. ***Round TEs as you've been taught in class. Round Variance to the nearest hundredths place.***

Activity ID	*Activity Name*	*Duration (weeks)*					
		Pred.	*Optimistic*	*Most Likely*	*Pessimistic*	*TE*	*Variance*
A	Fence-off Project Site	--	2	3	4	3.0	0.11
B	Replace Roofing Panels	A	5	7	10	7	0.69
C	Gut Interior	A	3	3	5	4.0	0.11
D	Install New Interior	C	4	5	8	6.0	0.44
E	Power Wash Exterior	A	2	2	4	2	0.11
F	Remove Site Fencing	B, D	2	2	2	2.0	0.00

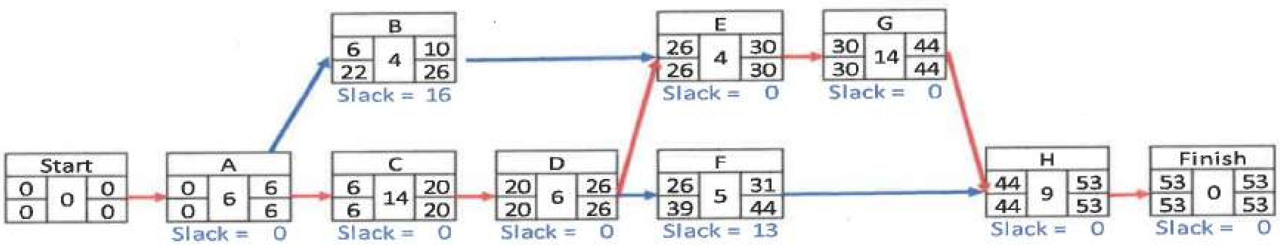
31. (18 pts) In the space provided *draw a legible network diagram per **DSE Standard**, for this project.* Use the Activity ID and the TE time from the table above ***(from Question 31)***.



Project Duration: 15
Critical Path: A, C, D, F

32. After further analysis, several additional tasks were added to the project, and task duration estimates were improved.

Activity ID	Duration (Days)		
	Pred.	TE	Variance
A	--	6.0	0.44
B	B	4.0	0.11
C	A	14.0	1.78
D	C	6.0	0.44
E	B, D	4.0	0.25
F	D	5.0	0.00
G	E	14.0	0.25
H	F, G	9.0	1.36



Answer the questions below, **given the updated table and network diagram shown above.**

a. (5 pts) What is the project variance (σ^2)?

$0.44 + 1.78 + 0.44 + 0.25 + 0.25 + 1.36 = 4.52$

b. (5 pts) What is the probability of completing the project in **53 days**?

$$\frac{53-53}{\sqrt{4.52}} = \frac{0}{2.13} = 0$$

Can't have zero

Answer: 50%

c. (5 pts) What is the probability of completing the project within **48 days**?

$$\frac{48-53}{\sqrt{4.52}} = \frac{-5}{2.13} = -2.35$$

Answer: -2.35

d. (5 pts) How many days will you need to complete the project if the CEO asks for a ***91%* assurance** that you'll meet your estimated date?

$$53 + (2.13 \cdot 8186) = 54.71$$

Answer: 55 days

33. (2 pts)______________ delays a successor activity and may result in an under-allocation of resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

Blank Page

EM411 WPR/MidTerm

Graded

Student

Amir Ali

Total Points

155 / 175 pts

Question 1

(no title) — **2** / 2 pts

- ✔ **– 0 pts** Correct

Question 2

(no title) — **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect

Question 3

(no title) — **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Click here to replace this description.

Question 4

(no title) — **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect

Question 5

(no title) — **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect

Question 6

(no title) — **0** / 2 pts

- **– 0 pts** Correct
- ✔ **– 2 pts** Incorrect

Question 7

(no title) — 0 / 2 pts

– 0 pts Correct

✔ – 2 pts Incorrect

Question 8

(no title) — 0 / 2 pts

– 0 pts Correct

✔ – 2 pts Incorrect

Question 9

(no title) — 2 / 2 pts

✔ – 0 pts Correct

– 2 pts Incorrect

Question 10

(no title) — 2 / 2 pts

✔ – 0 pts Correct

– 2 pts Incorrect

Question 11

(no title) — 4 / 5 pts

– 0 pts Correct

– 2 pts Math error. Formula is correct.

– 5 pts Incorrect formula, results, and reasoning.

✔ – 1 pt Incorrect or missing investment recommendation.

– 1 pt Solve the NPV.

– 4 pts Incorrect NPV but recommendation follows.

Question 12

(no title) — 4 / 4 pts

✔ – 0 pts Correct

– 2 pts One Correct

– 4 pts None Correct

Question 13

(no title) **4** / 7 pts

- **– 0 pts** Correct
- **– 1 pt** One Incorrect. Correct Answers: A, B, A, B, A, B, B
- **– 2 pts** Two Incorrect. Correct Answers: A, B, A, B, A, B, B
- ✔ **– 3 pts** Three Incorrect. Correct Answers: A, B, A, B, A, B, B
- **– 4 pts** Four Incorrect. Correct Answers: A, B, A, B, A, B, B
- **– 5 pts** Five Incorrect. Correct Answers: A, B, A, B, A, B, B
- **– 6 pts** Six Incorrect. Correct Answers: A, B, A, B, A, B, B
- **– 7 pts** Seven Incorrect. Correct Answers: A, B, A, B, A, B, B

Question 14

(no title) **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect. The correct answer is: b) Possesses high technical skills and analysis mindset.

Question 15

(no title) **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** Incorrect

Question 16

(no title) **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** Incorrect

Question 17

(no title) **6** / 6 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** One incorrect. Correct answers are C, A, B
- **– 4 pts** Two incorrect. Correct answers are C, A, B
- **– 6 pts** All incorrect. Correct answers are C, A, B

Question 18

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 1.5 pts Partially correct

– 3 pts Incorrect

Question 19

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 3 pts Incorrect

Question 20

(no title) **4** / 4 pts

✔ **– 0 pts** Correct

– 2 pts Partially Correct

– 4 pts Incorrect

Question 21

(no title) **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts 1 Incorrect

– 4 pts 2 Incorrect

– 6 pts 3 Incorrect

– 8 pts 4 Incorrect

Question 22

(no title) **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Incorrect

Question 23

(no title) **20** / 20 pts

23.1 **Part a** **9** / 9 pts

- ✔ **– 0 pts** Correct
- **– 9 pts** All incorrect
- **– 6 pts** RPNs incorrect
- **– 3 pts** Risk type incorrect

23.2 **Part b** **4** / 4 pts

- ✔ **– 0 pts** Correct
- **– 4 pts** None correct
- **– 2 pts** One correct

23.3 **Part c** **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** Incorrect

23.4 **Part d** **4** / 4 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** One correct
- **– 4 pts** None correct

Question 24

(no title) **9** / 12 pts

24.1 **Part a** **9** / 10 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect Indirect Costs
- ✔ **– 1 pt** Math error
- **– 2 pts** Incorrect contingency
- **– 1 pt** Incorrect direct cost

24.2 **Part b** **0** / 2 pts

- **– 0 pts** Correct
- ✔ **– 2 pts** Incorrect

Question 25

(no title) **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** 1 Correct
- **– 2 pts** Nothing Correct

Question 26

(no title) **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** 2 Correct

Question 27

(no title) **10** / 10 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Math errors
- **– 3 pts** Incorrect variables
- **– 3 pts** Incorrect equation
- **– 3 pts** "r" not calculated

Question 28

(no title) **8** / 8 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Wrong equation/approach
- **– 1 pt** Wrong # of iterations
- **– 1 pt** Math error

Question 29

(no title) **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect

Question 30

(no title)

6 / 8 pts

- **– 0 pts** Correct
- **– 2 pts** One mistake
- ✔ **– 2 pts** Rounding errors

Question 31

(no title)

18 / 18 pts

- ✔ **– 0 pts** Correct
- **– 8 pts** Multiple errors
- **– 2 pts** Slack not ID'd
- **– 2 pts** Incorrect backward/backward pass
- **– 2 pts** Incorrect relationships

Question 32

(no title) **17** / 20 pts

32.1 **Part a** **5** / 5 pts

- ✔ **– 0 pts** Correct
- **– 5 pts** Incorrect
- **– 1 pt** Partial credit

32.2 **Part b** **5** / 5 pts

- ✔ **– 0 pts** Correct
- **– 5 pts** Incorrect

32.3 **Part c** **3** / 5 pts

- **– 0 pts** Correct
- ✔ **– 2 pts** Partial credit for work shown
- **– 1 pt** Math or Z table error
- **– 5 pts** Blank

32.4 **Part d** **4** / 5 pts

- **– 0 pts** Correct
- ✔ **– 1 pt** Partial credit - equation/z value/math errors
- **– 5 pts** Blank or no meaningful work
- **– 2 pts** Partial credit - wrong approach

Question 33

(no title) **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Incorrect

Poor Student Work

Name: Andrew Berg
Section: H1
Time Complete:

EM411 – Project Management
AT26-1 Midterm Examination (Version A)

1. This is a 75-minute Written Partial Review worth 175 points. The exam consists of six topic areas and 13 pages, including this cover page. Ensure that your exam is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: This is a closed book exam. You may utilize a nongraphic calculator, one double-sided 5" x 8" handwritten notecard, and the formula reference provided. **Ensure your name is written on the notecard and turned in with the Exam.**

3. Feel free to disassemble the exam. A stapler is available to reattach the pages when you are finished. When reassembling your exam, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. EXTRA PAPER: Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name, and section.

6. GRADING SUMMARY:

	Section	*Possible*	*Earned*
I.	Project Selection and Initiation	25	
II.	Project Organization, Teams, & Stakeholders	25	
III.	Project Planning	20	
IV.	Risk in Project Management	20	
V.	Budgeting & Cost Estimation	35	
VI.	Scheduling	50	
	Total:	175	

This Midterm Examination will be released from academic security on Friday, 17 Oct 2025. Prior to that time, I will not discuss any aspects of this WPR with anyone except an EM411 instructor. By signing my name below, I acknowledge this requirement.

Signature: [signature]

Name Andrew Bers

I. Project Initiation and Selection (25 pts)

Questions 1-10: Multiple Choice (2pts each): Select the BEST answer to each question from the options provided.

1. In non-numeric project selection, the "Competitive Necessity" approach is typically used when:

a) A company must launch a project to stay relevant in the market
b) An executive insists on a project
c) A project must comply with safety laws
d) The payback period is short

2. Weighted scoring models are most often used to overcome the following disadvantage of profitability selection models.

a) The inability to account for other qualitative or strategic factors
b) The inability to account for project results beyond the payback period
c) The inability to account for time value of money
d) The inability to account for cash flow

3. There are three characteristics that **all** projects share. Which of the following is one of those characteristics?

a) The PM sometimes makes trade-offs
b) The PM must consider time, schedule, and budget
c) The project is one-time occurrence
d) The outcome of the project may change over time

4. When using Net Present Value (NPV)/Discounted Cash Flow (DCF), ____________________.

a) a rate of return or hurdle rate is required
b) the time value of money is not accounted for
c) risk is measured in the time it takes to recoup the initial investment
d) multiple criteria are considered in the evaluation

5. When faced with conflicting expectations among key stakeholders relating to a project's goals, project managers must make appropriate ___________ in order to balance competing demands related to these objectives.

a) weights
b) guidelines
c) trade-offs
d) tasks

6. The three direct goals of every project are ____________, ____________, and, ____________.

a) quality, scope, time
b) scope, time, cost
c) cost, time, budget
d) schedule, cost, time

Name Andrew Bers

7. Which of the following is true of the payback period model:

   a) It considers cash flow once the initial investment is recovered
   b) A shorter payback period is more attractive
   c) It accounts for annual changes in cash flows
   d) It is adequate for calculating risk for the duration of the project

8. True or False, all businesses which maintain a project portfolio employ a Project Management Office to oversee project delivery.

   a) True
   b) False

9. A project selection method that subjectively ranks projects in order to select the "best" project for the company to undertake.

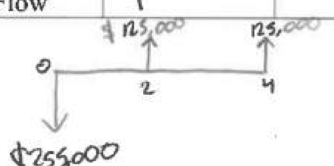
   a) sustainability
   b) product line extension
   c) comparative benefit
   d) operating necessity

10. A company must initiate a project to meet a change in government regulations. This project selection method is best classified as:

   a) sustainability
   ~~b)~~ product line extension
   ~~c)~~ comparative benefit
   d) operating necessity

11. (5 pts) Your company is considering a 4-year engineering project that requires an **immediate investment of $255,000**. You will only see cash flows in **year 2 and in year 4** – **which you estimate to be $125,000 for each year**. Your company expects an **3% rate of return** on any investment, and you estimate **4% annual inflation** for the next 4 years. **Using only DCF/NPV, should the company invest in this project?** Justify your answer. Show all work.

Year	0	1	2	3	4
Cash Flow	-$255,000				



$$NPV = -255{,}000 + \frac{125{,}000}{(1.07)^2} + \frac{125{,}000}{(1.07)^4} = -50{,}458.26$$

No you should not invest in the project because the negative NPV is showing that it is destroying the future cash flow.

Name Andrew Berg

II. Project Organization, Teams, and Stakeholders (25 pts)

12. (4 pts) What two items best describe a **Projectized Organization** **(select two):**

a) Focus on standalone projects
b) Project Managers retain significant authority
c) Maximum flexibility in the use of staff
d) Team members are mentored by Functional Managers

13. (7 pts) Match the following **advantages** to the correct Project Organization.

	Organization Description
A	*There is less anxiety about what happens post - project*
B	*High level of commitment and motivation*
A	*Specialists in the division can be grouped to share knowledge and experience*
B	*The lines of communication are shortened*
A	*Unity of command exists*
B	*The ability to make swift decisions is greatly enhanced*
B	*Supports a holistic approach to the project*

	Organization
A	*Functional Organization*
B	*Projectized Organization*

14. (2 pts) Desired traits for an effective Project Manager include all of the following EXCEPT:

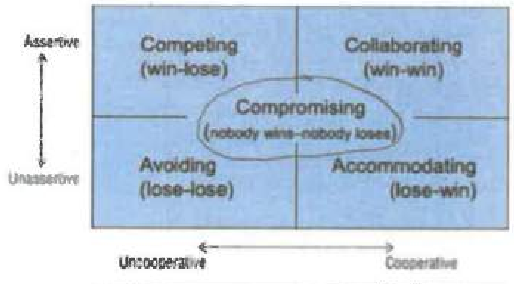
a) Uses a Systems approach
b) Possesses high technical skills and analysis mindset
c) Able to act as a facilitator and generalist
d) The ability to get the job done

15. (3 pts) True or False, traditional categories of conflict include (1) Interpersonal conflict, (2) Differing goals and expectations, and (3) Uncertainty regarding decision authority.

a) True
b) False

Name Andrew Berg

16. (3 pts) A contractor wants to use cheaper material to save money, but the project team insists on higher-grade material for long term durability. Both sides meet, analyze lifecycle costs, and agree on a mid-tier option that balances cost and quality. On the diagram below circle the conflict resolution strategy used.



17. (6 pts) A project lifecycle includes Initiation, Execution, and Closure phases. Match the following types of conflict with the Project Stage in which they are most commonly found.

	Type of Conflict
A	Scheduling Deadlines and Budget
C	Technical Objectives and Resource Commitment
B	Functional Manager Commitment and Scheduling

	Project Stage
A	Project Initiation
B	Project Execution
C	Project Closure

III. Project Planning (20 pts)

18. (3 pts) A Purpose allows the PM to discuss the project in sufficient detail with key stakeholders, in order to develop a general understanding of the technical scope, areas of responsibility, tentative schedules and budgets, and needed risk management group.

19. (3 pts) The objectives is a high-level document that helps define the scope and gives the PM authority to direct activities, request resources and spend funds.

20. (4 pts) A Work Breakdown Structure (WBS) is the main tool for managing project's scope and is typically broken down to the Stakeholders level, which can be managed by one individual.

Name Andrew Berg

21. (8 pts) Match the following descriptions of Project Charter Elements to the correct name from the list of options. ***(NOTE: You will only use 4 of the 8 elements listed.)***

	Project Charter Element Descriptions
B	*A more detailed statement of the projects goals, their priorities, what constitutes success, and measurable goals*
A	*High-level description of the project and its requirements, including managerial and technical approaches to the work.*
C	*A short summary of the project's overall scope and objectives.*
G	*This section covers potential problems as well as potential opportunities that could affect the project.*

	Elements
A	*Purpose*
B	*Objectives*
C	*Overview*
D	*Schedules*
E	*Resources*
F	*Stakeholders*
G	*Risk Management Plan*
H	*Evaluation Methods*

22. (2 pts) When developing a Linear Responsibility Matrix, a person who is ultimately answerable for the correct and thorough completion of a deliverable is given which role?

a) Responsible (R)
b) Accountable (A)
c) Consulted (C)
d) Informed (I)

Name Andrew Berg

IV. Risk in Project Management (20 pts)

23. During Project Planning, you utilize a quantitative risk analysis technique to identify two major risks to the project.

a. (9 pts) Complete the Risk Management Annex below, by filling in each blank in the Table.

Risk #	Risk	S	L	D	RPN	Risk Response Type	Risk Response Action
1	Custom Repair Parts are defective, which could cause a major mechanical failure	6	4	7	17	Procedural	Develop and implement an inspection and testing procedure for all mechanical parts delivered. This will improve our ability to detect a flaw prior to installation.
2	FAA Travel Restrictions delay key team members from traveling to competition	8	3	4	15		

b. (4 pts) There are three major types of risk. Identify which type(s) of risk have been identified in the table above.

Risk #1 = Procedural Risk #2 = Opperational

c. (3 pts) True or False, Initial Risk Assessments are focused mainly on externalities, while risk associated with schedule, budget, and resource allocation is addressed during the project planning phase.

a. True
b. False

d. (4 pts) If you were the PM, and you were **considering two approaches** of risk response to either remove the risk from the project to another stakeholder or do nothing, what **two responses** are you referring to? (select 2)

a) Avoid
b) Mitigate
c) Transfer
d) Accept

Name Andrew Berg

V. Budgeting and Cost Estimation (35 pts)

24. After completing an initial project plan, you begin the process of budgeting and cost estimation. You received the following information from your team members:

Resources	*Number required*	*Cost per unit*	*Total Cost*
Floor panels	10	$11,000 each	$110,000.00
Wall panels	12	$7,000 each	$84,000.00
Window panels	8	$3,000 each	$24,000.00
Roofing panels	8	$5,000 each	$40,000.00
Truss	4	$15,500 each	$62,000.00
Door	6	$2,500 each	$15,000.00
TOTAL MATERIALS COST			***$335,000.00***
Project Manager	300 hrs	$450/day	*$16,875.00*
Foreman	200 hrs	$250/day	*$6,250.00*
Document Control Mgr	100 hrs	$100/day	*$1,875.00*
TOTAL INTERNAL LABOR COST			***$25,000.00***
Inspector (Contractor)	48 hrs	$150/day	*$900.00*
Floor Sub-Contract	250 hrs	$24/hour	*$6,000.00*
Vertical Sub-Contract	200 hrs	$48/hour	*$9,600.00*
Roof Sub-Contract	150 hrs	$52/hour	*$7,800.00*
TOTAL CONTRACTED LABOR COST			***$24,300.00***

Organization Charges:
- Overhead Charges at 10% of internal labor costs,
- General Administrative costs at 5% of internal and contracted labor costs.
- Your organization includes a 15% contingency on all direct and indirect costs.

a) (10 pts) Given the project plan and organizational charges above, provide the Project Manager's **cost baseline or "fully costed" project budget** (SHOW YOUR WORK).

Direct Cost = 335,000 + 25,000 + 24,300 = 384,300

Indirect Cost = 25,000 (0.10) + (25,000 + 24,300)(0.15) = 9,895

Subtotal = 384,300 + 9,895 = 394,195

Contingency = (394,195)(0.15) = 59,129.25

Fully Costed = Subtotal + Contingency

= 394,195 + 59,129.25

Answer: = $453,324.25

b) (2 pts) The budget table above (Question 24, Page 9) is an example of _________________.

a. Activity-Oriented Budget
b. Top-Down Budget
c. Category-Oriented Budget
d. Parametric Budget

25. (2 pts) It costs the contractor \$6 million for its 26-person crew to complete a 4-month project. The Budget Office estimates that it will cost \$4.75 million for the crew to complete a 2.5-month project. What **cost estimating technique** is the Budget Office using to complete this estimate? What type of **budgeting strategy** is this?

Cost Estimating Tech: Analogous, Parametric, Scaling, Rule of Thumb/Expert Opinion **(circle one)**

Budget Strategy: Top-Down, Bottom-Up, Iterative **(circle one)**

26. (3 pts) There are 5 common sources of Budget Uncertainty. Select 3 of them from the list below.

1. Technological Shock
2. Change Management process
3. Poor Estimating Techniques
4. Personnel Turnover
5. Project Charter details

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27. (10 pts) In order to better estimate task completion times for elements in the WBS, you consider the impact of learning curves on the placement of floor panels. Your team calculates that the **learning rate of 90 percent** will be **fully realized after the 10th attempt**. They calculate the **10th attempt will take 6 hours to complete**. If this is true, how long would it take to place the **1st floor panel**?

$lr = 0.9$ $\quad T_n = 6$ $\quad T_1 = ?$ $\quad n = 10$

$r = \frac{\ln(0.9)}{\ln(2)} = -0.152$

$6 = T_1 (10)^{-0.152}$

$T_n = T_1 n^r$

$T_1 = \frac{6}{(10)^{-0.152}} = 8.51$ hours

Answer: 8.51 hrs

28. (8 pts) During project execution, you learn that the team actually only required 5 **hours to complete the 10th attempt, a new fully realized time.** If it took **80.3 hours** to complete the first 10 attempts, how long will it take to complete 30 floor panels.

$n_{10-1} + n_{30-11}$ = total time for 30 panels

80.3 + 100 = 180.3 hours to complete 30 floor panels

Answer: 180.3 hrs

VI. Scheduling (50 pts)

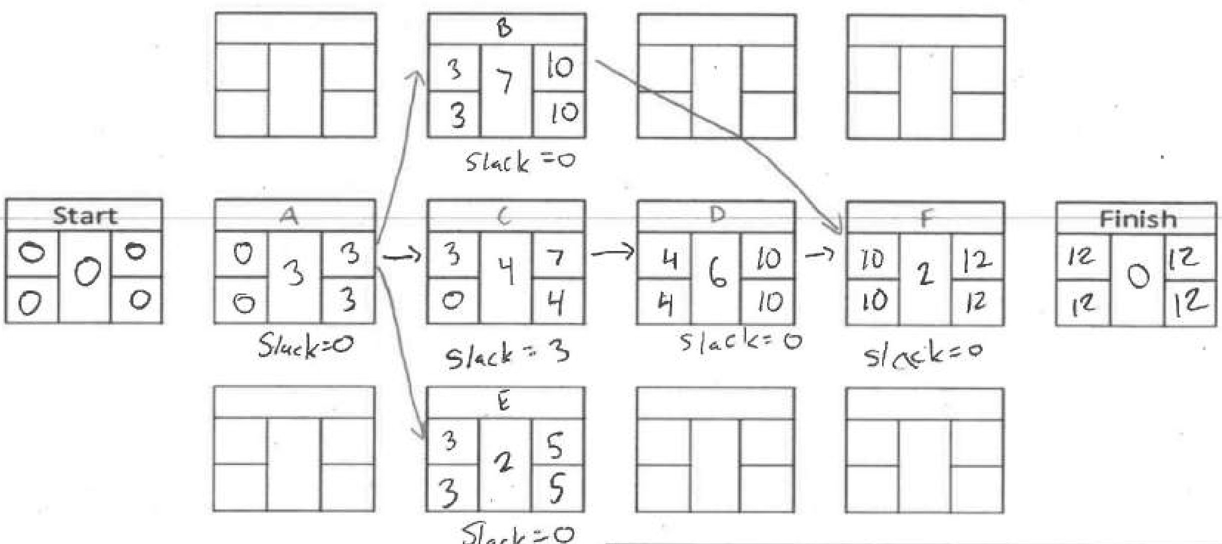
29. (2 pts) ________________ allows an acceleration of a successor activity but risks the over allocation of scarce resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

30. (8 pts) In the table below, ***complete*** the TEs and variances for activities B and E. Note that some answers have already been provided to you. ***Round TEs as you've been taught in class. Round Variance to the nearest hundredths place.***

Activity ID	Activity Name	Duration (weeks)					
		Pred.	a Optimistic	m Most Likely	b Pessimistic	TE	Variance
A	Fence-off Project Site	--	2	3	4	3.0	0.11
B	Replace Roofing Panels	A	5	7	10	7.0	0.69
C	Gut Interior	A	3	3	5	4.0	0.11
D	Install New Interior	C	4	5	8	6.0	0.44
E	Power Wash Exterior	A	2	2	4	2.0	0.11
F	Remove Site Fencing	B, D	2	2	2	2.0	0.00

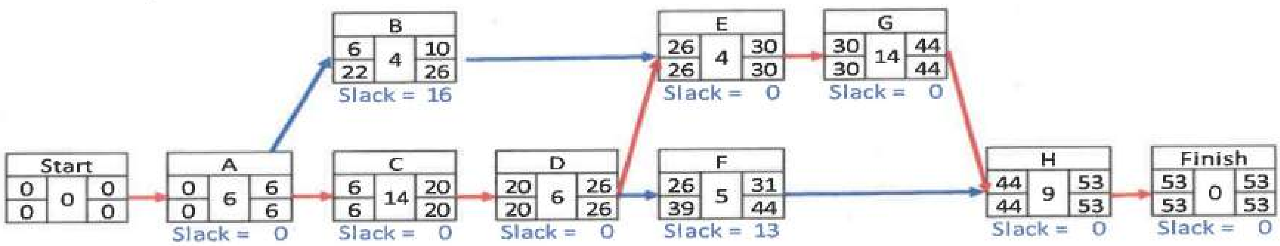
31. (18 pts) In the space provided *draw a legible network diagram per* ***DSE Standard****, for this project.* Use the Activity ID and the TE time from the table above ***(from Question 31)***.



Project Duration:________________
Critical Path: Gut Interior

32. After further analysis, several additional tasks were added to the project, and task duration estimates were improved.

Activity ID	*Duration (Days)*		
	Pred.	*TE*	*Variance*
A	--	6.0	0.44
B	B	4.0	0.11
C	A	14.0	1.78
D	C	6.0	0.44
E	B, D	4.0	0.25
F	D	5.0	0.00
G	E	14.0	0.25
H	F, G	9.0	1.36



Answer the questions below, **given the updated table and network diagram shown above.**

a. (5 pts) What is the project variance (σ^2)?

$\sigma^2 = 4.63$

b. (5 pts) What is the probability of completing the project in **53 days**?

$$z = \frac{53-53}{\sqrt{4.63}} = 0.5000$$

Answer: 0.5 probability of completing the project

c. (5 pts) What is the probability of completing the project within **48 days**?

$$z = \frac{48 - 53}{\sqrt{4.63}} = -2.32$$

$$= 0.9898$$

Answer: 0.9898 probability

d. (5 pts) How many days will you need to complete the project if the CEO asks for a ***91%* assurance** that you'll meet your estimated date?

$$D = 4.63 + 53 \cdot 0.91$$

$$D = 52.86 \text{ days}$$

Answer: $\approx$ 53 days for a 91% assurance

33. (2 pts)______________ delays a successor activity and may result in an under-allocation of resources.

a. Uncertainty
b. Lag time
c. Slack or Float
d. Lead time

Blank Page

EM411 WPR/MidTerm

Graded

Student

Andrew Berg

Total Points

106 / 175 pts

Question 1

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

Question 2

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 3

(no title) **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Click here to replace this description.

Question 4

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 5

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 6

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect

Question 7

(no title) 2 / 2 pts

✔ – **0 pts** Correct

– **2 pts** Incorrect

Question 8

(no title) 0 / 2 pts

– **0 pts** Correct

✔ – **2 pts** Incorrect

Question 9

(no title) 2 / 2 pts

✔ – **0 pts** Correct

– **2 pts** Incorrect

Question 10

(no title) 2 / 2 pts

✔ – **0 pts** Correct

– **2 pts** Incorrect

Question 11

(no title) 5 / 5 pts

✔ – **0 pts** Correct

– **2 pts** Math error. Formula is correct.

– **5 pts** Incorrect formula, results, and reasoning.

– **1 pt** Incorrect or missing investment recommendation.

– **1 pt** Solve the NPV.

– **4 pts** Incorrect NPV but recommendation follows.

Question 12

(no title) 2 / 4 pts

– **0 pts** Correct

✔ – **2 pts** One Correct

– **4 pts** None Correct

Question 13

(no title) **7** / 7 pts

✔ **– 0 pts** Correct

– 1 pt One Incorrect. Correct Answers: A, B, A, B, A, B, B

– 2 pts Two Incorrect. Correct Answers: A, B, A, B, A, B, B

– 3 pts Three Incorrect. Correct Answers: A, B, A, B, A, B, B

– 4 pts Four Incorrect. Correct Answers: A, B, A, B, A, B, B

– 5 pts Five Incorrect. Correct Answers: A, B, A, B, A, B, B

– 6 pts Six Incorrect. Correct Answers: A, B, A, B, A, B, B

– 7 pts Seven Incorrect. Correct Answers: A, B, A, B, A, B, B

Question 14

(no title) **2** / 2 pts

✔ **– 0 pts** Correct

– 2 pts Incorrect. The correct answer is: b) Possesses high technical skills and analysis mindset.

Question 15

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 3 pts Incorrect

Question 16

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 3 pts Incorrect

Question 17

(no title) **2** / 6 pts

– 0 pts Correct

– 2 pts One incorrect. Correct answers are C, A, B

✔ **– 4 pts** Two incorrect. Correct answers are C, A, B

– 6 pts All incorrect. Correct answers are C, A, B

Question 18

(no title) **0** / 3 pts

– 0 pts Correct

– 1.5 pts Partially correct

✔ **– 3 pts** Incorrect

Question 19

(no title) **0** / 3 pts

– 0 pts Correct

✔ **– 3 pts** Incorrect

Question 20

(no title) **0** / 4 pts

– 0 pts Correct

– 2 pts Partially Correct

✔ **– 4 pts** Incorrect

Question 21

(no title) **2** / 8 pts

– 0 pts Correct

– 2 pts 1 Incorrect

– 4 pts 2 Incorrect

✔ **– 6 pts** 3 Incorrect

– 8 pts 4 Incorrect

Question 22

(no title) **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Incorrect

Question 23

(no title) **0** / 20 pts

23.1 Part a **0** / 9 pts

– 0 pts Correct

✔ **– 9 pts** All incorrect

– 6 pts RPNs incorrect

– 3 pts Risk type incorrect

23.2 Part b **0** / 4 pts

– 0 pts Correct

✔ **– 4 pts** None correct

– 2 pts One correct

23.3 Part c **0** / 3 pts

– 0 pts Correct

✔ **– 3 pts** Incorrect

23.4 Part d **0** / 4 pts

– 0 pts Correct

– 2 pts One correct

✔ **– 4 pts** None correct

Question 24

(no title) **9** / 12 pts

24.1 Part a **9** / 10 pts

– 0 pts Correct

– 2 pts Incorrect Indirect Costs

✔ **– 1 pt** Math error

– 2 pts Incorrect contingency

– 1 pt Incorrect direct cost

Math error.

24.2 Part b **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Incorrect

Question 25

(no title) **0** / 2 pts

– 0 pts Correct

– 1 pt 1 Correct

✔ **– 2 pts** Nothing Correct

Question 26

(no title) **3** / 3 pts

✔ **– 0 pts** Correct

– 1 pt 2 Correct

Question 27

(no title) **10** / 10 pts

✔ **– 0 pts** Correct

– 2 pts Math errors

– 3 pts Incorrect variables

– 3 pts Incorrect equation

– 3 pts "r" not calculated

Question 28

(no title) **8** / 8 pts

✔ **– 0 pts** Correct

– 2 pts Wrong equation/approach

– 1 pt Wrong # of iterations

– 1 pt Math error

Question 29

(no title) **0** / 2 pts

– 0 pts Correct

✔ **– 2 pts** Incorrect

Question 30

(no title) **6** / 8 pts

– 0 pts Correct

– 2 pts One mistake

✔ **– 2 pts** Rounding errors

Question 31

(no title) **10** / 18 pts

– 0 pts Correct

✔ **– 8 pts** Multiple errors

– 2 pts Slack not ID'd

– 2 pts Incorrect backward/backward pass

– 2 pts Incorrect relationships

Incorrect forward and backward pass, incorrect duration, incorrect CP, slack not identified, DSE standard not met.

Question 32

(no title) **16** / 20 pts

32.1 **Part a** **4** / 5 pts

- **− 0 pts** Correct
- **− 5 pts** Incorrect
- ✔ **− 1 pt** Partial credit

32.2 **Part b** **5** / 5 pts

- ✔ **− 0 pts** Correct
- **− 5 pts** Incorrect

32.3 **Part c** **3** / 5 pts

- **− 0 pts** Correct
- ✔ **− 2 pts** Partial credit for work shown
- **− 1 pt** Math or Z table error
- **− 5 pts** Blank

32.4 **Part d** **4** / 5 pts

- **− 0 pts** Correct
- ✔ **− 1 pt** Partial credit - equation/z value/math errors
- **− 5 pts** Blank or no meaningful work
- **− 2 pts** Partial credit - wrong approach

Question 33

(no title) **2** / 2 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** Incorrect

TAB H TERM END EXAMINATION

Term End Examinations are kept in a secure location and can be accessed by request to the Program Director.

UNITED STATES MILITARY ACADEMY WEST POINT.

TAB I FINAL COURSE GRADES

Department of Systems Engineering

Final Grade Approval
AT26-1

EM411 Project Management

Department Head:	*COL Coxen*
Program Director:	*COL Thompson*
Course Director:	*Mr. Finch*

For Faculty Viewing Only

Secure When Not In Use
Caution: Not For Cadet Viewing!

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-SE 19 December 2025

MEMORANDUM THRU COL Brandon Thompson, Engineering Management Program Director

FOR COL Julia Coxen, Professor and Head, Department of Systems Engineering

SUBJECT: EM411, AT26-1 Project Management Final Grade Approval.

1. **RECOMMENDATION.** Recommend approval of final grade results. This report includes the Course Order of Merit List with recommended Cutlines in Tab A.

2. **COURSE DATA.**

a. Course Enrollment: 91

b. Course Average: 88.13%

c. Table 1 shows this semester's course average alongside the previous 13 semester averages. Table 2 shows this semester's course average by cadet major.

Academic Term	EM411 Average
26-1	88.13%
25-2	87.72%
25-1	85.77%
24-2	86.53%
24-1	85.59%
23-2	86.40%
23-1	85.89%
22-2	87.60%
22-1	87.11%
21-2	87.47%
21-1	86.27%
20-3	91.45%
20-2	88.51%
20-1	84.63%

Table 1: Historical Course Averages

Major	# of Cadets	Course AVG
SEN	9	92.36%
SDS	31	84.54%
ENM	24	88.10%
BEM	1	77.35%
MGN	3	92.88%
EVE	1	90.40%
EPS	1	85.30%
ORE	6	92.13%
NEN	1	93.60%
Other	14	91.07%

Table 2: Course Average by Major

d.

d. Table 3 and Figure 1 show this semester's Standard and Recommended Grade Distributions. Tab A contains the full OML with grade distribution cutlines.

Grade	Standard Cut	Cadets	% of Course	Recommended Cut	Cadets	% of Course
A+	97.50%	2	2%	97.50%	5	5%
A	92.50%	20	22%	92.50%	21	23%
A-	90.00%	16	18%	90.00%	13	14%
B+	87.50%	18	20%	87.50%	17	19%
B	82.50%	19	21%	82.50%	19	21%
B-	80.00%	8	9%	80.00%	8	9%
C+	77.50%	5	5%	77.50%	5	5%
C	72.50%	3	3%	72.50%	3	3%
C-	70.00%	0	0%	70.00%	0	0%
D	65.00%	0	0%	65.00%	0	0%
F	0.00%	0	0%	0.00%	0	0%

Table 3: Standard and Recommended Grade Distribution Table

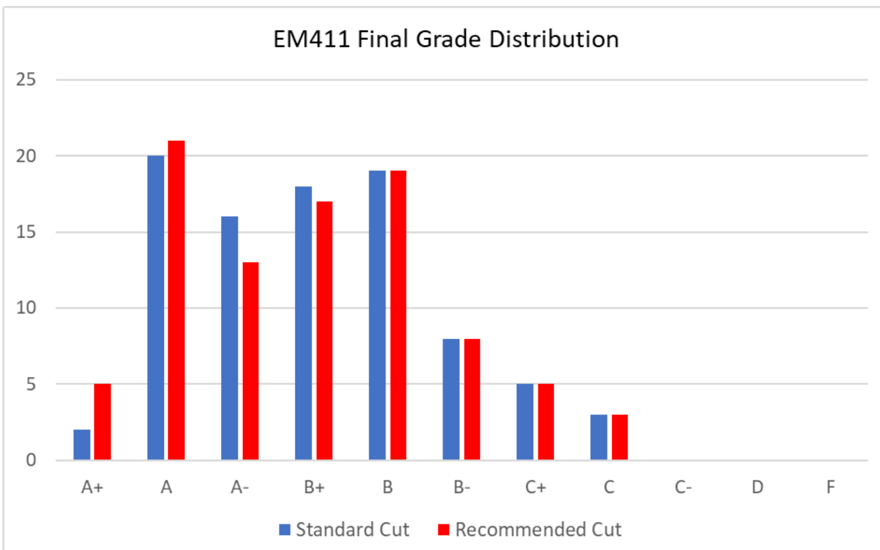


Figure 1: Histogram of Standard Grade Distribution

e. **FAILURES.** There are no course failures in the AT26-1 EM411 Course.

DANIEL FINCH
Senior Instructor
EM411 Course Director

Encl
Tab A. Course OML with Cut Lines

TAB A: Course OML with Standard and Recommended Cut Lines

Name	Maj	Grad Ye	OM	PS1 75	PS2 75	SG1 75	SG2 90	SG3 60	SG4 75	IP 50	Quiz 1 50	Quiz 2 50	WPR 175	TEE 225	Points Earne	Percen	Letter Grad	Standard Cut Line Recommended Cut Line
Dailey, Eric	ORE	2026		75.0	75.0	75.0	90.0	60.0	75.0	50.0	45.0	50.0	173.0	212.5	980.5	98.05%	A+	
Dosan, Logan	ENM	2026		75.0	70.0	75.0	90.0	58.0	75.0	50.0	50.0	50.0	168.0	215.0	976.0	97.60%	A+	A+
Lynch, Benjamin	SEN	2027		75.0	72.5	74.0	87.0	55.0	75.0	49.0	48.0	49.0	171.0	209.0	964.5	96.45%	A	A+
Hecker, Kennedy	SEN	2026		74.0	70.0	75.0	90.0	56.0	75.0	50.0	47.0	47.5	171.0	201.0	956.5	95.65%	A	A+
Swearingen, Jordann		2027		74.0	72.0	75.0	85.0	55.0	74.0	47.0	46.0	50.0	166.0	211.0	955.0	95.50%	A	A+
Lopez Bol, Rodrigo		2026		75.0	74.0	75.0	90.0	56.0	72.0	49.0	46.0	50.0	161.0	202.0	950.0	95.00%	A	
Matteson, Collin	MGN	2026		75.0	73.0	75.0	88.0	58.0	75.0	50.0	45.0	47.5	164.0	196.5	947.0	94.70%	A	
In, Jonathan	ORE	2027		75.0	71.0	66.0	81.0	55.0	75.0	50.0	45.0	49.0	170.0	207.5	944.5	94.45%	A	
Aycock, Walker	ENM	2026		65.0	72.0	74.0	86.0	59.0	73.0	50.0	44.0	49.0	168.0	200.5	940.5	94.05%	A	
Simon Crespo, Ignacio		2026		75.0	74.0	70.0	85.0	53.0	72.0	49.0	41.0	50.0	169.0	202.5	940.5	94.05%	A	
Schodrof, Cooper	ENM	2026		75.0	70.0	75.0	90.0	58.0	75.0	50.0	43.0	50.0	165.0	187.5	938.5	93.85%	A	
Mansfield, Zachary	ENM	2026		74.0	66.0	72.0	90.0	54.0	75.0	49.0	48.0	46.5	165.0	197.0	936.5	93.65%	A	
Cooper, William	NEN	2026		75.0	64.0	72.0	90.0	53.0	75.0	50.0	46.5	50.0	160.0	200.5	936.0	93.60%	A	
Guedes, Rafael		2026		75.0	74.0	75.0	89.0	58.0	75.0	45.0	46.5	49.0	159.0	190.5	936.0	93.60%	A	
Mott, Brady	SDS	2027		69.0	73.0	73.0	85.0	53.0	75.0	47.0	46.0	49.0	169.0	196.0	935.0	93.50%	A	
Chernik, Sarah	ENM	2026		75.0	75.0	75.0	90.0	55.0	75.0	49.0	47.0	50.0	155.0	187.5	933.5	93.35%	A	
Zink, Linus		2026		75.0	74.0	74.0	87.0	58.0	70.0	49.0	44.0	48.0	158.0	196.0	933.0	93.30%	A	
Parker, William	SEN	2026		74.5	73.0	75.0	90.0	60.0	75.0	50.0	42.0	46.0	168.0	177.5	931.0	93.10%	A	
Moffat, Jake	ORE	2026		75.0	70.0	73.0	85.0	53.0	73.0	50.0	45.0	49.5	155.0	200.5	929.0	92.90%	A	
Reynolds, Zachary	SEN	2026		71.0	75.0	73.0	86.0	59.0	75.0	50.0	43.0	49.0	155.0	191.5	927.5	92.75%	A	
Segat, Peter	SEN	2026		75.0	74.0	68.0	88.0	52.0	74.0	47.0	46.0	50.0	167.0	186.5	927.5	92.75%	A	
Sta Romana, Joseph	MGN	2026		68.5	72.0	71.0	87.0	60.0	72.0	45.0	47.0	48.0	150.0	205.5	926.0	92.60%	A	A
Ross, Jeffry	SDS	2027		75.0	73.0	72.0	81.0	55.0	71.0	50.0	41.0	50.0	168.0	188.0	924.0	92.40%	A-	A
Dorman, Lance	ENM	2026		75.0	68.0	70.0	89.0	56.0	74.0	49.0	46.0	48.0	159.0	188.0	922.0	92.20%	A-	A
Norris, Hunter	ENM	2026		75.0	73.5	74.0	61.0	58.0	73.0	50.0	46.0	47.0	157.0	206.5	921.0	92.10%	A-	A
Martinez Alvarez, Enrique		2026		73.0	70.0	74.0	90.0	53.0	73.0	50.0	47.0	50.0	156.0	184.5	920.5	92.05%	A-	A
Vanhessen, Zachary	MGN	2026		75.0	67.0	71.0	88.0	56.0	75.0	47.0	42.0	49.5	141.0	202.0	913.5	91.35%	A-	
Follenweider, Jordan	SEN	2026		73.0	74.0	75.0	90.0	58.0	75.0	50.0	44.0	46.0	168.0	160.0	913.0	91.30%	A-	
Mateos Monton, Jose Javier		2026		75.0	72.0	63.0	88.0	58.0	70.0	49.0	37.0	50.0	151.0	195.0	908.0	90.80%	A-	
Thabit, Sarah	ENM	2027		75.0	74.0	65.0	85.0	56.0	65.0	50.0	44.0	47.5	160.0	186.5	908.0	90.80%	A-	
Morote Rodriguez, Gonzalo		2026		75.0	70.0	73.0	86.0	53.0	75.0	50.0	40.0	50.0	149.0	186.0	907.0	90.70%	A-	
Soriano Ortega, Izan		2026		75.0	70.0	64.0	86.0	53.0	75.0	45.0	44.0	49.0	154.0	191.5	906.5	90.65%	A-	
Gologorsky, Daniel	ORE	2026		65.0	68.5	70.0	86.0	53.0	71.0	50.0	46.0	45.0	163.0	188.5	906.0	90.60%	A-	
Montanez, Jonah	EVE	2026		68.0	68.0	70.0	90.0	53.0	75.0	50.0	40.0	47.5	155.0	187.5	904.0	90.40%	A-	
Vo, Vincent	SEN	2026		72.0	73.0	73.0	87.0	60.0	75.0	45.0	40.0	50.0	151.0	177.0	903.0	90.30%	A-	
Izquierdo Barba, Pablo		2026		73.0	75.0	74.0	89.0	58.0	68.0	45.0	41.0	49.0	140.0	190.0	902.0	90.20%	A-	
Won, Justin	SEN	2026		75.0	65.0	71.0	88.0	58.0	75.0	50.0	39.0	48.5	158.0	174.0	901.5	90.15%	A-	
Nnadozie, Stephen	SDS	2027		73.0	71.0	64.0	86.0	59.0	71.0	48.0	43.0	43.0	168.0	174.0	900.0	90.00%	A-	A-
Baron, Emma	ORE	2028		74.0	73.0	66.0	89.0	56.0	71.0	50.0	39.0	46.5	166.0	167.5	898.0	89.80%	B+	A-

Kanta, Lucas	SDS	2026		68.0	70.0	73.0	85.0	52.0	70.0	46.0	46.0	45.0	167.0	172.5	894.5	89.45%	B+	
Laguna Morilla, Javier		2026		75.0	73.0	73.0	85.0	52.0	72.0	48.0	34.0	46.0	146.0	190.5	894.5	89.45%	B+	
Lewless, James	SDS	2027		75.0	65.0	66.0	74.0	55.0	73.0	45.0	44.0	50.0	156.0	191.5	894.5	89.45%	B+	
Kloepper, Jacob	SDS	2027		64.0	65.0	65.0	85.0	55.0	70.0	47.0	47.0	44.0	169.0	181.0	892.0	89.20%	B+	
Dimapo, Bophelo	SDS	2026		70.0	67.0	58.0	87.0	60.0	69.0	45.0	43.0	48.0	141.0	203.0	891.0	89.10%	B+	
McDaniel, Trae	ENM	2026		75.0	71.0	65.0	80.0	51.0	75.0	50.0	42.0	49.0	164.0	168.5	890.5	89.05%	B+	
Whitaker, Justus	SEN	2026		69.0	71.0	67.0	82.0	48.0	68.0	44.0	42.0	48.5	152.0	197.0	888.5	88.85%	B+	
Strange, Anderson	SDS	2027		72.0	72.5	75.0	90.0	58.0	75.0	45.0	34.5	50.0	137.0	179.0	888.0	88.80%	B+	
Straw, Brendan	ENM	2026		75.0	75.0	69.0	64.0	48.0	70.0	44.0	47.0	49.5	162.0	182.0	885.5	88.55%	B+	
Faulkner, Harrison	ENM	2026		68.5	73.0	57.0	61.0	58.0	72.0	50.0	44.0	43.0	166.0	192.0	884.5	88.45%	B+	
Pointer, Kavon	SDS	2027		71.0	75.0	71.0	83.0	58.0	75.0	45.0	37.0	46.0	147.0	176.5	884.5	88.45%	B+	
Smith, Cooper	SDS	2027		66.0	65.0	71.0	87.0	58.0	66.0	46.0	48.0	48.0	145.5	183.0	883.5	88.35%	B+	
Dabkowski, Gabriel	ENM	2026		70.0	70.8	68.0	81.0	55.0	75.0	50.0	47.0	43.0	143.5	179.5	882.8	88.28%	B+	
Brownfield, Georgia	ENM	2026		67.0	71.0	74.0	87.0	55.0	73.0	49.0	43.0	50.0	141.0	172.0	882.0	88.20%	B+	
Ali, Amir	SDS	2027		55.0	68.0	69.0	85.0	52.0	72.0	47.0	41.0	47.5	155.0	189.0	880.5	88.05%	B+	
Bagley, Joshua	ENM	2026		58.0	62.0	68.0	90.0	56.0	66.0	45.0	41.0	43.0	150.5	198.0	877.5	87.75%	B+	
Rambo, Chloe	SDS	2027		73.0	67.0	74.0	78.0	57.0	66.0	50.0	44.0	43.0	150.0	175.5	877.5	87.75%	B+	B+
Hubert, Ashley	ENM	2026		59.0	68.0	69.0	80.0	48.0	70.0	44.0	42.0	40.5	166.0	183.5	870.0	87.00%	B	
Orr, Marianne	ORE	2027		74.0	67.0	73.0	78.0	56.0	68.0	46.0	39.5	46.5	161.0	161.0	870.0	87.00%	B	
Painter, John	SDS	2027		67.0	63.0	68.0	90.0	56.0	70.0	47.0	44.0	50.0	146.0	168.5	869.5	86.95%	B	
Embree, Omari	SDS	2027		68.0	74.0	70.0	78.0	56.0	70.0	45.0	41.0	49.0	144.0	173.0	868.0	86.80%	B	
De Meer Canon, Miguel		2026		69.0	74.0	65.0	78.0	56.0	71.0	49.0	37.0	50.0	124.0	193.5	866.5	86.65%	B	
Parra Martinez, Antonio		2026		73.0	74.0	71.0	87.0	55.0	66.0	46.0	42.0	50.0	144.5	156.5	865.0	86.50%	B	
Spoerl, James	ENM	2026		65.0	73.0	72.0	89.0	58.0	72.0	45.0	41.0	43.0	145.0	162.0	865.0	86.50%	B	
Steinmetz, Silas		2026		75.0	72.0	74.0	90.0	56.0	70.0	49.0	39.0	46.0	121.0	173.0	865.0	86.50%	B	
Carter, Terrence	ENM	2026		69.0	70.5	68.0	78.0	57.0	69.0	45.0	40.0	43.0	145.0	169.0	853.5	85.35%	B	
Bocis, Samantha	EPS	2027		71.0	65.0	75.0	90.0	58.0	75.0	45.0	39.0	47.0	122.0	166.0	853.0	85.30%	B	
McKeon, Daniel	SDS	2027		59.0	68.0	63.0	85.0	52.0	61.0	45.0	37.0	48.0	154.0	179.5	851.5	85.15%	B	
Hellums, Cale	SDS	2027		72.0	73.0	67.0	83.0	58.0	72.0	45.0	41.0	33.0	155.0	151.5	850.5	85.05%	B	
Miller, Noah	ENM	2027		69.0	66.0	63.0	85.0	56.0	70.0	47.0	39.0	48.5	134.0	172.5	850.0	85.00%	B	
Kirby, Kendrick	SDS	2027		69.0	62.0	53.0	90.0	58.0	75.0	45.0	37.0	47.0	139.0	173.5	848.5	84.85%	B	
Foreman, Chadon	ENM	2026		68.0	62.0	64.0	61.0	58.0	71.0	45.0	45.0	44.0	152.0	177.0	847.0	84.70%	B	
Nixon, Noah	SDS	2027		75.0	73.0	64.0	87.0	55.0	69.0	44.0	45.0	47.0	135.0	146.5	840.5	84.05%	B	
Coats, Natas	SDS	2027		75.0	67.0	60.0	74.0	55.0	65.0	48.0	45.0	46.0	142.0	160.5	837.5	83.75%	B	
Hummel, Edward	ENM	2027		67.0	72.0	69.0	87.0	60.0	64.0	45.0	32.0	42.0	135.5	154.5	828.0	82.80%	B	
Saulsberry, Taylor	ENM	2026		57.0	73.0	71.0	85.0	52.0	56.0	45.0	41.0	43.5	137.0	166.5	827.0	82.70%	B	B
Perry, Regal	SDS	2027		73.0	60.5	66.0	81.0	55.0	68.0	50.0	39.0	41.0	142.0	148.5	824.0	82.40%	B-	
Appleton, Henry	SDS	2027		73.0	66.0	60.0	78.0	57.0	68.0	43.0	37.5	41.0	144.0	154.5	822.0	82.20%	B-	
Garcia, David	SDS	2025		53.0	53.0	60.0	89.0	58.0	0.0	50.0	47.0	48.0	168.0	190.5	816.5	81.65%	B-	
Williams, Ashlon	ENM	2025		70.0	64.0	69.0	80.0	48.0	55.0	38.0	38.5	37.5	159.0	156.0	815.0	81.50%	B-	
Thompson, Kurt	SDS	2027		73.0	71.0	52.0	80.0	51.0	64.0	40.0	31.5	44.0	159.0	146.5	812.0	81.20%	B-	
Werntz, Bryson	ENM	2026		75.0	68.0	59.0	64.0	48.0	69.0	47.0	39.0	42.5	122.0	175.5	809.0	80.90%	B-	
Hernandez, Austin	ENM	2026		64.0	72.0	65.0	74.0	55.0	67.0	46.0	34.5	46.0	131.0	147.5	802.0	80.20%	B-	
Andrews, Aatavion	SDS	2027		75.0	69.0	65.0	80.0	48.0	63.0	40.0	45.5	41.5	115.0	158.0	800.0	80.00%	B-	B-
Brungard, Caden	SDS	2027		73.0	69.0	57.0	80.0	51.0	72.0	40.0	34.0	45.0	140.0	137.5	798.5	79.85%	C+	
Berg, Andrew	SDS	2026		59.0	67.0	57.0	87.0	58.0	70.0	43.0	38.0	42.5	106.0	169.5	797.0	79.70%	C+	
Penswick, Robert	SDS	2027		59.0	67.0	56.0	80.0	55.0	70.0	44.0	36.0	44.5	122.0	154.0	787.5	78.75%	C+	
Rodgers, Owen	SDS	2027		69.0	41.0	56.0	82.0	48.0	70.0	43.0	38.0	47.0	118.0	168.0	780.0	78.00%	C+	
Nash, Chadwick	SDS	2027		59.0	56.0	62.0	90.0	54.0	61.0	46.0	42.0	39.5	137.0	128.5	775.0	77.50%	C+	C+
Malbon, Jeremiah	BEM	2026		66.0	51.0	61.0	82.0	48.0	62.0	38.0	36.0	47.5	131.0	151.0	773.5	77.35%	C	
Mayes, Jaydan	SDS	2027		62.0	71.0	34.0	83.0	58.0	0.0	45.0	36.5	38.0	155.0	162.0	744.5	74.45%	C	
Engler, Henry	SDS	2026		69.0	0.0	67.0	64.0	48.0	46.0	43.0	43.5	45.0	158.0	157.5	741.0	74.10%	C	C

SECTION III

ASSIGNMENTS

TAB J PROBLEM SETS

PROBLEM SET #1

Instructor Solution

Name: SOLUTION_________
Section: ______

EM411 Project Management
AT26-1

Project Planning Problem Set #1 (75 points)

DOCUMENTATION. This deliverable is an individual assignment. Any assistance received must be documented in detail. Document all sources in accordance with the Office of the Dean Pamphlet "Documentation of Academic Work," Appendix E, and course guidance. Documentation must be turned in at the time of submission. The deliverable is considered late until all portions of the assignment and the documentation are submitted.

1. ***Scope***: Problem Set submissions will total 150 points over 2 assignments. Problems will be assigned when the associated content has been covered.

2. ***Turn in requirement***: Problem Set #1 will be submitted as a hard copy or via MS Teams Assignments **NLT 2359 on 29 September 2025**. Late submissions are subject to a 5% penalty per 24-hour period.

3. ***Guidelines for Collaboration***: For this assignment, individual work is highly encouraged, but collaboration between individuals is allowed. All collaboration must be documented. Any discussion of this Problem Set with anyone other than an EM411 instructor requires documentation. Documentation **must be specific** and detail the topics discussed and actions taken.

4. ***Acknowledgement Statement:*** Your submission **MUST** include an acknowledgement cover sheet.

5. ***SHOW ALL YOUR WORK!*** Unsupported or missing work earns NO credit. If Excel is used, submit the Excel file with your work.

6. ***Grading:* Problem Set #1 is worth 75 points.**

Name: SOLUTION_________
Section: _______

1. Net Present Cost Analysis (22 points)

The U.S. Army has launched its Future Portable Radio Communications (PRC) Competition. The PRC program office has narrowed the competition to two companies: A Harris Radio Team and a Thales Team. The two competitors have unique designs with requirements that differ in maintenance and power requirements. As part of the evaluation process, the Army required each company to provide cost data for outfitting, operating, and maintaining a radio package for an Infantry Company for **ten years** (see table below for notional data). A company radio package consists of **66 radios**. Under the Army's current battery contract, each Future PRC Battery costs **$21.50**, and both companies were required to design radios that use this battery. Each company must have **three (3)** soldiers trained on 15-level radio maintenance. Assume that the Army uses a **5%** hurdle rate and a **9%** annual inflation rate. The details of the two bids are below.

	Harris	**Thales**
Initial Cost per Radio	$26,564	$16,500
Batteries per Radio per Year	294	441
Repair Parts per Radio per year	$4,750	$4,020
Required Maintenance Training per Soldier (Executed at Years, 0, 3, 6, 9)	$7,500	$4,500

Calculate the Net Present Value for each PRC proposal and identify the best option (Your work must include Cash Flow Diagrams).

Cash Flow

Year	0	1	2	3	4	5	6	7	8	9	10
Option A	($1,775,724.00)	($730,686.00)	($730,686.00)	($753,186.00)	($730,686.00)	($730,686.00)	($753,186.00)	($730,686.00)	($730,686.00)	($753,186.00)	($730,686.00)
Option B	($1,102,500.00)	($891,099.00)	($891,099.00)	($904,599.00)	($891,099.00)	($891,099.00)	($904,599.00)	($891,099.00)	($891,099.00)	($904,599.00)	($891,099.00)

Annual NPV

Year	0	1	2	3	4	5	6	7	8	9	10
Option A	($1,775,724.00)	($640,952.63)	($562,239.15)	($508,379.10)	($432,624.77)	($379,495.41)	($343,141.41)	($292,009.40)	($256,148.59)	($231,610.68)	($197,098.03)
Option B	($1,102,500.00)	($781,665.79)	($685,671.75)	($610,578.56)	($527,602.14)	($462,808.90)	($412,123.14)	($356,116.42)	($312,382.82)	($278,171.38)	($240,368.44)

Harris NPV: -$5,619,423.16 Thales NPV: -$5,769,989.33

Best Option: Option A, Harris. This option has the lowest (least negative) NPV and therefore is the best value for the Government.

Name: SOLUTION__________
Section: ______

2. Budgeting and Cost Estimation (33 points)

Your company was just awarded the PRC contract. It will be your job to oversee the Initial Field Evaluation Production (IFEP) contract for the radio. Following a successful Initial Field Evaluation, your company can then move into full scale production. Your company's Vice President for Tactical Communications worked on your company's last radio fielding program and based on her experience, the cost of each radio during IFEP is about 26% greater than the cost during Full-Scale Production (FSP). An FSP PRC radio will cost $47,950.00. The Army intends on purchasing 3 lots of 22 radios during the IFEP period.

a. What is your budget estimate for the IFEP contract/project? (**4pt**)

Budget = Lots * Radios / Lot * IFEP Production Cost
= 3 * 22 * (1.26 * $47,950)
= $3,987,522.00

b. What ***cost estimation technique*** are you using? What type of ***budget strategy*** are you using? Name two advantages of this type of budgeting strategy. (**4 pts**)

Technique: Analogous / Expert Judgement
Strategy: Top-Down Budgeting

1. This method allows aggregate budgets to be determined quite accurately, even though individual elements may be in error.
2. No need to ID every small task; adds speed; no fear of overlooking tasks; relies on historic files and experience

Name: SOLUTION_________
Section: ______

Cost estimation information:

One of the main contributors to the IFEP's higher price is the cost of the required Signal Frequency and Interference Test. You decide to get together with your Company's Field Test Evaluator and Spectrum Manager to develop a more detailed cost estimate for the IFEP Contract.

Material and labor costs are provided in the table below. For labor costs, you're assuming an 8-hour day. Internal labor consists of the Project Manager, Operations Manager, Radio Manager, Field Test Operator, and Spectrum Manager. All other labor is external subcontracted labor. Your organization charges **overhead at 70% of internal labor costs**, and **general administrative costs at 24% of the total direct costs**. The overhead charges pay for benefits, training, and administration costs related to personnel. The administrative costs pay for the company's legal, billing, and contract management expenses across all projects. To account for potential weather and maintenance delays, you will apply a **12% contingency fee to your cost subtotal**.

Description	Resources	Number Required	Type	Cost per unit	Total Cost
Materials	Radio Body	66	Each	$9,700.00	$640,200.00
	Processor Chips	132	Each	$17,500.00	$2,310,000.00
	Antennas, Long	36	Pair	$129.00	$4,644.00
	Antennas, Short	36	Pair	$149.00	$5,364.00
	Testing Area Rental	14	Days	$800.00	$11,200.00
	Signals Test Kit	2	Each	$18,000.00	$36,000.00
	Batteries	10	boxes	$258.00	$2,580.00
				Total Materials	$ 3,009,988.00
Internal Labor	Project Manager	360	Hours	$220.00	$79,200.00
	Operations Manager	320	Hours	$180.00	$57,600.00
	Radio Manager	280	Hours	$165.00	$46,200.00
	Field Test Operator	80	Hours	$260.00	$20,800.00
	Spectrum Manager	80	Hours	$320.00	$25,600.00
				Total Internal Labor	$ 229,400.00
External Labor	Signals Data Analyst	300	Hours	$325.00	$97,500.00
	Materials Evaluator	1,600	Hours	$280.00	$448,000.00
	Battery Waste Transport & Disposal	40	Hours	$180.00	$7,200.00
	Auditor	160	Hours	$320.00	$51,200.00
				Total External Labor	$603,900.00
				Total Direct Costs	$3,843,288.00

Name: SOLUTION__________
Section: _______

c. Determine the Direct costs for the project. (**5 pts**)

Direct = Materials + Internal Labor + External Labor
= $ 3,009,988.00 + $ 229,400.00 + $ 603,900.00
= $ 3,843,288.00

d. Determine the Indirect costs for the project. (**5 pts**)

Indirect Costs = Overhead + G&A
= 0.70 * Internal labor + 0.24 * Direct costs
= 0.70 ($ 229,400.00) + 0.24 ($ 3,843,288.00)
= ($160,580.00) + ($922,389.12)
= $1,082,969.12

e. Determine the Subtotal estimate for the project budget.(**5 pts**)

Project Subtotal = Direct Costs + Indirect Costs
= $3,843,288.00 + $1,082,969.12
= $4,926,257.12

f. Determine the Contingency Estimate for the project budget. **(5 pts)**

Contingency = Direct Costs + Indirect Costs * Contingency Rate
= ($3,843,288.00 + $1,082,969.12) * .12
= $591,150.85

g. Determine the Fully Costed / Baseline for the project budget. (**5 pts**)

Fully costed budget = Project Subtotal + Contingency Estimate
= $4,926,257.12 + $591,150.85
= $5,517,407.97

Name: SOLUTION__________
Section: _______

3. Improving Cost Estimation (20 points)

Looking closer at your cost estimate, you see that you've estimated that it takes approximately 80 hours for the Field Test Operator and Spectrum Manager to conduct their final evaluation (consisting of 10x 8-hour Test Iterations). You based your cost estimate on historical evaluation data from the company's veteran test operators. However, you learn that your company just hired new test operators; and you're now concerned that your cost estimates are no longer accurate. The company's new test operator team completed their first of ten Test Iterations in **12 hours**. Based on company data, their learning rate should be **88%** and the team will reach their full efficiency after the 5th Iteration. Estimate the following.

a. How long will it take the Test Team to complete the 3rd Iteration? (**5 pts**)

$T_n = T_1 * n^r$; Hrs to complete the 3rd CFA: n = 3, $T_1 = 12, lr = 0.88$

$$T_3 = 12 * 3^{\frac{\ln(0.88)}{\ln(2)}}$$

$$T_3 = 12 * 3^{-0.1844}$$

$$T_3 = 12 * 0.8166$$

$$T_3 = 9.80\ hours$$

b. How long will it take the Test Team to complete the first 5 Iterations (cumulative)? (**5 pts**)

Using this formula: $Total\ Time = T_1 * \sum_{n=1}^{N} n^r$

$$Total\ Time = 12 * \sum_{1}^{5} n^r = 12 * (1 + .88 + .82 + .77 + .74) = 50.57$$

See Excel file in folder.

c. How long will it take the Test Team to complete all 10 Iterations (cumulative)? (**5 pts**)

$Total\ Time = (10 - 5) * T_5 + Total\ Time\ for\ first\ 5\ Iterations$

$Total\ Time = (5 * 8.92) + 50.57$

$Total\ Time = 44.6 + 50.57 = 95.16$

See Excel file in folder.

d. Do you need to adjust your cost estimate? If so, what is the new labor cost estimate for the Field Test Operator and Spectrum Manager? (**5 pts**)

Yes. The original time estimate for this activity was 80 hours for the Field Test Operator and Spectrum Manager. I need to adjust the time estimate to 95.16 hours and adjust the cost estimate accordingly.

New cost will be:
= (95.16 hours * $260/hour) + (95.16 hours * $320/hour)
= $24,741.80 + $30,451.44
= $55,193.24 → $8,793.24 MORE THAN the original cost.

PROBLEM SET #2

Instructor Solution

EM411 Project Management
AT26-1

Project Planning Problem Set #2 (75 points)

DOCUMENTATION. This deliverable is an individual assignment. Any assistance received must be documented in detail. Document all sources in accordance with the Office of the Dean Pamphlet "Documentation of Academic Work," Appendix E, and course guidance. Documentation must be turned in at the time of submission. The deliverable is considered late until all portions of the assignment and the documentation are submitted.

1. ***Scope***: Problem Set submissions will total 150 points over 2 assignments. Problems will be assigned when the associated content has been covered.

2. ***Turn in requirement***: Problem Set #2 is to be submitted as a Word or PDF document on CANVAS NLT **2359, 22 OCT 2025**. Late submissions are subject to a 5% penalty per 24-hour period.

3. ***Guidelines for Collaboration***: For this assignment, individual work is highly encouraged, but collaboration between individuals is allowed. All collaboration must be documented: any discussion of the deliverable with anyone other than an EM411 instructor requires documentation. Documentation **must be specific** and detail the topics discussed and actions taken.

4. ***Acknowledgement Statement:*** Your submission **MUST** submit an acknowledgement cover sheet and a works cited page with this assignment.

5. ***SHOW ALL YOUR WORK!*** Unsupported or missing work earns NO credit. If Excel is used, submit Excel file on CANVAS or email the file to your instructor.

6. ***Grading:* Problem #2 is worth 75 points.**

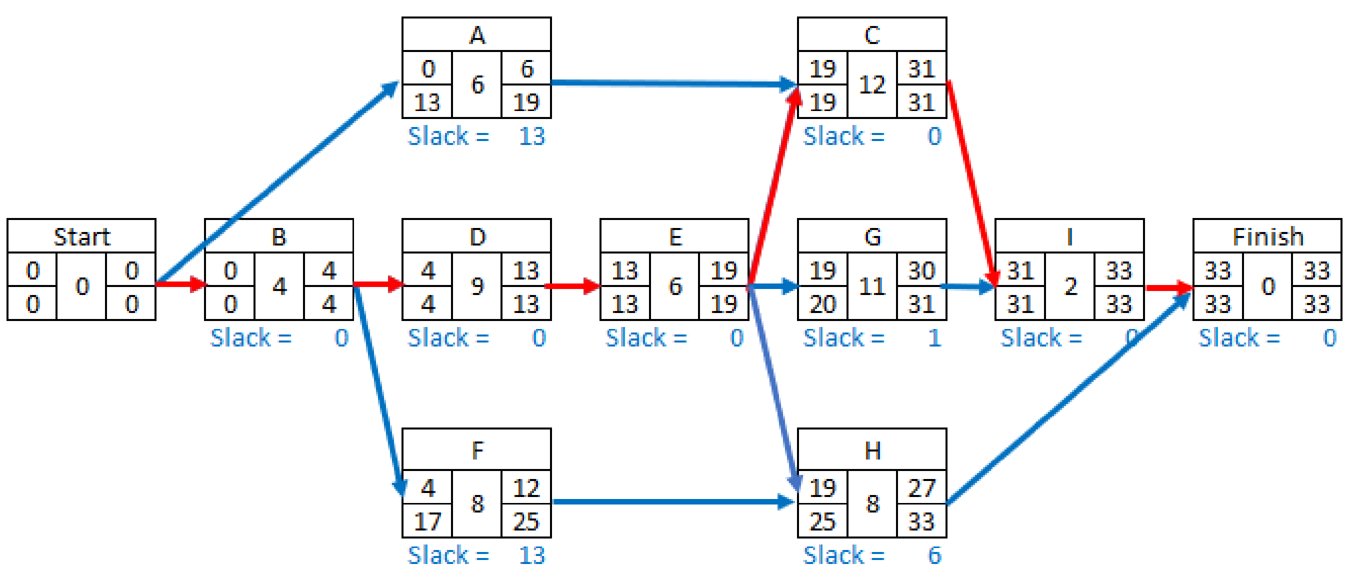
1. PERT and CPM (10 pts)

Before you coordinated with the Army Test Evaluation Command (ATEC) Test Officer, you did some initial planning for the operational test activities of XM30 Mechanized Infantry Combat Vehicle (MICV) competition in the table below. Your next task as the Project Manager is to calculate expected times, solve the network diagram, and calculate the probabilities of completing the project with the desired time.

Activity ID	*Activity Name*	*Pred.*	*Duration (weeks)* *Optimistic*	*most likely*	*Pessimistic*	*TE*	*Variance*
A	**Risk Assessment Plan**	--	4	6	8	6.0	**0.44**
B	**Initial Inspection**	**--**	**3**	**3**	**6**	**4.0**	**0.25**
C	**New Equipment Training**	**A,E**	**7**	**12**	**16**	**12.0**	**2.25**
D	**Automotive Performance Testing**	**B**	**6**	**8**	**11**	**9.0**	**0.69**
E	**RAM Testing**	**D**	**3**	**5**	**9**	**6.0**	**1.00**
F	**Weapons Firing**	B	6	7	9	**8.0**	**0.25**
G	**Armor Impact Testing**	E	8	11	12	**11.0**	**0.44**
H	**Test Incident Reporting**	E,F	6	7	9	**8.0**	**0.25**
I	**Final Inspection**	**C,G**	**1**	**2**	**3**	**2.0**	**0.11**

a. Complete the table above by calculating the Expected Time (TE) and Variance for each task.

b. Complete the Activity-on-Node Network Diagram, identify each task with Slack/Float time and identify/highlight the Critical Path. In the space provided ***complete the network diagram, per DSE Standard, for this project*** (*hint: you will not use all the boxes*).



c. How many days will you need to complete the project if your boss asks for a 90% assurance that you'll meet your estimated date?

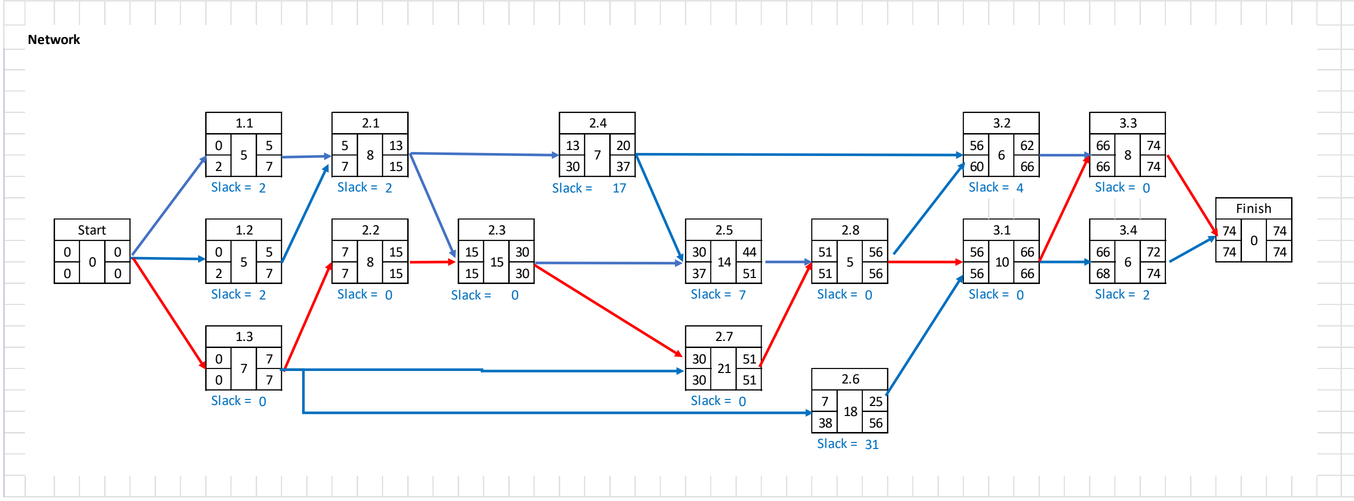
D = 35.65 Round up to 36

2. Activity Duration (20 pts)
You work with the ATEC Test Officer and you realized the activity durations are underestimated. The ATEC Test Officer provides valuable input, and you refine the optimistic, most likely, and pessimistic durations for the XM30 Test Plan in the table below. You then need to use these estimates to refine the operational test project schedule.

a. Complete the table below by calculating the Expected Time (TE) and Variance for each task.

WBS	Task Name	Predecessors	Optim. (a)	Most Likely (m)	Pessimistic (b)	TE	Var
0	**XM30 Test Plan**						
1	**Project Planning**						
1.1	Test Plan Design		3	5	7	5	0.44
1.2	Site Visit/Range Recon		1	4	8	5	1.36
1.3	Risk Assessment Plan		5	6	8	7	0.25
2	**Execution**						
2.1	Initial Inspection	1.1, 1.2	8	8	8	8	0.00
2.2	New Equipment Training	1.3	6	7	11	8	0.69
2.3	Automotive Performance Testing	2.1, 2.2	14	14	17	15	0.25
2.4	RAM Testing	2.1	5	6	9	7	0.44
2.5	Weapons Firing	2.3, 2.4	14	13	15	14	0.03
2.6	Armor Impact Testing	1.3	14	17	22	18	1.78
2.7	Test Incident Reporting	1.3, 2.3	18	21	24	21	1.00
2.8	Final Inspection	2.5, 2.7	3	5	7	5	0.44
3	**Post Execution**						
3.1	Post Test Analysis	2.6, 2.8	7	9	15	10	1.78
3.2	Ammunition Disposal	2.4, 2.8	3	5	10	6	1.36
3.3	Post Test Operations	3.1, 3.2	5	8	10	8	0.69
3.4	Final Report	3.1	4	6	8	6	0.44

Given your calculated activity durations, draw an AON diagram using the DSE Standard and determine the Critical Path. Consider rotating this page 90º so your diagram can fit!



b. List the critical path. _1.3-2.2-2.3-2.7-2.8-3.1-3.3__________________________________

c. Given your calculated activity durations, identify the variable values needed to solve for the probability of completing the Operational Test Project in 70 days. Also, determine the probability of success for completing the project in 70 days and 72 days.

μ = ___74 days___ σ^2 = ___5.11_

P (70 Days) = _3.8%_ P (72 Days) = 18.8%

d. You are concerned that cutting the project time is too high risk. What is the desired project time to ensure a 95% successful completion of the operational test on time?

D = 77.7186, Round up to 78 Days

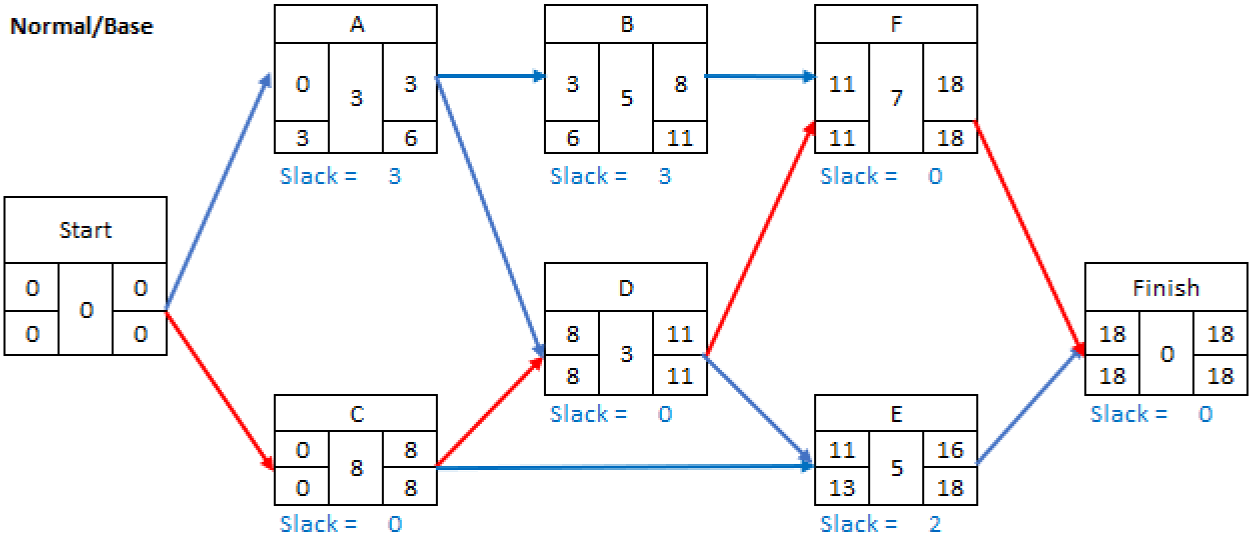
3. Crashing (25 pts)

The XM30 operational vehicle testing is underway, and the Armor Impact Test kits are not functioning properly. As a result, you order additional kits to complete this test. The shipping delay results in the Armor Impact Testing becoming a critical task. You deep dive activity durations to determine how you can prevent an impact on the project duration and at what cost. Given the following partial project plan, find the lowest cost to complete the project earlier than the expected completion time. Durations in the table, below, are in days.

Activity	Predecessor	Expected Duration	Crash Duration	Normal Cost	Crash Cost	Activity Slope
A – Calibration	--	3	2	9000	18000	-9000.00
B – Test Driving	A	5	5	7000	7000	N/A
C – Installation/Simulation Tests	--	8	6	4000	5600	-800.00
D – Driving Impact	A, C	3	1	6000	9500	-1750.00

E – Armor Piercing	C, D	5	4	6000	7000	-1000.00
F – Small Arms	B, D	7	6	7000	9500	-2500.00

a. Complete the AON diagram, using DSE Standard, using Expected Durations for each activity.



b. Identify the critical path and base expected project completion time.

C-D-F, 18 days.

c. Determine the activity slopes and fill in these values in the table above.

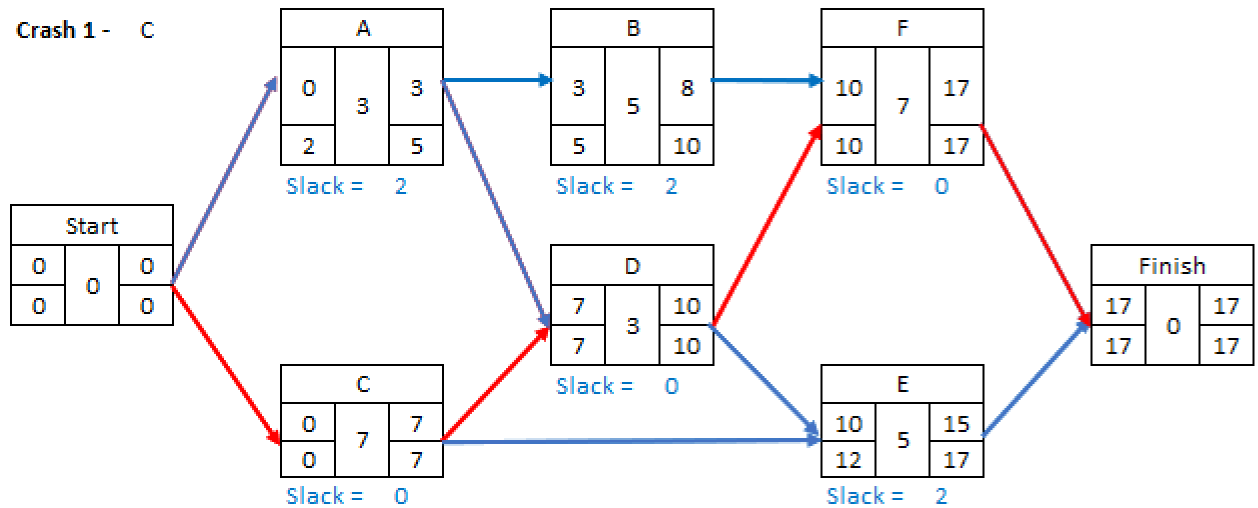
d. You realize the Armor Impact Test must be completed in **14 days** to not extend the project duration. Crash the most cost-efficient critical activities to determine the costs associated with completing the test in 14 days. Populate the table below with values for each crash and draw the AON diagrams below and on the next page for each crashing event. *Note: you many not require all rows in the table or diagrams, depending on your crashing plan!*

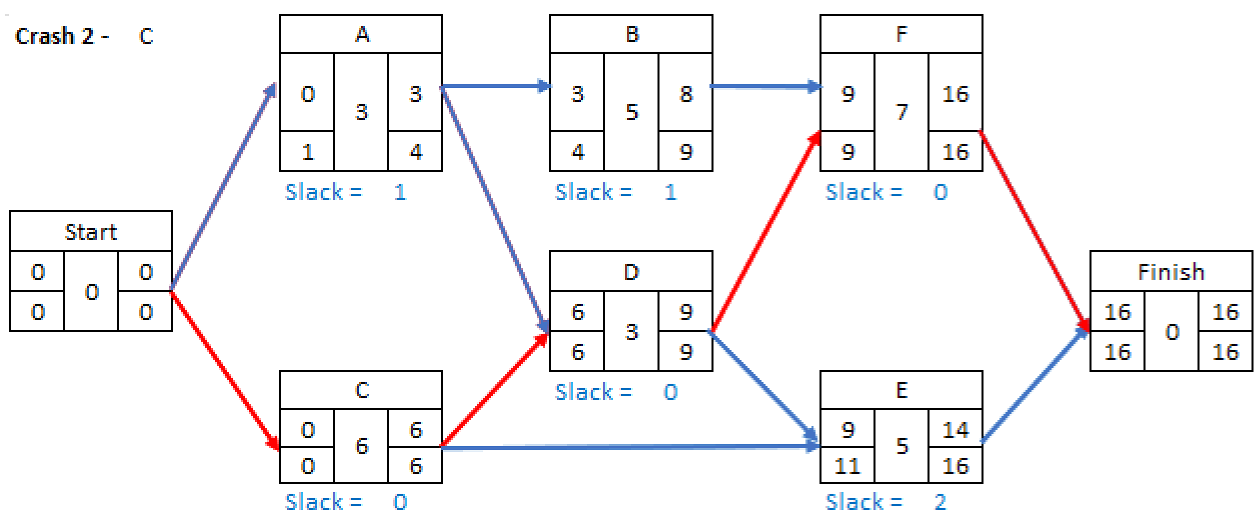
Project Plan	Activity Crashed	Additional Cost per Time Period Saved	Critical Path	Project Completion Time	Total Cost
Base			C-D-F	18 days	$39,000.00
1	C	$800	C-D-F	17 days	$39,800.00
2	C	$800	C-D-F	16 days	$40,600.00
3	D	$1,750	C-D-F	15 days	$42,350.00

			A-B-F, A-D-F		
4	F	$2,500	C-D-F A-B-F, A-D-F	14 days	$44,850.00

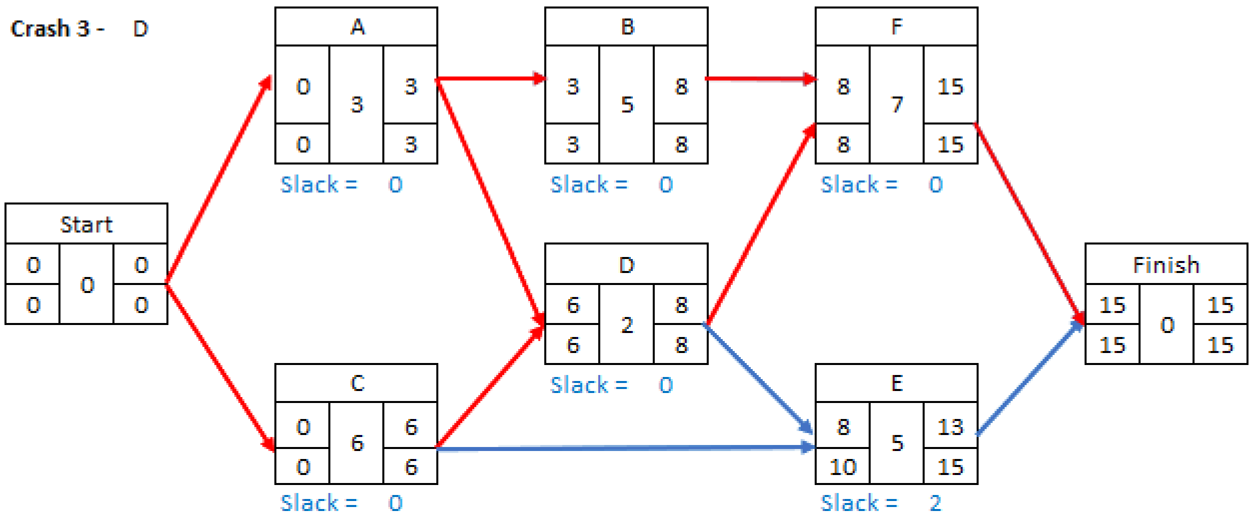
e. What is the extra cost for completing the project in 14 days? You have a 10% contingency reserve for this project (use the base project cost to determine your contingency reserve). Is it feasible to crash the project duration to 14 days? Explain.

Extra cost to complete in 14 days is $5,850. NO, since the contingency reserve is 10% of $39,000 = $3,900. The cost to expedite the project to 14 days is $1,950 over the contingency reserve and not feasible to expedite.

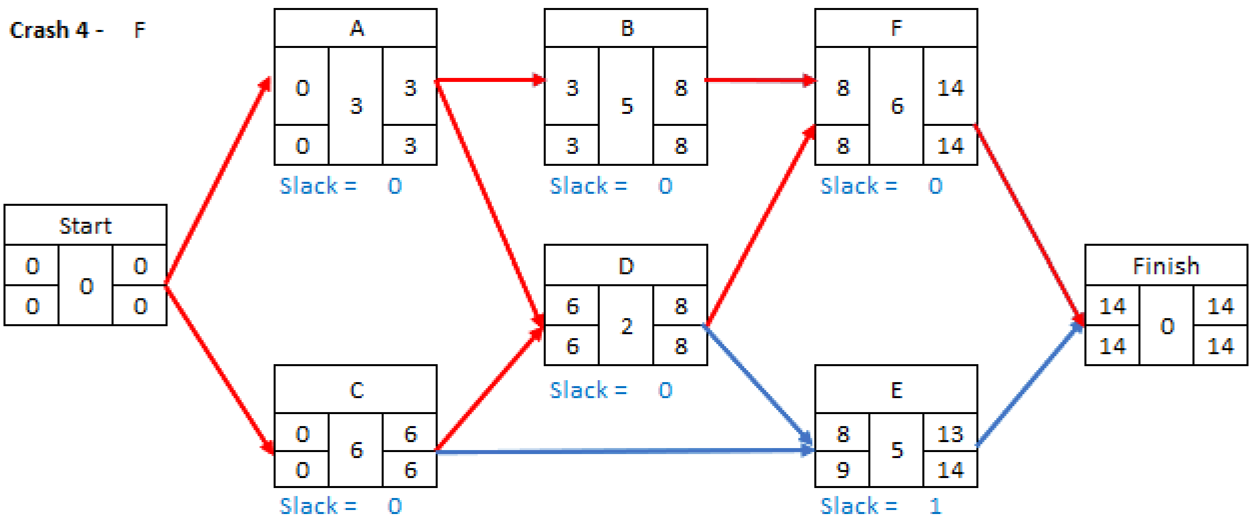




Crash 3 - D



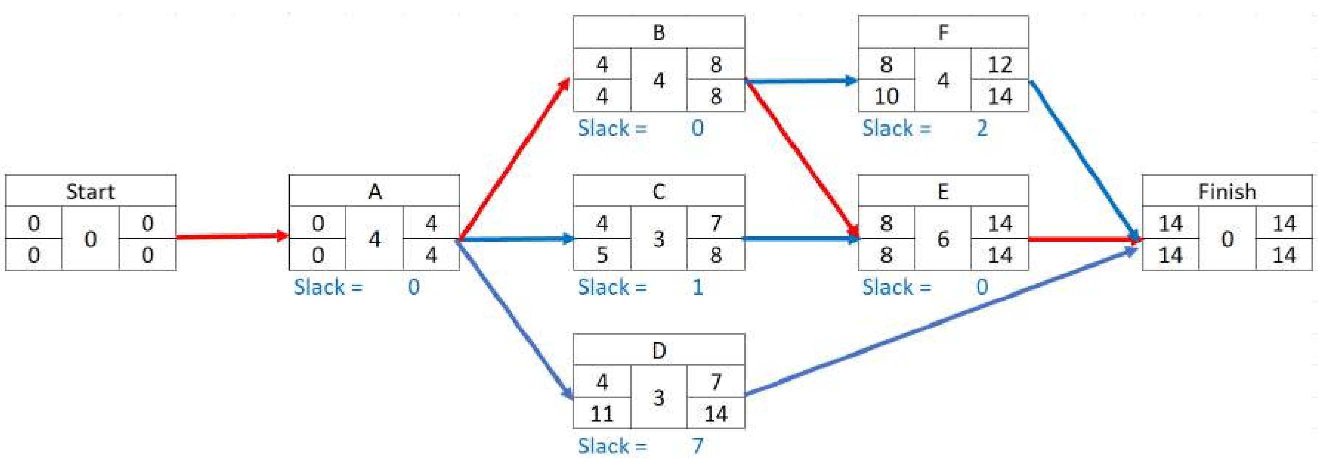
Crash 4 - F



4. **Resource Loading and Leveling (20 pts)**
Prior to the Weapons Firing Test of the XM30 operational test, you realize you only have **four XM30 vehicles available** for the testing. Given the following project plan, determine the Expected Time (TEs), and analyze the network for critical path and slack. Then conduct a resource loading and leveling analysis to help you understand how utilizing the weapon resources affect your project. The durations in the table below are in days.

Activity	*Predecessor*	*Optimistic*	*Most Likely*	*Pessimistic*	*Expected Duration*	*Vehicles Required*
A – Table II (Dry) Qual	--	3	3	4	4	3
B – Table III Qual	A	1	3	7	4	1
C – Table IV Qual	A	2	2	3	3	2
D – Table V Qual	A	1	3	4	3	1
E – Table VI Qual	B, C	4	5	8	6	4
F – Design Limit Range	B	2	4	6	4	2

a. Calculate the Expected Duration (TEs) and complete the AON diagram using the DSE Standard.



b. What is the critical path and expected duration without Leveling?

A-B-E, 14 days

c. Fill in the table below to determine Weapon resource Loading, then Level the resources as necessary across the activities that require them. There are **4 XM30 vehicles available**.

Activity	XM30	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
		Time Period (days)																	
A	3	3	3	3	3														
		3	3	3	3														
B	2					2	2	2	2										
						2	2	2	2										
C	2					2	2	2	]										
						2	2	2											
D	1					1	1	1							]				
									1	1	1								
E	3									3	3	3	3	3	3				
										3	3	3	3	3	3				
F	2									2	2	2	2		]				
																2	2	2	2
Required:		3	3	3	3	5	5	5	2	5	5	5	5	3	3	0	0	0	0
Leveled:		3	3	3	3	4	4	4	3	4	4	3	3	3	3	2	2	2	2

d. Does Leveling the XM30 usage change the critical path? If so, how?

Yes. The critical path is now A-B-E and A-B-F.

e. Does Leveling change your project duration? If so, how?

Yes. It increases the project duration to 18 days.

TAB K Quizzes

Quiz #1

Instructor Solution

EM 411 Project Management
AY26-1 Quiz 1

Quiz 1 – SOLUTION

1. This is a 30 minute quiz worth 50 points. The quiz consists of true/false, multiple choice and short answer. Ensure that your exam is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: This is a closed book quiz. You may use one handwritten 3x5 card (1 side), the formula reference sheet below, and your authorized calculator.

3. Feel free to disassemble the quiz; a stapler is available to reattach the pages when you are finished. Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name and section. When reassembling your quiz, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. GRADING SUMMARY:

Problem	***Possible***	***Earned***
Definition of Project Management	12	
Project Initiation and Selection	20	
The PM, the Profession, and Stakeholders	10	
The Project Organization and Teams	8	
Total	**50**	

Payback Period:	$PaybackPeriod = \frac{InitialFixedInvestment}{EstimatedNetCashInflows}$
Discounted Cash Flow:	$NPV_{project} = A_0 + \sum_{t=1}^{n} \frac{F_t}{(1+k+p_t)^t}$
Weighted Factor Scoring Model:	$S_i = \sum_{j=1}^{n} s_{ij} w_j$ $\sum_{j=1}^{n} w_j = 1$

EM 411 Project Management
AY26-1 Quiz 1

Definition of Project Management

1. (3 Points) Identify and circle the three universal characteristics of a project:

a. Conflict
b. Limited Resources
c. Finite Duration
d. Unique
e. Regular Interactions, internal and external
f. One-time occurrence

2. (3 Points) List the three project goals/objectives (also known as the Triple Constraint or Iron Triangle of any Project) that a PM might have to manipulate during the life cycle of a project.

Time Cost Scope

3. (3 Points) The Project Life Cycle S-Curve could best explain which of the following projects:

a. a battalion dining in
b. the CEAC
c. a new orthopedics center for Keller Army Community Hospital
d. planning and execution of a wedding
e. a presidential political convention
f. the Army's Next Generation Combat Vehicle

4. (3 Points) The PM closely manages the triple constraint because (select all that apply)

a. Production Line technologies must constantly be assessed and modernized
b. Tradeoffs to achieve the proper Quality of the project can vary between stakeholders
c. The three characteristics are interdependent
d. The performance of the task must be routine
e. Resources are limited
f. Training the workforce for efficiency is paramount to success

EM 411 Project Management
AY26-1 Quiz 1

Project Initiation and Selection

5. (10 Points) Apple is trying to decide between two projects: research and development for iPhone 17X or for Apple Watch 2026. They have hired you as a consultant to help make a recommendation on which project to select. A colleague completed the analysis on Project 1 (shown in the table below). You've been asked to complete the discounted cash flow, profitability index and payback period analysis for Project 2 and make a recommendation with a brief justification on which they should pursue.

Company Required Rate of Return: 8.7%

Expected Inflation: 2.6%

Project 1 (iPhone 17X)

Initial investment: $360,000

Cash inflows year 1-3: $150,000 per year

Payback Period: 2.40 years

NPV is: $4,653.05

Project 2 (Watch 2026)

Initial investment: $315,000

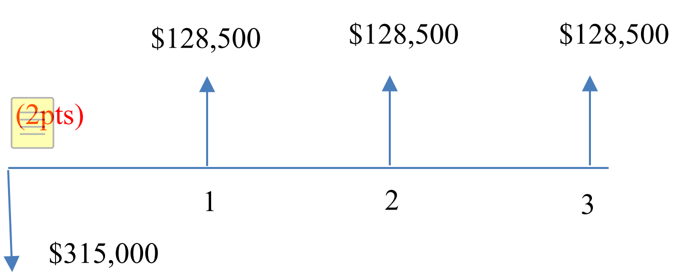
Cash inflows year 1-3: $128,500 per year

<u>You must show ALL WORK for credit! (round to nearest hundredth place)</u>

Payback Period: 2.45 years = $315,000 / $128,500 (2 points)

Draw your Cash Flow Diagram:

(2pts)



(Must show calculations for NPV for full 4pts)

Enter your answer in this table:	
NPV	-$2,613.89
Payback Period	2.45

Which project do you recommend? (2pts) <u>Choose the IPhone 17x as its slightly better in payback, and positive in its NPV (1 point if calcs are wrong, but right conclusion given incorrect answers)</u>______________

6. (2 Points) Numeric and Non-numeric project selection models are considered mutually exclusive processes by which projects might be evaluated and selected for an organization's portfolio.

(Circle answer) True / False

7. (3 Points) For the following scenarios, identify which type of project selection model is being utilized and/or described.

a. Recently, the five year old company PriDev has decided to add a real time package tracking feature similar to a new one Amazon launched in early 2024.

Selection Model: Operating Necessity (will accept Competitive Necessity as well)

b. Apple is deciding between launching a new in-house daycare service for its employees, or providing stock option bonuses.

Selection Model: Comparative Benefit

c. SpaceX CEO has decided on his own that it will pursue a project to put a man on Mars by 2035 with its Starship Heavy. Which selection model did the company likely use?

Selection Model: Sacred cow

8. (5 Points) Solve the Weighted Scoring Model in the yellow highlighted fields, and identify which project you would select.

Value Criteria	Objective	Value	Relative Wt.	Raw Score		Weighted Score	
				Project 1	Project 2	Project 1	Project 2
Innovation	Maximize	10	**.40**	8		**3.2**	
Interoperability	Maximize	4	**.16**	6		**.96**	
Maintainability	Maximize	8	**.32**	8		**2.56**	
Producibility	Maximize	3	**.12**	3		**.36**	
Total		25				**7.08**	**6.72**

Which project would you select (circle): **Project #1** **or** **Project #2** **(.5pts per table response and .5 point for project selection answer)**

The Project Manager, the Profession, and Stakeholders

9. (3 Points) The Project Manager's responsibilities fall primarily into three separate areas:

a. Parent Organization
b. Media
c. Suppliers
d. Creditors
e. Client and Stakeholders
f. The Project Team

10. (1 Point) The project manager should be more skilled at ___________, whereas the functional manager should be more skilled at ______________________. ***(circle the correct answer)***

a. analysis; synthesis.
b. communication; managing.
c. synthesis; analysis.
d. managing; analysis.

11. (1 Point) The PM attribute which encompasses the ability to handle and navigate political, interpersonal, and technical complexity and challenges is: ***(circle the correct answer)***

a. Credibility
b. Sensitivity
c. Leadership, ethics and management style
d. Ability to Handle Stress

12. (1 Point) The stakeholder tool which helps determine you strategy to influence stakeholders degree of support for the project is ***(circle the correct answer)***

a. The Power-Interest Grid

b. The Linear Responsibility Chart (RACI chart)

c. The Commitment Assessment Matrix

d. The Conflict Resolution Matrix

13. (2 Points) Which one of the below is NOT a main point of principled negotiation

a. Point 1: Separate the people from the problem (leave emotion out of it)

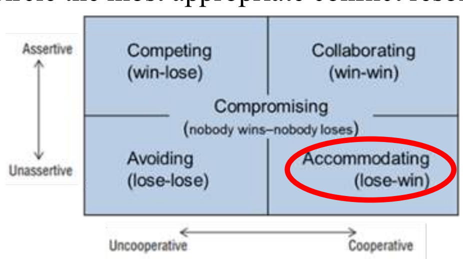
b. Point 2: Focus on interests not positions (don't do positional bargaining)

c. Point 3: Accommodate where possible to support other stakeholders

d. Point 4: Insist on objective criteria *(reduces "battle of wills" tendency)*

14. (2 Points) A stakeholder insists on including a minor reporting feature in the software that isn't in the original scope. The project manager agrees to include it, knowing the small cost to the program is worth building goodwill with the stakeholder.

In the diagram below circle the most appropriate conflict resolution strategy for this scenario.



EM 411 Project Management
AY26-1 Quiz 1

The Project Organization and Teams

15. (3 points) Identify which type of project organization is being described below

a. The project organization has the disadvantage that no individual is given full responsibility for the project, however the advantage is specialists can be grouped to share knowledge and experience easily.

Circle the right answer: Functional / Matrix / Projectized

b. The project organization has the disadvantage of team members worrying 'life after the project', however the advantage is in short lines of communication to deliver the project.

Circle the right answer: Functional / Matrix / Projectized

c. The project organization that does not take a holistic approach to the project, yet the advantage is maximum flexibility in the use of its staff.

Circle the right answer: Functional / Matrix / Projectized

16. (2 Points) Because PMs typically have no direct authority over project team members, and in particular in a matrix organization, they should motivate good performance how? ***(circle the best answer)***

a. By using cash awards of various types
b. By posting progress charts with goals and targets for each week.
c. By promoting recognition, achievement, challenging assignments, responsibility, a chance to learn new skills, and the work itself
d. By threatening to expose team members' faults, errors, and poor performance

17. (1 Point) A Weak Matrix project organization is closer in nature to a Functional Organization. The PM owns a lesser degree of authority and autonomy over team members.

Circle Answer: True / False

18. (1 Point) The Project Management Office is an organizational structure that standardizes the project-related governance processes and facilitates sharing of resources, methods, tools and techniques.

Circle Answer: True / False

19. (1 Point) All businesses which maintain a project portfolio employ a Project Management Office to oversee project delivery.

Circle Answer: True / False

Quiz #2

Instructor Solution

EM411 Quiz 2 **NAME: ___________________________**

SECTION: ________________________

1. This is a 75 minute quiz worth 50 points. The quiz consists of true/false, multiple choice and short answer. Ensure that your quiz is complete and place your name on each page in the space provided.

2. AUTHORIZED REFERENCES: Earned Value Management reference sheet and hand written 3x5 card.

3. Feel free to disassemble the quiz; a stapler is available to reattach the pages when you are finished. Use only the additional paper that is available at the front of the room. Attach the additional pages you use to your exam and turn them in. Please label additional pages with the problem number, your name and section. When reassembling your quiz, ensure you have all pages (including this cover page).

4. SHOW ALL YOUR WORK! Unsupported or missing work earns little or no partial credit.

5. GRADING SUMMARY:

Problem	***Possible***	***Earned***
Monitoring – Earned Value Management	40	
Project Monitoring, Controlling, Auditing, and Termination	10	
Total	**50**	

This Quiz will be released from academic security on Wednesday, 22 April 2026. Prior to that time, I will not discuss any aspects of the Quiz with anyone except an EM411 instructor. By signing my name below, I acknowledge this requirement.

Signature:____

EM411 Project Management Name___________________________
AT26-2 – Quiz 2 (SOLUTION)

[40 Total Points] You are the Project Manager for the Apex Raider Future Vertical Lift (FVL) prototype. The stakeholder deliverables are two prototypes for Army Aviation's next-generation armed reconnaissance aircraft. You owe your program director an update on the project status with relation to the direct goals of time and cost. **The Planned Completion % in the table is 'as of' 1 September 2027 (Month 18 of the project). The Critical Path is A – B – D – E with a total project duration of 38 months.**

ID	Activity	Pred.	Duration (months)	Budget (Millions)	Actual Cost (Millions)	Actual % Completion	Planned % Completion	Planned Value (PV)	Earned Value (EV)	ST
A	Advanced Engine Dev	--	12	$180	$210	100%	100%	$180	$180	12
B	Airframe Composite Build	A	12	$240	$110	20%	50%	$120	$48	2.4
C	Avionics Architecture	A	10	$150	$130	70%	60%	$90	$105	7
D	Structural Stress Testing	B	6	$60	$0	0%	0%	$0	$0	0
E	Final Assembly & Flight Test	D, C	8	$120	$0	0%	0%	$0	$0	0
	Total			$750	$450			$390	$333	14.4

1. (12 Points – 2 for each) Determine the following EVM Measures:

   BAC – $180 + $240 + $150 + $60 + $120 = $750M

   AC – $210 + $110 + $130 = $450M

   PV – $180 + $120 + $90 = $390M

   EV – $180 + $48 + $105 = $333M

   AT – 18 months

   ST – 12 + 2.4 = 14.4

2. (6 Points) Calculate and analyze the following EVM Metrics:

   CV – EV – AC = $333 - $450 = ($117M)

   SV – EV – PV = $333 - $390 = ($57M)

   TV – ST – AT = 14.4 – 18 = (3.6 months)

3. (6 Points) Calculate all EVM indices:

   SPI – EV/PV = $333/$390 = .85

   CPI – EV/AC = $333/$450 = .74

   CSI – SPI*CPI = .85*.74 = .63

4. (8 points) Based on the results of your earned value analysis, what can you say about the Apex Raider Project in terms of budget and schedule?

The project is behind schedule and over budget

5. (8 points) Using project completion metrics of ETC and EAC, how would you inform your Program Director about budget? Your contingency reserve was calculated at 25% of the direct costs (planned budget) of your project.

ETC = (BAC-EV)/CPI = ($750-$333)/.74 = $563.51M

EAC = (ETC+AC) = $563.51 + $450 = $1013.51M

Contingency = $750*.25 = $187.50M

Based on the EAC, the final project will be $750 - $1013.51 = ($263.51M) over budget without intervention.

The contingency of $187.50M is not sufficient to cover the over budget amount of $263.51M

(Multiple Choice and T/F are worth 1 point each)

6. Your boss requests a monitoring report on a regular basis as the project reaches each milestone. This type of report is referred to as:

   a. Ad hoc Report
   b. Milestone Report
   c. Narrative Report
   d. Routine Report

7. What is considered the most critical document to distribute before a meeting to ensure it remains productive and focused?

   a. A list of attendees
   b. An agenda
   c. A technical project charter
   d. The previous meetings minutes

8. Earned Value Management monitors the performance of a project based on aggregate performance measure called Earned Value, and addresses all performance in time, cost and scope.

   a. True
   b. False

9. If a project EVM analysis reports an SPI of 1.0 and a CPI of 1.0, but the deliverables fail to meet required technical safety standards, what is the most accurate assessment of the project's status?

   a. The project is officially successful because the "Triple Constraint" (Time, Coat, Scope) has been met
   b. The project is healthy on paper, but the EVM metrics are failing to reflect a critical quality deficiency
   c. The CPI will automatically drop to 0.5 to account for the failed safety standards
   d. The project is ahead of schedule because the work was completed on time despite the errors

10. In a project life cycle, where do Go/No Go control points most commonly occur?

   a. Daily during the morning stand-up meetings
   b. Only during the initial project chartering
   c. Exclusively after a project failure has occurred
   d. At the conclusion of a major project phase or milestone

11. An effective change control process includes all the following except:

   a. Implementation of requested changes as soon as they are received to maintain momentum
   b. The PM must be consulted during the process
   c. Change must be approved in writing
   d. Changes to the project plan must include a description of how it will be managed
   e. Project plan and schedules must be updated to reflect the change

12. An audit performed at the end of a project is primarily beneficial for capturing lessons learned, though its ability to positively influence the current project's budget is significantly diminished.

   a. True
   b. False

EM411 Project Management Name_________________________
AT26-2 – Quiz 2 (SOLUTION)

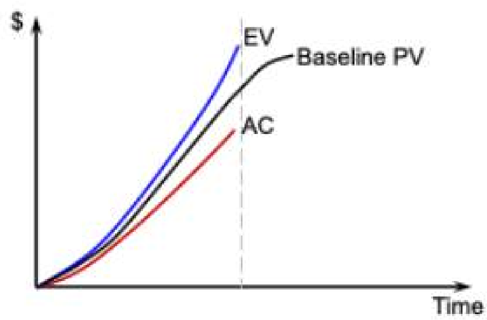
13. Which of the following is a major difference between a 'Status Report' and a 'Project Audit'?

    a. Status reports are optional, while audits are legally required
    b. Audits are performed daily, while status reports are performed annually
    c. Status reports only involve the client, while audits only involve the project team
    d. Status reports focus on project progress, while audits focus on proper processes.

14. A project to develop a new software application is completed, the final product is delivered to the customer, and the project team is disbanded to return to their functional departments. This is an example of project termination by:

    a. Starvation
    b. Integration
    c. Addition
    d. Extinction

15. The figure below shows the PV, EV, and AC curves for a project at the current status date (dashed vertical line).



Based on the chart, which of the following best describes the project's status?

    a. Over budget and behind schedule
    b. Over budget and ahead of schedule
    c. Under budget and behind schedule
    d. Under budget and ahead of schedule

TAB L Course Project

Course Project - Stage Gate #1

Instructions/Rubric/Student Example

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT26-2 12 January 2026

MEMORANDUM FOR RECORD

SUBJECT: Project Stage Gate #1 Instruction Sheet

1. **Purpose.** The purpose of Stage Gate #1 is to conduct a Project Selection Analysis and provide a recommendation on which Course of Action should be pursued. Three courses of actions will be evaluated. You will also provide an initial risk management and stakeholder management plan.

2. **References.** No change from the EM411 Course Instructional Memorandum.

3. **Summary**. ***This deliverable is worth a total of 75 points.*** Each subsection will detail the breakout of the point total. Each Cadet will submit their project deliverables **NLT 2359 (EDT) on 2 February 2026** via Canvas Assignments. Detailed submission requirements are outlined in paragraph 5.

4. **Background.** The United States Military Academy (USMA) has hired Old Grad Research and Engineering, Inc. (OGRE) to provide oversight for the Thayer Hall Renovation Project. ORGE has selected your Project Team to lead this Project. As part of the contract for the Thayer Hall Renovation, the Academy is requesting a detailed project selection analysis of their three possible COAs.

5. **Stage Gate Instructions.** Complete a Project Selection Analysis, utilizing a weighted scoring model, the information provided in these instructions, and Annex A COA Narratives. The three COAs to be analyzed are: COA1: Expansion and Renovation, COA2: Major Renovation, and COA3: Minor Renovation. The details of the analysis will be submitted in a memorandum along with your final COA recommendation (***i.e. "…I recommend COA X, because…"***).

a. ***Weighted Scoring Analysis (40 pts).*** Each Cadet will complete a weighted scoring analysis given the four (4) following value criteria, corresponding scoring guidance, and information provided in Annex A.

(1) Instruction & Research, (Maximize). This is the project's ability to deliver classrooms, labs, collaboration rooms, and auditoriums equipped with the latest technology to promote cadet learning and research. This criterion has the highest average stakeholder value of 10. On a scale of 1-10, a complete renovation along with expansion (i.e. a new floor or building wing) scores a 9, while a purely cosmetic touch-up (i.e. no additional renovations or new equipment) scores a 1.

(2) Historical Authenticity, (Maximize). The Academy would like to maintain as much historical authenticity as possible. Stakeholders valued this criterion at a 3.5. A project that completely maintains the historical authenticity of the building (both internal and external) would score a 10. A building that is completely modern and does not blend with the existing traditional academy architecture would score a 1.

(3) Disruption to Academy Operations, (Minimize). This is the impact to routine Academy operations. On average, stakeholders valued this criterion's importance as a 7. Less disruption is better. One a scale of 1-10, no construction (i.e. no disruption) would be a 10 and a major construction project with major, long-duration impact on Academy and traffic operations would be a 1.

(4) Discounted Cash Flow, (Maximize). The Academy wants to maximize returns on the project over a 40-year time period. On average, stakeholders valued this criterion as a 5

a) A Discounted Cash Flow above $55,000,000 scores a 10 with a Discounted Cash Flow below $15,000,000 scoring a 1.

b) The U.S. Treasury is directing a 5.1% rate of return. Using U.S. Bureau of Labor and Statistics' Consumer Price Index data, OGRE will use a 3.9% Inflation Rate for its calculations.

b. ***Memorandum (35 pts)***. Each Cadet will provide a memorandum to the Superintendent, United States Military Academy, regarding the project and its analysis. The memorandum will be written in the Army Memorandum Format (AR 25-50) and should include the following:

(1) Identify what the project is, the purpose of the project, the measurable objectives, and your role in the project.

(2) *S*ummarize your project selection analysis, including your recommendation for the COA with which to proceed. Ensure you justify your value criteria scores (i.e. how you calculated the value raw scores for each COA).

(3) List the key stakeholders and provide your initial stakeholder analysis. This section should also discuss the initial plan for managing those stakeholders. Use a Power-Interest Grid and/or Commitment Assessment Matrix to plot and support your answer.

(4) Discuss the major risks that may impact the overall health of the project and your project team. Include initial risk response plans for the identified risks. Utilize a FMEA and highlight the initial and residual Risk Priority Numbers (RPNs).

(5) Include enclosures that contain your weighted scoring model, scoring justifications, and DCF/NPV calculations.

(6) Annex B is provided as a template for developing your memorandum. (Note: There are comments within the template that provide additional clarification. Delete these comments prior to finalizing and submitting your memorandum.)

6. **Submission requirements.** Project Stage Gate 1 is **due NLT 2359 (EDT) on 2 February 2026**. The memorandum, with cover sheet and appendices, will be submitted as a single Acrobat PDF via Canvas Assignments. Supporting files (ex: Excel Spreadsheet, etc.) will be submitted along with the PDF, but as separate files in their native formats.

a. File Labeling Format Example:

(1) "EM411_H11_'YOUR LAST NAME'_SG1_Project Recommendation Memo.pdf"

(2) "EM411_H11_'YOUR LAST NAME'_SG1_XXXXX" – For supporting files.

7. **List of Supporting Annexes.**

a. Annex A COA Narratives (pdf)

b. Annex B Memorandum Template (word doc)

c. AR 25-50_Prep & Manage Correspondence (pdf)

8. The point of contact for this memorandum is the undersigned at daniel.finch@westpoint.edu.

///ORIGINAL SIGNED///
DANIEL FINCH
Senior Instructor
EM411 Course Director

COA 1: Expansion & Renovation

This is the most ambitious Course of Action and includes major exterior and interior renovations, along with adding another classroom floor and an additional parking lot level to the top of Thayer Hall. From the financial analysis, the initial cost of the project is estimated to be $460,000,000. Given the expansive new construction and additional weight on the load-bearing columns, the building will need major scheduled maintenance at years 5, 10, 15, 25, 30, and 35. Each scheduled maintenance will cost $5,200,000. In year 20, a minor update is planned for IT systems and will surge maintenance costs to $6,100,000. With the new floorspace and major IT improvements, the academy expects to bring in an additional $45,000,000 in research funding and alumni donations annually.

This project will greatly enhance the Academy's instruction and research capabilities. This COA, with its additional floorspace and major interior upgrades, will give USMA valuable new collaboration, classroom, lab, and conference spaces. New electrical, HVAC, and IT infrastructure will meet or exceed the Academy's needs for the next 40 years.

The drawbacks of this project are in the "historical authenticity" and "disruption to operations" criteria. It maintains the least historical authenticity since the new design will reshape the West Point Skyline and the interior will lose its traditional appearance. It's expected that this project will take 5-6 years to complete, meaning that there will be significant, multi-year impacts to vehicle and pedestrian traffic, as well as Academy operations. An ambitious renovation and expansion on a historic building also brings a moderate level of risk of major delays.

COA 2: Major Renovation

The second COA under consideration is to conduct a major renovation of Thayer Hall. The project will gut the interior of the building all the way down the structural steel and concrete and renovate it from the ground up. The interior will be updated and modernized, and will include all new heating, ventilation, air conditioning, plumbing, electrical, and IT infrastructure, as well as all new flooring, furniture, and fixtures. From the financial analysis, the initial cost of the project is estimated to be $365,000,000. Major scheduled maintenance will be needed at years 5, 10, 15, 25, 30, and 35, each costing $4,500,000. In year 20, a minor update is planned for IT systems and will surge maintenance costs to $5,300,000. This COA will markedly improve the Academy's prestige and bring in an estimated positive cash flow of $41,500,000 in additional research funding and alumni donations annually.

The Academy's instruction and research capabilities will still be significantly improved, thanks to the new electrical, HVAC, and IT infrastructure. The gutting and redesign of the interior will produce a better blend of classrooms, collaboration rooms, labs, offices, and conference rooms.

This project maintains a moderate amount of historical authenticity. While the interior will experience major renovations and changes, the exterior will only be cleaned and repaired. Since the majority of the project will be interior renovations, exterior disruptions to vehicle and pedestrian traffic should only last 1.5 years. Overall, the project should be completed in 3-3.5 years.

COA 3: Minor Renovation

The third COA under consideration is to conduct a minor renovation of Thayer Hall. In this COA, the project will do a targeted renovation of the interior, focusing on the installation of new heating, air conditioning, and IT infrastructure, as well as new furniture and fixtures. Like, COA2, the exterior will be cleaned and repaired. From the financial analysis, the initial cost of the project is estimated to be $175,000,000. Minor scheduled maintenance will be needed at years 5, 10, 15, 25, 30, and 35, each costing $3,000,000. In year 20, a minor update is planned for IT systems and will surge maintenance costs to $3,500,000. This COA will improve the Academy's prestige modestly; however, the Academy believes it will still bring in an additional $19,000,000 in research funding and alumni donations annually.

The Academy's instruction and research capabilities will be moderately improved, thanks to the new IT infrastructure, furniture, and fixtures. However, the limited interior renovations under this COA does not include any floorplan updates. Thayer Hall will keep its current classroom, office, and conference room layout.

This project maintains a moderately high level of historical authenticity. The exterior will only be cleaned and repaired, and the interior will retain the traditional look that it has had over the last 50 years. The entire renovation will only take 2-2.5 years, making this the least disruptive COA.

Course Project Stage Gate #1 Grading Rubric

Instructor Score

0 pts

SG#1 Grading Rubric

Criteria Ratings Points

1. NPV Calculations (15 points)

Superior

Complete answer with all work shown, all 3 COA NPVs calculated correctly. Excel file submitted. Free of errors.

15 to >13 pts

Good

At least 2 of 3 NPVs calculated correctly with all work shown. Supporting file submitted.

13 to >10 pts

Okay

Only 1 of 3 NPVs calculated correctly with all work shown. Supporting file submitted.

10 to >8 pts

Poor

All calculations are incorrect, or work is not shown. Supporting file submitted, but with major errors.

8 to >0 pts

2. Purpose and Background (10 points)

Superior

Complete answer with all work shown. All key questions posed in Annex B template answered.

10 to >9 pts

Good

One or more questions left unanswered, or purpose and background not clear and concise.

9 to >7 pts

Okay

Multiple questions from Annex B left unanswered. Purpose and/or background unclear.

7 to >5 pts

Poor

Poorly written and/or most questions from Annex B left unanswered.

5 to >0 pts

3. Selection Analysis (15 points)

Superior

Complete answer with all work shown and interpretation justified. Weighted Scoring model correct, value scoring criteria table used to justify scores.

15 to >13 pts

Good

Minor errors in the weighted scoring model. A value scoring criteria table was used to justify scores. Values interpreted mostly correctly.

13 to >10 pts

Okay

Some larger errors with formulation of weighted scoring model or interpretation. Weight for criteria may be incorrect. Value scoring criteria table used, and interpreted correctly.

10 to >8 pts

Poor

Major errors in formulating weighted scoring model and/or interpretation. Weights or scores were calculated incorrectly, and the incorrect COA was recommended.

8 to >0 pts

4. Stakeholder Analysis (15 points)

Superior

Complete answer supported by a PIG and/or CAM. All questions posed in Annex B answered clearly and concisely. Key stakeholders identified leave out those with low power or low interest.

15 to >13 pts

Good

Most key stakeholders were identified, with some possible low-interest/power stakeholders included. Work supported by a PIG and/or CAM.

13 to >10 pts

Okay

Stakeholders misidentified, or significant key stakeholders missing from analysis. Work poorly supported.

10 to >8 pts

Poor

Major errors to Stakeholder analysis. Work may be unsupported by a PIG and/or CAM. Poorly written and justified in MFR.

8 to >0 pts

5. Risk Management (15 points)

Superior

Complete answer that includes an initial and residual RPNs using a FMEA. Several major risks identified, with appropriate responses planned.

15 to >13 pts

Good

More than a few risks were identified, with proper FMEA conducted. Includes initial/residual RPNs.

13 to >10 pts

Okay

Less than a few risks identified and supported by a FMEA. May be missing adequate risk response mitigations.

10 to >8 pts

Poor

Risk analysis was not supported by a FMEA, and only a couple of risks were identified. Answers may be missing risk response mitigations or residual RPNs.

8 to >0 pts

6. Memo Format (5 points)

Superior

Neat and free of errors, Professional, Ready to present to the client.

5 to >4 pts

Okay

Minor errors to formatting of AR 25-50 format.

4 to >3 pts

Poor

Major errors to AR 25-50 format, or incorrectly followed submission instructions for the assignment.

3 to >0 pts

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT26-2 31 January 2026

MEMORANDUM FOR Superintendent, United States Military Academy West Point, NY 10996

SUBJECT: Thayer Hall Renovation

1. **Purpose:** The United States Military Academy intends to renovate Thayer Hall, and several courses of action (COAs) for the project are being considered. COA 1 represents a complete overhaul of existing infrastructure, renovating both the interior and exterior and adding additional classroom and parking space. COA 2 represents a scaled down renovation, focusing on upgrading all aspects of the interior and cleaning and repairing the exterior. COA 3 is the most conservative course of action, cleaning and repairing the exterior, upgrading furniture, and fixing HVAC and IT systems. The Academy has listed historical authenticity, limited disruption to Academy operations, discounted cash flow, and maximization of instructional and research capabilities as the primary criteria by which to evaluate the success of the Thayer Hall Renovation Project. This memo will explain the careful evaluation of the different COA's with these criteria in mind, leading to the conclusion that COA 2, a major renovation of the interior of the existing infrastructure, will be the best COA for the Academy.

2. **Background:** The Project Team will oversee the selection, planning, execution, and completion of the Thayer Hall Renovation Project. Funding availability, and a demand for improved institutional academic resources has made the Thayer Hall Renovation Project a priority for the Academy. Through this project the United States Military Academy aims to provide state of the art academic facilities in order to enhance development of future Army leaders. Some of the measurable objectives of the project will be the installation of new HVAC and IT systems, renovated classroom spaces, and the amount of increased external funding. Measurable criteria also include the Academy's listed criteria for project success as mentioned in paragraph 1. This project aims to turn Thayer Hall into a modern and valuable center of learning, and the project team will be critical to ensure that the right project is selected, that it is executed efficiently, and that it meets the Triple Constraints of cost, budget, and schedule.

3. **Selection Analysis:** COA 2 was selected based on the weighted scoring matrix shown in Table 2. The Academy provided the project success criteria and the value for each criterion allowing the calculation in Table 2. Analysis of each COA led to a raw score for each category as shown in Table 1. COA 1 maximizes Instruction and Research benefits and received a 10 for the category due to its expansion of classroom and parking capacity. However, it has the largest impact on Historical Authenticity due to significant interior and exterior alterations, earning it a 1 in this category. COA 1 also fails to Minimize Disruption, rendering the Thayer Hall area unusable for over 5 years (2). Finally, COA 1 received a 2 for DCF, with an estimated $15 million return over 40 years. COA 2 represents a balanced approach, earning a 5 in Instruction and Research by modernizing current classroom capacity, and both a 7 in Minimizing Disruption with a project length of 3.5 years, and a 7 in Historical Authenticity, preserving the exterior look

of Thayer Hall. COA 2 has the highest DCF score, a 10, with an expected $76 million return over 40 years. COA 3 scored best in Historical Authenticity, earning a 10 due to its preservation of the interior and exterior of the existing building, it scored a 9 in Minimizing Disruption with a project length under two years. However, COA 3 scored a 3 in Instruction and Research, bringing limited modernization to current classrooms. This COA also scored moderately in DCF, with an estimated $24 million return earning a score of 4.

Value Criteria	Raw Scores		
	COA 1	COA 2	COA 3
Instruction & Research	9	5	3
Historical Authenticity	1	7	10
Disruption to Academy Operations	2	7	9
Discounted Cash Flow	2	10	4

Table 1. Value Criteria and Raw Scores

Using a weighted scoring matrix based on the objectives of each COA and the value of each criterion expressed by the Academy, COA 2 was chosen as the best option with an overall weighted score of 7.04 as opposed to 4.61 for COA 1 and 5.41 for COA 2. See Table 2 for details about weighted scores, and Appendix A for more details about DCF calculation.

Value Criteria	Value	Relative Weight	Raw Scores			Weighted Score		
			COA 1	COA 2	COA 3	COA 1	COA 2	COA 3
Instruction & Research (Max)	10	0.392156863	9	5	3	3.53	1.96	1.18
Historical Authenticity (Max)	3.5	0.137254902	1	7	10	0.14	0.96	1.37
Operational Disruption (Min)	5	0.196078431	2	7	9	0.39	1.37	1.76
Discounted Cash Flow (Max)	7	0.274509804	2	10	4	0.55	2.75	1.10
Totals	25.5					**4.61**	**7.04**	**5.41**

Table 2. Weighted Scoring Matrix

4. **Stakeholder Analysis:** Some key stakeholders in this project are cadets (both present and future), faculty, Academy leadership, the US Army, the Department of Defense, and the Association of Graduates. The most important stakeholders are the USMA leadership team, the US Army Corps of Engineers, and the US Army, fulfilling the roles of client, service provider, and sponsor respectively. Each of these stakeholders has high influence over the success of the project and a high interest in its success, and ought to be managed closely. On a larger scale, it is important to keep the Department of Defense and Congress satisfied with the project as they may have limited interest but certainly extensive influence. Current and future cadets, as well as the association of graduates are certainly very interested in the project, but do not have much direct influence and as such should be kept well informed. The power Interest grid in Table 3 breaks down the stakeholders into different action categories.

POWER	LOW INTEREST	HIGH INTEREST
HIGH	**HIGH POWER, LOW INTEREST** 1. Congress 2. Department of Defense **KEEP SATISFIED**	**HIGH POWER, HIGH INTEREST** 1. US Army Corps of Engineers (USACE) 2. US Army 3. USMA Leadership **MANAGE CLOSELY**
LOW	**LOW POWER, LOW INTEREST** 1. New York State 2. NY Dept of Environmental Conservation 3. Local Population **MONITOR**	**LOW POWER, HIGH INTEREST** 1. Current Cadets 2. Future Cadets 3. Future Faculty & Staff 4. Association of Graduates (AoG) **KEEP INFORMED**
	LOW — INTEREST	HIGH

Table 3. Power Interest Grid

The commitment assessment matrix in Table 4 shows where current key stakeholders are in their commitment (X) and where the project team would like them to be (O). Overall, the team wants all influential stakeholders to strongly support the project, and all stakeholders regardless of influence to at least be neutral about the project.

Commitment Level	**Stakeholder**						
	USACE	USMA Leadership	Current Cadets	Future Cadets	Local Population	AoG	Federal Government
Strongly Support	O	XO	O			O	O
Support	X		X				X
Neutral				XO	XO	X	
Opposed							
Strongly Opposed							

Table 4. Commitment Assessment Matrix

5. **Risk Management:** The two primary risk categories for this project are schedule delays, and scope changes. There is also a risk that Congress withdraws funding which the team has decided to accept upfront. Schedule delays can result from logistical issues, weather

conditions, delays in obtaining permits and approvals, or equipment failures. Scope issues arise from changes in Academy project objectives, or unforeseen structural or geotechnical conditions that increase the project workload The Failure Mode and Effects Analysis (FMEA) in Tables 5 and 6 analyzes the initial and residual risk for each of risk category after implementing the following response plans.

a. **Withdrawal of Funding** (Accept) – The project is unviable without Congressional funding, and this risk is beyond the control of the project team.

b. **Logistics Delays** (Mitigate) – Careful analysis of USMA's extensive construction history will allow for identification of common logistical challenges and solutions, preparing the project team to implement individual strategies ensuring equipment and materials are timely and sufficient.

c. **Weather Delays** (Avoid/Mitigate) – Project initiation during late spring maximizes favorable warm weather conditions. Interior renovation will be scheduled for the winter months as weather will have a less pronounced impact. Continuous monitoring of weather forecasts will allow USACE to prepare for severe conditions regardless of season, limiting project disruption.

d. **Permit Delays** (Accept/Mitigate) – Permit delays can be mitigated by coordinating early and often with authorities to ensure plans and designs meet regulatory standards.

e. **Equipment Failure** (Transfer) – Renting equipment and contracting equipment services to an outside organization transfers the risk of equipment failure outside the project team.

f. **Client Scope Changes** (Avoid) – A project charter signed by all stakeholders will prevent scope changes after the project has begun.

g. **Process Scope Changes** (Accept) – Unforeseen conditions will likely alter the scope of the project during construction. Mitigation can occur by building finances into the budget, and extra time into the schedule for incidental events, but little can be done to avoid these events in the first place.

Risk	Initial Risk				Response
	Severity, S	Likelihood, L	Ability to Detect, D	RPN	
Withdrawal of Funding	10	1	1	**10**	*Accept*
Logistics Delays	5	8	3	**120**	*Mitigate*
Weather Delays	4	8	3	**96**	*Avoid/Mitigate*
Permit Delays	8	5	1	**40**	*Accept/Mitigate*
Equipment Failure	9	4	7	**252**	*Transfer*
Client Scope Change	5	4	1	**20**	*Avoid*
Process Scope Change	7	8	6	**336**	*Accept*

Table 5. Initial Risk Analysis

Risk	Response	Residual Risk			
		Severity, S	Likelihood, L	Ability to Detect, D	RPN
Withdrawal of Funding	*Accept*	10	1	1	**10**
Logistics Delays	*Mitigate*	5	5	3	**75**
Weather Delays	*Avoid/Mitigate*	2	8	2	**32**
Permit Delays	*Accept/Mitigate*	8	2	1	**16**
Equipment Failure	*Transfer*	9	2	1	**18**
Client Scope Change	*Avoid*	5	1	1	**5**
Process Scope Change	*Accept*	7	8	6	**336**

Table 6. Residual Risk Analysis

Process Scope Change has the highest RPM value and is most difficult risk to address. A contingency plan for this issue will anticipate probable process scope changes by studying previous USMA projects to examine how such changes affect schedule, budget, and overall project scope, building those costs and time considerations into the total budget and schedule for the project.

6. **Recommendation:** Overall, the Project Team recommends COA 2 as the best option. It provides a balance approach to meeting the Academy's success criteria and directly addresses the need that led to the project itself – expanding Academy educational capability. Additionally, COA 2 will bring in an estimated $76 million in returns over 40 years, well over the $55 million bar the Academy set as a "perfect" score for Discounted Cash Flow. This option satisfies the most stakeholders as well, preserving the exterior look of Thayer Hall for the AoG and other interested conservationists, and upgrading educational capability for the Academy, and by extension for the Army as a whole. It satisfies cadets by reasonably limiting disruption to academy operations, while also bringing the new facility into use relatively soon. Risk analysis shows that process scope change, which is changes to project scope based on unforeseen conditions at Thayer Hall, are the biggest threat to the project. By implementing a contingency plan that anticipates such issues the project team can help lessen the impact of such risks if they become a reality. In conclusion, the project team strongly recommends COA 2 as the best approach for the Thayer Hall Renovation Project.

7. The point of contact for this memorandum is the undersigned at john.afariaikins@westpoint.edu

JOHN AFARI-AIKINS
CDT, C2
Project Manager

Appendix
1. Appendix A – NPV Calculation

UNITED STATES MILITARY ACADEMY
WEST POINT.

Course Project - Stage Gate #2

Instructions/Rubric/Student Example

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-2 | 2 Feb 2026

MEMORANDUM FOR RECORD

SUBJECT: Project Stage Gate #2 Instruction Sheet

1. **PURPOSE.** The purpose of Project Stage Gate #2 is to complete a project plan summary, leveraging tools learned since the submission of Project Stage Gate #1.

2. **REFERENCES.** No change from EM411 Course Instructional Memorandum.

3. **SUMMARY.** ***This Deliverable is worth a total of 90 points.*** Project Managers will coordinate with instructors and conduct a deskside Project Update Meeting to your stakeholder ***NLT 11/12 March 2026*** and submit their Project Plan Summary ***NLT 2359 (EDT) on 20 March 2026*** via Canvas Assignments. Deliverables are outlined in Paragraph 5. Detailed submission requirements are outlined in Paragraph 6.

4. **BACKGROUND.** After reviewing your project selection analysis, the United States Military Academy (USMA) has selected Course of Action #2: Major Renovation of Thayer Hall. The Superintendent has reviewed and signed a Thayer Hall Renovation Project Charter. A detailed project plan must now be developed. Students will develop a Project Plan Summary utilizing Microsoft Project and then analyze the plan in terms of cost, schedule, and scope.

5. **DELIVERABLES:**

   a. Project Manager Meeting (deskside brief) with Instructor.

   b. Project Plan Summary. See Annex B for the template and requirements.

   c. MS Project Baseline Schedule (MSP File).

   d. MS Project Scope Change Schedule (MSP File).

   e. MS Excel Spreadsheet with Project Budget and other required calculations.

6. **STAGE GATE INSTRUCTIONS.**

   a. ***Develop a Project Baseline (20 pts).***

   (1) Create a **BASELINE Project Schedule** in *Microsoft Project* using the Project Task Data provided in Annex A. Note that some tasks use the same resource. The project is

scheduled to begin on 17 Feb 2026. Use a six-day work week, Monday through Saturday, 0800-1700, with a one-hour lunch from 1200-1300. Plan initially to not work on any Federal Holiday. *Use the list of federal holidays found at: https://www.opm.gov/policy-data-oversight/pay-leave/federal-holidays/.* Make sure to use all the formatting and documentation standards for the Microsoft Project (MSP) file you learned in class, including creating and using a new project calendar specific to your project. Use a clean version of the DSE Template as a starting point for entering project data.

(2) ***Before*** leveling any resources, crashing any tasks, or undertaking any other project plan improvements, ***STOP!*** Save this file. Label it, using the guidance in paragraph 7. Continue from this point using a copy of this file.

(3) After saving your "pre-leveled" MSP file and creating a copy of this file, continue by leveling your schedule. Use this "leveled" MSP Schedule to complete your schedule analysis and updated budget for your Project Plan Summary.

b. ***Scope Change Request (30 pts).*** After completing your Project Baseline, address the following Scope Change Request.

(1) **Request:** The Superintendent has requested that the Renovations be completed before 1 May 2028. With the threat of budget cuts looming, the Superintendent wants to highlight the importance of donor contributions to the academy's mission. Specifically, the Superintendent wants to know what it will take to complete the project by 1 May 2028, so the newly renovated building can be showcased during graduation week.

(2) **Requirement:** Determine how to accomplish this and provide your recommendation, along with identifying any changes or impacts to cost, schedule, or risk.

(3) **Additional Instructions:**

a) Possible methods to address this scope change include working on Federal Holidays (at additional cost), utilizing your contingency funds to add more resources, and/or crashing tasks.

b) If you choose to crash tasks, follow the rules outlined in class regarding which tasks to crash and when.

c) If you choose to work Federal Holidays, this will cost an additional $12,500/day.

d) Resolve all resource overallocations that may result from the changes you make to the plan. Your recommendation to the Superintendent should not include any resource overallocations.

e) After implementing your solution and leveling your new schedule, save this file. Label using guidance in paragraph 7.

c. ***Project Update Meeting (*****10 pts*****).*** A progress update is required from every project team. This meeting will be coordinated with your instructor **NLT 11/12 March 2026**. The project manager <u>must submit</u> the BASELINE MS Project file to the Instructor <u>at least 2 working days prior</u> to your scheduled meeting with your Instructor. **During the meeting, the Project Manager will brief the status of the project and discuss the BASELINE project plan in detail**. The expectation is that the BASELINE Project File is complete with all data and appropriate formatting when the meeting occurs.

d. ***Budget and Analysis (20 pts).*** Complete a budget estimate that accurately accounts for all direct, indirect, and contingency costs. Use the information in Annex A, Project Task Data, and your MS Project Baseline Schedule to complete an updated budget estimate. The OGRE Accounting Office is directing an Overhead Rate of 12% on Labor, a General and Administrative (G&A) Rate of 5% on all direct costs, a Contingency of 15% of the Subtotal, and a Profit margin of 6% of the Subtotal.

e. ***Stakeholder and Risk Management Plans (10 pts).*** Update your stakeholder and risk management plans with your team's new understanding of the Thayer Hall renovation project. How has your team's plans to manage stakeholders and risks changed? Discuss your key stakeholders and updates to major risks facing your project and project team.

7. **Submission requirements.** Complete a Project Plan Summary (see Annex B Project Plan Template for content/formatting guidance) and include all relevant supporting documents/data. Ensure you <u>***include a cover sheet with a documentation statement with your submission***</u>. Project Stage Gate 2 is due **NLT 2359 (EDT) on 20 March 2026** and should be submitted via Canvas Assignments. Required Documents include:

a. Project Plan Summary (PDF).

1) Label Format Example: "EM411_H13_ROMEO_SG2_Project Plan Summary.pdf"

b. Project Baseline Schedule (MS Project File).

1) Label Format Example: "EM411_H13_ROMEO_SG2_Schedule, Baseline.mpp"

c. Scope Change Schedule (MS Project File).

1) Label Format Example: "EM411_H13_ROMEO_SG2_Schedule, Scope Change.mpp"

d. Project Budget & Analysis.

1) Label Format Example: "EM411_H13_ROMEO_SG2_Budget & Analysis.xlsx"

8. **List of Supporting Annexes.**

   a. Annex A – Project Task Data (xlsx)

   b. Annex B – Project Plan Summary Template (docs)

DANIEL FINCH
Senior Instructor
EM411 Course Director

WBS	Activity Task List	Predecessor (ID)	Predecessor (WBS)	Optimistic Time (Days)	Most Likely Time (Days)	Pessimistic Time (Days)	TE (Days)	Materials Total	Labor	Labor $/hr
0	**EM411 Thayer Renovation Project**									
1	**Exterior**									
1.1	**Site work**									
1.1.1	Fencing			9	12	25		$ 14,000.00	Labor Crew	$2,400.00
1.1.2	Scaffolding	3	1.1.1	20	24	28		$ 73,500.00	Labor Crew	$2,400.00
1.2	**Exterior Walls**									
1.2.1	Exterior Wall Cleaning	4	1.1.2	24	42	54		$ 195,000.00	Mason Crew	$1,900.00
1.2.2	Remove old grout	6	1.2.1	51	54	70		$ 18,500.00	Mason Crew	$1,900.00
1.2.3	Install new grout	7	1.2.2	10	35	44		$ 415,000.00	Mason Crew	$1,900.00
1.2.4	Masonry Repair	8	1.2.3	50	60	75		$ 1,950,000.00	Mason Crew	$1,900.00
1.3	**Windows**									
1.3.1	Remove old windows	6	1.2.1	30	34	36		$ 78,000.00	Carpenter Crew	$1,750.00
1.3.2	Install new windows	11	1.3.1	40	41	55		$ 127,500.00	Carpenter Crew	$1,750.00
1.4	**Rooftop**									
1.4.1	Strip Rooftop	9	1.2.4	18	22	28		$ 152,000.00	Roof Crew	$1,125.00
1.4.2	Repair Drain System	14	1.4.1	30	35	48		$ 490,000.00	Plumbing Crew	$1,050.00
1.4.3	Pave Rooftop	15,34	1.4.2, 3.1.1	3	8	17		$ 1,420,000.00	Roof Crew	$1,125.00
1.4.4	Paint Rooftop	15,16	1.4.3, 1.4.2	4	5	22		$ 15,500.00	Roof Crew	$1,125.00
2	**Interior**									
2.1	**Furniture and Fixtures**									
2.1.1	Remove old FFE	3	1.1.1	12	15	17		$ 27,500.00	Labor Crew	$2,400.00
2.1.2	Remove FFE Electric	20	2.1.1	18	22	26		$ 18,530,000.00	Carpenter Crew	$1,750.00
2.1.3	Install new fixtures	28	2.2.4	26	30	40		$ 52,500,000.00	Carpenter Crew	$1,750.00
2.1.4	Install new furniture	22	2.1.3	24	29	44		$ 9,725,000.00	Labor Crew	$2,400.00
2.2	**Walls**									
2.2.1	Demo	21,34,43	2.1.2, 3.1.1,3.3.2	84	90	96		$ 225,000.00	Labor Crew	$2,400.00
2.2.2	Frame New Walls	30,25	2.3.1, 2.2.1	55	80	92		$ 11,523,000.00	Carpenter Crew	$1,750.00
2.2.3	Install Drywall	36,40,44	3.1.3, 3.2.2, 3.3.3	38	50	58		$ 1,955,000.00	Carpenter Crew	$1,750.00
2.2.4	Paint Interior Walls	27	2.2.3	22	34	36		$ 579,500.00	Paint Crew	$1,100.00
2.3	**Floors**									
2.3.1	Remove old flooring	25	2.2.1	34	38	45		$ 12,100,000.00	Labor Crew	$2,400.00
2.3.2	Install new flooring	30,27	2.3.1, 2.2.3	75	80	98		$ 32,980,000.00	Floor Crew	$1,300.00
3	**Utilities**									
3.1	**HVAC**									
3.1.1	Remove old HVAC system	21	2.1.2	36	37	39		$ 119,000.00	Labor Crew	$2,400.00
3.1.2	Remove old ductwork	34	3.1.1	18	21	38		$ 12,500,000.00	Mechanical Crew	$1,900.00
3.1.3	Install new ductwork	26,35	2.2.2, 3.1.2	50	65	75		$ 17,658,020.00	Mechanical Crew	$1,900.00
3.1.4	Install new HVAC system	36,40	3.1.3, 3.2.2	24	48	66		$ 19,874,000.00	Mechanical Crew	$1,900.00
3.2	**Plumbing**									
3.2.1	Remove Old Plumbing	25,21	2.2.1, 2.1.2	42	50	58		$ 163,000.00	Plumbing Crew	$1,050.00
3.2.2	Install New Plumbing	39,26	3.2.1, 2.2.2	55	70	82		$ 8,426,000.00	Plumbing Crew	$1,050.00
3.3	**Electrical & IT**									
3.3.1	Remove old electrical gear	21	2.1.2	36	37	40		$ 49,500.00	Electrical Crew	$2,240.00
3.3.2	Demo old wiring	42	3.3.1	44	60	70		$ 54,350.00	Electrical Crew	$2,240.00
3.3.3	Install new wiring and CAT	40,26	3.2.2, 2.2.2	70	90	135		$ 9,129,000.00	Electrical Crew	$2,240.00
3.3.4	Install new electrical gear	44,28,23	3.3.3, 2.2.4, 2.1.4	32	34	48		$ 19,800,000.00	Electrical Crew	$2,240.00
3.3.5	Install new IT gear	45	3.3.4	50	59	66		$ 6,688,234.00	IT Crew	$1,723.00

Course Project Stage Gate #2 Grading Rubric

Instructor Score

0 pts

SG2 Rubric

Criteria Ratings Points

1. MS Project Baseline (25 points)

Superior

MSP Baseline file complete, All tasks accounted for and relationships correct. All Federal Holidays accounted for through all of CY 2027. All material and labor costs accounted for in resource sheet. Correct project duration and cost.

25 to >22 pts

Good

Minor errors to MSP Baseline file to cost and/or schedule. May be missing 1-2 Federal Holidays. Labor/material cost had minor error to entry. Duration and cost overall within 1% of correct baseline.

22 to >20 pts

Okay

More significant errors to MSP baseline file cost/and or schedule. Missing more than a few Federal Holidays, Had larger errors with labor/material cost entry. Duration and cost overall within 5% of correct baseline.

20 to >13 pts

Poor

Major errors to MSP baseline file cost/and or schedule. Missing many Federal Holidays, Major errors with labor/material cost entry. Duration and/or cost overall exceeds 5% margin of error from correct baseline.

13 to >0 pts

2. Scope Change Request (20 points)

Superior

Supe's request accommodated and thoroughly documented the methodology and thought process of achieving the scope change request. Appropriate techniques to meet the request and minimize impacts to both cost and risk are discussed. The schedule request was met, and the cost was no greater than 1% of the baseline.

20 to >17 pts

Good

Supe's request accommodated and sufficiently documented the methodology and thought process of achieving the scope change request. Techniques to meet the request are discussed. The schedule request was met, and the cost was no greater than 3% of the baseline.

17 to >15 pts

Okay

Supe's request accommodated, but lacking in discussion of methods and recommendation . Techniques to meet the request may not have been appropriate. The schedule request was met and the cost was no greater than 5% of the baseline.

15 to >10 pts

Poor

Supe's request was not accommodated, and the methodology and thought process of achieving the scope change request was not sufficiently documented. Techniques to meet the request are not discussed The schedule request was not met, and the cost was greater than 5% of the baseline.

10 to >0 pts

3. Project Update Meeting (10 points)

Superior

The meeting was coordinated in advance of 22 Oct 25, and the MSP Baseline file was e-mailed to instructor NLT 2 days ahead of the scheduled meeting. At least one Cadet from the team briefed the baseline plan and received approval to continue from the instructor. Baseline plan had no issues to improve and Cadet clearly briefed their plan.

10 to >8 pts

Good

The meeting was coordinated before 12 MAR 25, and the MSP Baseline file was e-mailed to instructor NLT 2 days before the meeting. At least one Cadet from the team briefed the baseline plan and received approval to continue from the instructor. Baseline plan had minor issues to improve and/or Cadet briefing may have had issues clearly briefeing the plan.

8 to >7 pts

Okay

The baseline plan had major issues to improve, and/or the Cadet briefing may have had major issues clearly briefing the plan. Meeting was coordinated late, and/or MSP baseline file may not have been received 2 days ahead of meeting.

7 to >5 pts

Poor

The baseline plan had significant issues to improve, and/or the Cadet briefing may have had major issues clearly briefing the plan. Meeting was coordinated late, and/or MSP baseline file was received late.

5 to >0 pts

4. Formatting (5 points)

All document files were named properly, and submitted proper file format, no issues with internal file formatting. Easy to follow, free of gross spelling and grammatical errors.

5 to >4 pts

Okay

Minor errors to formatting of memo, file naming convention may not have been followed. Minor errors to spelling or grammar.

4 to >3 pts

Poor

Major errors to formatting of memo and file naming convention may not have been followed. Major errors to spelling or grammar.

3 to >0 pts

5. Project Budget and Analysis (20 points)

Superior

All direct, indirect, contingency costs calculated and accounted for properly in Excel file. All OGRE Accounting rates properly applied IAW Annex B. No errors.

20 to >18 pts

Good

Minor errors to calculating direct costs, indirect costs, contingency costs, or final budget not to exceed 1% of correct baseline.

18 to >15 pts

Okay

Major errors to calculating direct costs, indirect costs, contingency costs, or final budget that exceed 3% of correct baseline.

15 to >10 pts

Poor

Significant errors to calculating direct costs, indirect costs, contingency costs, or final budget that exceed 5% of correct baseline.

10 to >0 pts

6. Stakeholder and Risk Management (10 points)

Superior

Thorough SH management plan that discusses how to engage with key stakeholders (method/frequency). Complete and updated risk management plan that includes a FMEA, risk mitigations, and recalculated RPNs.

10 to >8 pts

Good

Adequate SH management plan that discusses how to engage with key stakeholders (method/frequency). Updated risk management plan that includes a FMEA, risk mitigations, and recalculated RPNs.

8 to >7 pts

Okay

Lacking SH management or risk management plan. Method/frequency may not be sufficiently discussed on SH management. Risk management plan did not apply a FMEA or had major issues in doing so.

7 to >5 pts

Poor

Poorly updated SH management or risk management plan. Does not appear to

understand major changes to project risk/SHs. Method/frequency not sufficiently discussed on SH management. Risk management plan did not apply a FMEA.

5 to >0 pts

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT26-2 20 March 2026

MEMORANDUM FOR Superintendent, United States Military Academy West Point, NY 10996

SUBJECT: Project Plan Summary, Thayer Hall Renovation

1. **Purpose.** This memorandum presents the detailed project plan for the Thayer Hall Renovation Project following the decision to proceed with the second Course of Action (Major Renovation of Thayer Hall). It outlines the project baseline schedule, budget analysis, scope change timeline, and stakeholder and risk management assessment. The original timeline is estimated to take 760 days from 17FEB2026-19AUG2028. Following approval of the scope change the project is estimated to take 668 days from 17FEB2026-01MAY2028 at a total cost of $262,413,724.

2. **Objectives.** This memorandum provides an update following Stage Gate #1 and presents a detailed project plan and budget. The project seeks to modernize Thayer Hall's amenities through full reconstruction and modernization of the interior. The objective is to align resources to deliver a feasible plan that meets the Superintendent's target completion date while managing cost, schedule, and scope without overextending personnel or assets.

3. **Project Baseline.**

a. **Schedule.**

1) Old Grad Research and Engineering, Inc. (OGRE) expects the project to begin on 17 February 2026. ORGE will execute it on a six-day workweek, Monday through Saturday, from 0800 to 1700, with a one-hour lunch period from 1200 to 1300. This project excludes Federal holidays from the baseline schedule. ORGE structured the project to begin during a period of relatively reduced cadet activity in order to minimize disruption to academy operations during site mobilization, facility assessment, and initial interior demolition. Construction will then proceed through successive phases over multiple academic and summer cycles. Analysis in Microsoft Project identified a 16-task critical path, consisting of activities listed in Appendix A.

2) This memorandum anticipates periods of elevated schedule risk during the academic year and major academy events, to include graduation, reorganization week, and home football weekends. These events create traffic congestion and access limitations that may disrupt material deliveries and site operations. Other concurrent projects on post create additional risk, including the Michie Stadium renovation and CEAC construction, as well as limited installation access points. Seasonal weather conditions, particularly during the winter months, may also disrupt construction. The baseline schedule accounts for these constraints through project calendar assumptions, schedule margin, and resource leveling to

eliminate over-allocations. Contingency funds and management reserve are included to address potential delays and associated cost growth.

b. **Budget.** Based on the estimates provided for material and labor, the team determined that the baseline project's total baseline fully costed value of the project is $334,314,108.82. This estimate includes direct labor and materials, 12% overhead on labor, 5% G&A on all direct costs, 15% contingency on subtotal, and 6% profit on subtotal. The baseline cost, accounting for labor and materials, is $260,717,320. Refer to Appendix B Figure 3 for a detailed cost breakdown.

4. **Scope Change Request.** The Superintendent of USMA requested that renovations be completed before 1 May 2028. The Superintendent also expressed concern about the threat of budget cuts.

a. **Plan of Action:** The baseline schedule does not meet the Superintendent's request, so the team evaluated three acceleration methods to meet the new completion date. The first option involved working federal holidays. OGRE rejected this method because it did not meaningfully reduce the critical path and would impose a toll on morale. The second option involved hiring additional crews. The team rejected this course of action due to high expenses in relation to overtime work. This memorandum therefore recommends using selective overtime hours on critical-path tasks. This option reduced the schedule by 92 days and increased the direct cost by $1,696,404. To mitigate burnout, each crew only works one task over the entire project of overtime work. Finally, OGRE re-leveled to eliminate all resource overallocations.

b. **Overtime.** Throughout the project, each crew will work an additional 4 hours per day over the crashed task duration. This concludes the workday at 9pm (not including additional breaks). ORGE mitigates morale risks by only assigning one overtime task per crew. See Appendix E for calculations, and Appendix B Figure 3 for specific crashed tasks on the critical path. Team selection consisted of finding the lowest cost crew on the critical path, adding overtime hours, leveling the project, and repeating the process for the next lowest cost crew. This method successfully crashed the project to the deadline at the minimum overtime cost, while abiding to the one-overtime detail per crew.

c. **Budget.** To satisfy the scope change, the team also developed two feasible expedited schedules. OGRE recommends Option 1 to achieve project completion by 1MAY2028. Overtime work on selected critical path tasks increases the budget to $262,413,724. Option 2 provides an earlier completion date of 22APR2028 by crashing a longer duration critical task at the end (same crew, two different possible tasks to crash) This option creates a week of schedule buffer before the Superintendent's requirement at a higher cost of $264,036,074. Select Option 2 if the stakeholder values schedule assurance over minimizing project cost.

Baseline Budget (19AUG)	Expedited Budget (1MAY)	Accelerated Buffer Plan (22APR)
$260,717,320.00	$262,413,724.00	$264,036,074.00

Table 1: Budget Comparison

5. **Stakeholder Management Plan.** The primary stakeholders for the Thayer Hall Renovation Project include Congress, the Superintendent, the Dean, USMA leadership, and the academic faculty. The project priorities shifted from general modernization to schedule and cost control. Congress and donors value visible results and project cost. The Superintendent primarily desires finishing the project before graduation so the First Class cadets can walk the stage in a finished stadium. The Dean and faculty seek to limit academic disruption, while USMA leadership wants effective coordination and safety during accelerated construction.

a. **Stakeholder Considerations.** Due to the accelerated schedule, there will be greater sensitivity to delay and coordination problems. To mitigate this and address priorities, OGRE prioritizes frequent stakeholder communication.

b. **Stakeholder Involvement.** To address these priorities, the team will provide regular milestone updates to the Stakeholders. Additionally, OGRE will address future disruptions to faculty and will coordinate meetings with the contractors to provide plan and status updates.

6. **Risk Management Plan.** The main risks with the acceleration of construction shifts to labor fatigue, reduced productivity, and cost growth caused by crashing selected critical-path tasks through overtime. Because the expedited plan relies on specific crews working overtime on specific tasks, the primary risk is fatigue. This could reduce efficiency and increase errors, ultimately causing task delays. A second major risk is the increased project cost due to overtime pay. To mitigate these risks, the team limited each crew to one overtime task, kept holidays non-workdays, and included the 15% contingency reserve for unforeseen expenses. The project team will monitor crashed tasks closely through the biweekly coordination meetings to prevent delays from affecting the expeditated schedule A detailed summary of the updated risk assessment and mitigation measures is provided in Appendix C.

7. The point of contact for this memorandum is the undersigned at ethan.ellis@westpoint.edu

ETHAN ELLIS
CDT LT
Project Manager

Appendix
1. Appendix A – Baseline Schedule & Critical Path
2. Appendix B – Budget & Cost Analysis
3. Appendix C – Stakeholder Analysis
4. Appendix D – Updated Risk Assessment
5. Appendix E – Link to excel

Appendix A –Schedule & Critical Path

	Duration	Start	Finish	Cost
Baseline:	**760 days**	**Feb 17 '26**	**Aug 19 '28**	**$260,717,320.00**
Expedited:	**668 days**	**Feb 17 '26**	**May 1 '28**	**$262,413,724.00**

Figure 1: Baseline & Expedited Schedule

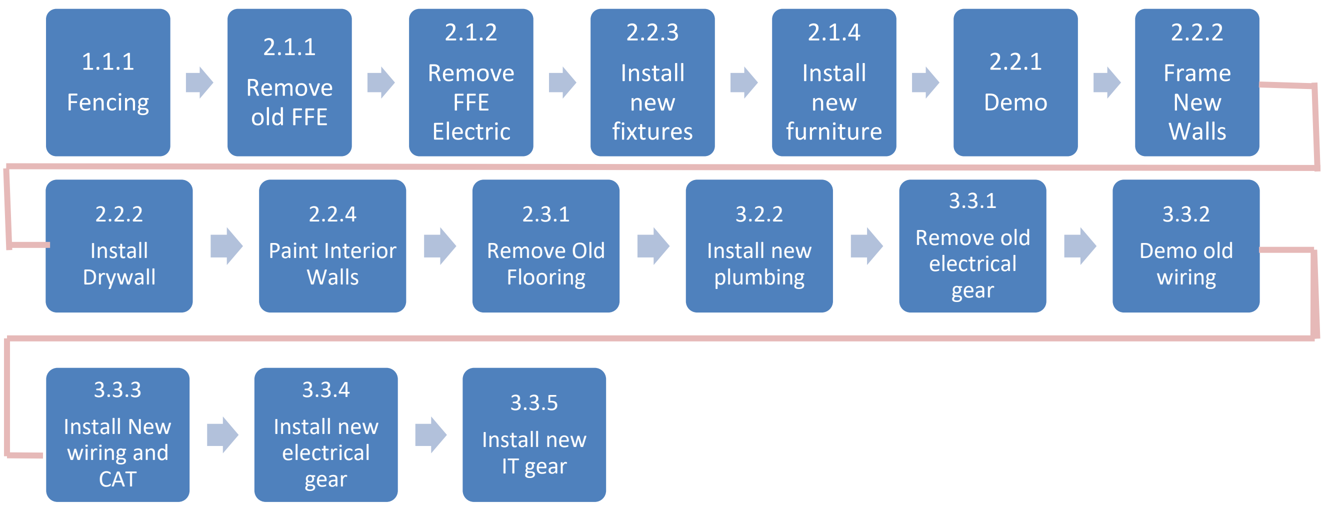


Figure 2: Critical Path Activities for Thayer Renovation

Appendix B – Budget & Cost Analysis

Costs	Amount
Direct	$260,717,320.00
Labor	$21,162,216.00
Indirect	$15,575,331.92
Overhead	$2,539,465.92
G & A	$13,035,866.00
Subtotal	$276,292,651.92
Contingency	$41,443,897.79
Fully Costed/ Baseline	$317,736,549.71
Profit Magin	$16,577,559.12
Project Budget	$334,314,108.82
Sum of Additional Overtime Cost	$2,297,840.00
Expedited Project Cost	$262,413,724.00
Baseline Project Cost	$260,717,320.00
Most Expedited Project Cost	$264,036,074.00

Figure 3: Overall Budget Overview

	Crashed Task	Crew Worked	Original Cost	Original Days	New Days	Additional Pay if Overtime	Additional Pay if Extra Crew	Total Cost with Overtime	Total Cost with Extra Crew
	Install New Plumbing	Plumbing	$9,014,000.00	70	47	$289,800.00	$394,800.00	$9,303,800.00	$9,408,800.00
	Paint Interior Walls	Paint	$869,900.00	33	22	$145,200.00	$193,600.00	$1,015,100.00	$1,063,500.00
	Install New IT Gear	IT	$7,501,490.00	59	40	$392,844.00	$551,360.00	$7,894,334.00	$8,052,850.00
	Frame New Walls	Carpenter	$12,615,000.00	78	52	$546,000.00	$728,000.00	$13,161,000.00	$13,343,000.00
	Install New Electrical Gear	Electrical	$20,445,120.00	36	24	$322,560.00	$430,080.00	$20,767,680.00	$20,875,200.00
(ALTERNATE)	Demo Old Wiring	Electrical	$1,111,630.00	59	40	$510,720.00	$716,800.00	$1,622,350.00	$1,828,430.00

Figure 4: Crashed Schedule Costs

Appendix C – Stakeholder Analysis

POWER INTEREST GRID

POWER	Low INTEREST	High INTEREST
High	**Keep Satisfied** - taxpayers (general public) - Congress	**Closely Manage** - Superintendant / USMA High Brass - DPW - USACE - Major Donors
Low	**Monitor** - Old Grads - Highland Falls Community	**Keep Informed** - Instructors / faculty / Department Heads - Research Centers - USCC Cadets

Table 1: Power Interest Grid of Key Stakeholders

Commitment Assessment Matrix

Level of Commitment	Stakeholder Groups: Taxpayers	Congress	USMA High Brass	DPW	USACE	Major Donors	Old Grads	Highland Falls	Department Heads	Instructors	Research Centers	USCC Cadets
Strongly Support			C		C	C					D	
Support		D	D	D	D	D	C D		C D	D	C	
Neutral	C D	C		C				C D				D
Opposed										C		C
Strongly Opposed												

Table 2: Commitment Assessment Matrix of Key Stakeholders

Appendix D – Updated Risk Analysis FMEA Matrix

Failure Mode and Effect Analysis

Potential Failure modes	Severeity	Likelihood	Ability to I	RPN	comments	Risk Response	Severeity	Likelihood	Ability to Detect	New RPN
Tight Schedule	8	7	6	336	not fully id task	Mitigate. Create a long range calendar with key dates and backwards plan. Fully defined tasks at each sprint	8	5	4	160
Overtime Fatigue	9	6	5	270	overtime may reduce productivity	Mitigate: Limit each crew to one overtime task, avoid working holidays, monitor performance	9	4	4	144
Costs Escalate	9	7	6	378	innacurate early estimates, inflation, unforseen situations uncover	Mitigate: Conduct an iterative budgeting process to work out an accurate budget, conducting both top down and bottom up approach	9	5	4	180
Worker Injury	10	5	5	250	safety not enforced, fatigue, time pressure, unsafe shortcuts, demolition	Avoid: Enforce safety and require job hazard analysis for all high risk tasks before work starts	10	3	3	90
Discovery of Hazardous Material	9	5	7	315	lead, undocumented surveys, uranium	Mitigate: pre-demo hazmat surveys	9	3	5	135
Structural/Design Conflicts	8	5	7	280	architecture clash with structure	Mitigate. Conduct early field scanning/selection and verification	8	3	5	120
Code Compliance Delays	8	4	6	192	fire/life safety requirements, inspections schedule	Mitigate: Engage with code authority early and track the inspection times	8	3	4	96
Quality Defects	8	5	6	240	military grade	Mitigate: Create clear acceptance critieria from vendors	8	3	4	96
Supply Chain Disruption	7	6	6	252	extended lead times	Mitigate: identify long-lead items early and pre-approve alternates	7	4	4	112

Potential Failure modes	Severeity	Likelihood	Ability to [	RPN
Tight Schedule	8	7	6	336
Overtime Fatigue	9	6	5	270
Costs Escalate	9	7	6	378
Worker Injury	10	5	5	250
Discovery of Hazardous Material	9	5	7	315
Structural/Design Conflicts	8	5	7	280
Code Compliance Delays	8	4	6	192
Quality Defects	8	5	6	240
Supply Chain Disruption	7	6	6	252

Risk Response	Severeity	Likelihood	Ability to Detect	New RPN
Mitigate. Create a long range calendar with key dates and backwards plan. Fully defined tasks at each sprint	8	5	4	160
Mitigate: Limit each crew to one overtime task, avoid working holidays, monitor performance	9	4	4	144
Mitigate: Conduct an iterative budgeting process to work out an accurate budget, conducting both top down and bottom up approach	9	5	4	180
Avoid: Enforce safety and require job hazard analysis for all high risk tasks before work starts	10	3	3	90
Mitigate: pre-demo hazmat surveys	9	3	5	135
Mitigate. Conduct early field scanning/selection and verification	8	3	5	120
Mitigate: Engage with code authority early and track the inspection times	8	3	4	96
Mitigate: Create clear acceptance critieria from vendors	8	3	4	96
Mitigate: identify long-lead items early and pre-approve alternates	7	4	4	112

Table 3: Failure mode and Effect Analysis with Risk Responses

Risk Assessment Matrix				
Severity/ Probability	**Catastrophic (1)**	**Critical (2)**	**Marginal (3)**	**Negligible (4)**
Frequent (A)	Fire in construction zone	Recurring cost growth from overtime	Regular Minor Equipment Failures	Routine minor stakeholder complaints
Probable (B)	Serious Worker injury	Budget overrun from cost escalation	Supply chain delays on long-lead equipment	Minor temporary utility disruption
Occasional (C)	Hazardous material exposure	Schedule Delays	Permitting/inspection delays	Minor classroom relocation friction
Remote (D)	Partial Strutural failure	Quality defects from rushed work	Localized parking disruption (faculty)	Rework punch-list
Improbable (E)	Major utility outage affecting construction	Miss 1MAY deadline	Labor fatigue cause productivity loss	Routine admin delay
Eliminated (F)	Eliminated			

Table 4: Qualitative Risk Matrix

Appendix E – Link to Excel

EM411_H13_ROMEO_SG2_Budget & Analysis.xlsx

Acknowledgements

ChatGPT. Assistance given to the author, AI. I used ChatGPT revise draft memorandum language for clarity, active voice, and concision. The outputs were used to support my understanding and editing process; all final writing, calculations, and verification of sources and requirements were my own. OpenAI, (https://chat.openai.com/chat). West Point, NY, 19MAR2026.

Chernik, Sarah CDT G2 ‘26. Assistance given to the author, visual reference and verbal discussion. CDT Chernik permitted me to review her project plan summary and I utilized her framework for the critical task timeline to create our own. West Point, NY, 19MAR2026.

UNITED STATES MILITARY ACADEMY
WEST POINT.

Course Project - Stage Gate #3

Instructions/Rubric/Student Example

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-2 23 March 2026

MEMORANDUM FOR RECORD

SUBJECT: Project Stage Gate #3 Instruction Sheet

1. **Purpose.** The purpose of Project Stage Gate #3 is to review the project's progress, conduct earned value analysis and provide a progress assessment report to the program manager and client.

2. **References.** No change from EM411 Course Instructional Memorandum.

3. **Summary.** ***This Deliverable is worth a total of 60 points.*** Project Managers will submit their project deliverables **NLT 2359 (EDT) on 15 April 2026** via Canvas Assignments. Detailed submission requirements are outlined in paragraphs 5 and 6. One submission per group.

4. **Background.** Imagine the Thayer Hall Renovation Project is underway. The Supe's initial request for earlier delivery has been rescinded. After your analysis and understanding the risks to scope and costs, he decided to stick with the baseline project plan. Your approved project plan included a requirement to produce periodic progress assessment reports at specific dates throughout the project, to include 60-day, 90-day, and 120-day milestones. Future assessment dates will depend on your progress at the 120-day milestone. The U.S. Army Corps of Engineers has clarified that the critical ratio (Cost-Schedule Index) should be above 0.85 at each assessment date. You've captured project performance data through your monitoring systems and must now analyze and report your project's performance at the three assessment dates.

5. **Stage Gate Instructions.** The team will assess the project's progress at the three milestones, using Earned Value Analysis (EVA), and provide a progress report in the form of a ***Memorandum for Record and Dashboard.***

a. **60-Day and 90-Day EVA.** Conduct an Earned Value Analysis (EVA) on your project using the Approved Project Plan in Annex A and the Project Analysis Data in Annex B. (Note: <u>Do not</u> include Overhead, G&A, Contingency, Profit, or any other indirect costs in this analysis. ***<u>USE DIRECT COSTS ONLY!)</u> For the 60-day EVA, assume that the project has completed work at the end of the working day on 1 July 2026. For 90-Day EVA, 1 Aug 2026. For 120-Day EVA, 1 Sep 2026.***

b. **120-Day EVA.** Based on your 90-day analysis, the Superintendent is concerned that the project will not be completed within time and/or cost. As a result, he is anxious to see the

120-Day Analysis. Conduct an Earned Value Analysis at the 120-day mark, compare it to your 60-day and 90-day assessments, and estimate the project's future progress and health. ***For the 120-day EVA, assume that the project has completed work at the end of the working day on 1 Sep 2026. Again, <u>use Direct Costs Only</u>, and reference Annex B for actual and planned assessments.***

c. **Project Management Dashboard.** Create a professional Project Management Dashboard to convey your analysis of the performance and health of the project. The Dashboard should include all pertinent project status information, including your EVA results for the 60, 90, and 120-day timeframes. You may develop your Dashboard using Microsoft Word, PowerPoint, or Excel. The Dashboard can be attached to your memorandum (as an appendix) or be submitted as a stand-alone document (as an Annex).

d. **Memorandum.** Analyze the status of your project at the 60, 90, and 120-day milestones, and summarize your results as they pertain to ***Cost, Schedule, and Scope*** in a short memorandum (2-3 pages) to the Superintendent. Discuss any activities, events, or processes that could have heavily influenced your EVA results and the performance changes between the milestones. Also, describe any concerns you have and what you plan to do about it. The memorandum should be written in the Army Memorandum Format (AR 25-50). Annex C is provided as a template for developing your memorandum.

6. **Submission requirements.** Project Stage Gate #3 is **due NLT 2359 (EDT) on 15 April 2026 via Canvas** and will consist of the following deliverables:

a. Memorandum. See Annex C for format and content requirement details.
b. Dashboard (Either as an appendix or annex).
c. Excel file with your 60, 90, and 120-day EVA Data and calculations.
d. Ensure a cover sheet and works cited are submitted with your memorandum.

7. **List of Supporting Files.**

a. Annex A, Approved Project Plan (mpp)
b. Annex B, Project Analysis Data (xlsx)
c. Annex C, Memorandum Template (docx)

8. The Point of Contact for this memorandum is the undersigned.

DANIEL FINCH
Senior Instructor
EM411 Course Director

			Day 1	15-Apr-26					Day 60		1-Jul-26	Day 90		1-Aug-26	Day 120		1-Sep-26
ID	WBS	Task Name	Predecessors	Start	Finish	Duration (Days)	Critical	Cost	Actual %	Planned %	Actual Cost	Actual %	Planned %	Actual Cost	Actual %	Planned %	Actual Cost
0	**0**	**EM411 Thayer Renovation Project**		**Apr 15 '26**	**Oct 5 '28**	**760 days**	**Yes**	**$260,717,320.00**									
1	**1**	**Exterior**		**Apr 15 '26**	**Nov 9 '27**	**479 days**	**Yes**	**$10,359,800.00**									
2	**1.1**	**Site work**		Apr 15 '26	Aug 25 '26	113 days	Yes	$817,100.00									
3	1.1.1	Fencing		Apr 15 '26	Apr 30 '26	14 days	Yes	$282,800.00	100%		$290,000.00	100%		$290,000.00	100%		$290,000.00
4	1.1.2	Scaffolding	3	**Jul 29 '26**	**Aug 25 '26**	24 days	**No**	**$534,300.00**							85%		$531,000.00
5	**1.2**	**Exterior Walls**		Aug 26 '26	Apr 15 '27	192 days	No	$5,496,900.00									
6	1.2.1	Exterior Wall Cleaning	4	Aug 26 '26	Oct 14 '26	41 days	No	$818,200.00	10%		$25,000.00	20%		$175,000.00	100%		$860,000.00
7	1.2.2	Remove old grout	6	Oct 15 '26	Dec 22 '26	57 days	No	$884,900.00				15%		$95,000.00	96%		$770,000.00
8	1.2.3	Install new grout	7	Dec 23 '26	Feb 2 '27	33 days	No	$916,600.00									
9	1.2.4	Masonry Repair	8	**Feb 3 '27**	**Apr 15 '27**	61 days	**No**	$2,877,200.00									
10	**1.3**	**Windows**		Oct 15 '26	Jan 19 '27	78 days	No	$1,297,500.00									
11	1.3.1	Remove old windows	6	Oct 15 '26	Nov 24 '26	34 days	No	$554,000.00				29%		$236,000.00	78%		$390,000.00
12	1.3.2	Install new windows	11	**Nov 25 '26**	**Jan 19 '27**	44 days	**No**	**$743,500.00**							55%		$40,000.00
13	**1.4**	**Rooftop**		Apr 16 '27	Nov 9 '27	174 days	No	$2,748,300.00									
14	1.4.1	Strip Rooftop	9	Apr 16 '27	May 12 '27	23 days	No	$359,000.00									
15	1.4.2	Repair Drain System	14	Sep 7 '27	Oct 20 '27	37 days	No	$800,800.00									
16	1.4.3	Pave Rooftop	15	Oct 21 '27	Oct 30 '27	9 days	No	$1,501,000.00									
17	1.4.4	Paint Rooftop	16	**Nov 1 '27**	**Nov 9 '27**	**8 days**	**No**	$87,500.00									
18	**2**	**Interior**		**May 1 '26**	**Jun 16 '28**	651 days	**Yes**	**$147,192,600.00**									
19	**2.1**	**Furniture and Fixtures**		May 1 '26	Jun 16 '28	651 days	Yes	$82,407,700.00									
20	2.1.1	Remove old furniture	3	May 1 '26	May 18 '26	15 days	Yes	$315,500.00	100%		$315,500.00	100%		$315,500.00	100%		$315,500.00
21	2.1.2	Remove old fixtures	20	May 19 '26	Jun 13 '26	22 days	Yes	$18,838,000.00	20%		$17,500,000.00	100%		$19,255,000.00	100%		$19,255,000.00
22	2.1.3	Install new fixtures	28	Apr 6 '28	May 11 '28	31 days	Yes	$52,934,000.00									
23	2.1.4	Install new furniture	22	**May 12 '28**	**Jun 16 '28**	31 days	**Yes**	$10,320,200.00									
24	**2.2**	**Walls**		Oct 9 '26	Apr 5 '28	455 days	Yes	$18,092,900.00									
25	2.2.1	Demo	21,34,43	Oct 9 '26	Jan 28 '27	90 days	Yes	$1,953,000.00							25%		$462,000.00
26	2.2.2	Frame New Walls	30	Mar 17 '27	Jun 16 '27	78 days	Yes	$12,615,000.00									
27	2.2.3	Install Drywall	36,40,44	Dec 31 '27	Feb 26 '28	50 days	Yes	$2,655,000.00									
28	2.2.4	Paint Interior Walls	27	**Feb 28 '28**	**Apr 5 '28**	**33 days**	**Yes**	$869,900.00									
29	**2.3**	**Floors**		Jan 29 '27	Jun 2 '28	415 days	Yes	$46,692,000.00									
30	2.3.1	Remove old flooring	25	Jan 29 '27	Mar 16 '27	39 days	Yes	$12,848,800.00									
31	2.3.2	Install new flooring	30,27	**Feb 28 '28**	**Jun 2 '28**	**83 days**	**No**	$33,843,200.00									
32	**3**	**Utilities**		**Jun 15 '26**	**Oct 5 '28**	**709 days**	**Yes**	**$103,164,920.00**									
33	**3.1**	**HVAC**		Jun 15 '26	Nov 4 '27	421 days	No	$52,947,820.00									
34	3.1.1	Remove old HVAC system	21	Jun 15 '26	Jul 30 '26	38 days	No	$848,600.00	20%		$601,000.00	75%		$635,000.00	100%		$920,000.00
35	3.1.2	Remove old ductwork	34	Jul 31 '26	Aug 27 '26	24 days	No	$12,864,800.00							100%		$12,500,000.00
36	3.1.3	Install new ductwork	26,35	Jun 17 '27	Sep 2 '27	65 days	No	$18,646,020.00									
37	3.1.4	Install new HVAC system	36,40	**Sep 10 '27**	**Nov 4 '27**	**47 days**	**No**	$20,588,400.00									
38	**3.2**	**Plumbing**		Jan 29 '27	Sep 9 '27	187 days	Yes	**$9,597,000.00**									
39	3.2.1	Remove Old Plumbing	25,21	Jan 29 '27	Mar 29 '27	50 days	No	$583,000.00									
40	3.2.2	Install New Plumbing	39,26	**Jun 17 '27**	**Sep 9 '27**	**70 days**	**Yes**	$9,014,000.00									
41	**3.3**	**Electrical & IT**		Jun 15 '26	Oct 5 '28	709 days	Yes	**$40,620,100.00**									
42	3.3.1	Remove old electrical gear	21	Jun 15 '26	Jul 30 '26	38 days	Yes	$730,460.00	31%		$402,200.00	95%		$668,000.00	100%		$755,000.00
43	3.3.2	Remove old wiring	42	Jul 31 '26	Oct 8 '26	59 days	Yes	$1,111,630.00	5%		$210,000.00	54%		$631,850.00	100%		$950,000.00
44	3.3.3	Install new wiring	40,26	Sep 10 '27	Dec 30 '27	95 days	Yes	$10,831,400.00									
45	3.3.4	Install new electrical gear	44,28	Jun 17 '28	Jul 28 '28	36 days	Yes	$20,445,120.00									
46	3.3.5	Install new IT gear	45	Jul 29 '28	Oct 5 '28	59 days	Yes	$7,501,490.00									

Course Project Stage Gate #3 Grading Rubric

Instructor Score

0 pts

SG3 Rubric

Criteria Ratings Points

1. EVM Data (25 points)

Superior

All 60/90/120-day EVA Calculations Correct, work shown and files submitted

25 to >22 pts

Good

Minor errors to 60/90/120-day EVA Calculations. Work shown and files submitted

22 to >18 pts

Okay

Some larger errors to 60/90/120-day EVA Calculations. Work shown and files submitted

18 to >15 pts

Poor

Significant errors to 60/90/120-day EVA Calculations. Not all work may be shown and/or files not submitted.

15 to >0 pts

2. Dashboard (15 points)

Superior

60/90/120 Dashboard Correct & Easily Readable (titles, labels, color-scheme, scales, etc)

15 to >13 pts

Good

60/90/120 Dashboard may have minor errors and/or formatting may have minor issues.

13 to >10 pts

Okay

Some larger errors with 60/90/120 Dashboard. Larger formatting errors.

10 to >8 pts

Poor

Significant errors with 60/90/120 Dashboard. Format is difficult to read or interpret. Dashboard may be incomplete or missing.

8 to >0 pts

3. Analysis (15 points)

Superior

Analysis of EVA Data (Current Trend and Expectations moving forward). Course Concepts correctly applied. Clear understanding of course concepts/EVM metrics & measures.

15 to >13 pts

Good

Minor errors with analysis of EVA Data. Course Concepts mostly correctly applied. Discussion shows an understanding of course concepts/EVM metrics & measures.

13 to >10 pts

Okay

Major errors with analysis of EVA Data. Course Concepts may not be correctly applied. Discussion shows some understanding of course concepts/EVM metrics & measures.

10 to >8 pts

Poor

Significant errors with analysis of EVA Data. Course Concepts are misunderstood or applied incorrectly. Discussion lacks an understanding of course concepts/EVM metrics & measures.

8 to >0 pts

4. Memo and Formatting (5 points)

Superior

Correct format. Neat and free of errors. Professional. Ready to present to the client. Files named appropriately with instruction sheet.

5 to >4 pts

Good

Minor errors to formatting of AR 25-50 format.

4 to >3 pts

Poor

Major errors to AR 25-50 format, or incorrectly followed submission instructions for the assignment.

3 to >0 pts

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-1 17 November 2025

MEMORANDUM FOR Superintendent, United States Military Academy, West Point, NY 10996

SUBJECT: Project Assessment Report, 120-day

1. **BLUF.** Earned value analysis at the 60-day mark showed an unhealthy project that was both behind schedule and over budget. However, by 120 days ample recovery was made, and the project was ahead of schedule and under budget. While we are trending toward a healthy project, we plan to implement control measures which ensure that quality is not being sacrificed to achieve such favorable metrics.

2. **60/90 DAY ASSESSMENT.** We conducted earned value analyses 60 days after the start of the project, and then 90 days. These analyses assessed the health of the project in terms of scope, schedule, and budget.

a. **60 Day EVA.** At 60 days, the project was severely behind schedule and over budget. This is made clear by the Cost Variance and Schedule Variance dumbbell plots in Appendix A, Figures 1 and 2. The schedule performance and cost performance indices (SPI and CPI) were 0.245 and 0.253, respectively. This yielded a cost-schedule index (CSI) of 0.062, which is significantly below the critical ratio of 0.85 clarified by the US Army Corps of Engineers. These drastically low performance measures were caused by the progress on the "Remove Old Fixtures" task. This task has a budget of $18,838,000 and was planned to be 100% complete at 60 days. However, the task was only 20% complete, and $17,500,000 had been spent. This led to an extremely low earned value and high actual cost, prompting a low CPI. The effect of this task is made clear by the "Cost Efficiency By Task" section of the dashboard, showing that at 60 days, this task contributes to 94.3% of the total cost variance. While "Remove Old HVAC System" and "Remove Old Electrical Gear" were also over budget and behind schedule, these activities were not nearly as expensive nor as far behind, making it evident that "Remove Old Fixtures" was the main driving force in returning poor performance measures for schedule and budget. The scope of the project was not altered.

b. **90 Day EVA.** At 90 days, the project was ahead of schedule and just slightly over budget. The SPI and CPI were 1.037 and 0.979, respectively. This yielded a CSI of 1.015, which is well above the critical ratio. The project was ahead of schedule because we started three tasks before their planned start date. Therefore, these tasks had planned values of zero while their earned values totaled over $450,000, thus driving a high SPI. We remained slightly over budget because "Remove Old Fixtures" ended up costing $417,000 more to complete than anticipated, and several other partially completed tasks were on pace to exceed their budget, as shown by "Cost Efficiency By Task" in the dashboard. As a result, actual costs outweighed earned value, yielding the CPI below 1. The scope of the project was not altered.

c. **What Caused Rapid Change?** The metrics assessing cost and schedule of the project experienced significant change between the 60- and 90-day milestones. The scope of the project did not change.

(1) **Cost.** The drastic change in CPI between the 60- and 90-day assessments was due to the most expensive task, “Remove Old Fixtures”. The task was budgeted to cost $18,838,000, and at 60 days we had spent $17,500,000 while only 20% of the work was completed. Therefore, the discrepancy between earned value and actual cost from this single task was strong enough to lower the entire CPI. But, at the 90-day milestone the task was 100% complete, and the total cost was $19,255,000, only $417,000 over budget. The EAC at the 60-day mark estimated that the total cost for the task would be $76,999,833. The problem with this is that it assumes work completion increases at the same rate as cost. Based on the data, we can infer that a strong majority of this task was material costs, and at 60 days the materials had all been acquired, but the work was just yet to be done.

(2) **Schedule.** SPI changed massively between 60 and 90 days also because of the “Remove Old Fixtures” task. Since this task was so expensive and we planned to complete 100% but only did 20%, planned value significantly outweighed earned value and thus lowered the entire SPI. At 90 days when the task reached 100% completion, this was resolved. Furthermore, at 90 days we started three new tasks that were not planned to begin yet. This allowed earned value to get ahead of the planned value of zero and contributed to pushing the SPI above 1.

3. **120-DAY ASSESSMENT.** We conducted an earned value analysis 120 days after the start of the project. This analysis assessed the health of the project in terms of scope, schedule, and budget. Only the cost and schedule are assessed below, as the project scope has not been altered from the Project Charter.

a. **Cost.** At 120 days, the project is ahead of schedule and under budget. The SPI and CPI are 1.097 and 1.011, respectively. This yields a CSI of 1.108, which is well above the critical ratio of 0.85. The earned value exceeds the actual cost by $404,444 – this is the cost variance, and it tells us that we have spent $404,444 less than we allocated to complete the amount of work that we have done. These savings are illustrated in Appendix A, Figure 1 by the dumbbell plot for cost variance. This was caused specifically by major savings from the “Install New Windows” and “Remove Old Ductwork” tasks which saved $368,925 and $364,800, respectively. This is shown in the dashboard under “Cost Efficiency by Task” when 120 days is selected as the assessment to view. This positive cost variance yielded the CPI which measures earned value divided by actual cost.

b. **Schedule.** Earned value exceeds the planned value by $3,383,287.90, as shown by the dumbbell plot in Appendix A, Figure 2. This means that if we quantify the amount of work that we have completed ahead of schedule as a dollar amount, it equals this. This is known as the schedule variance. This high schedule variance stems from the fact that we have made substantial progress on four tasks that were not even scheduled to begin yet, and we have completed another task that was only planned to be 15% complete at 90 days. However, only one of these tasks, “Demo” is on the critical path. This positive schedule variance yielded the SPI which measures earned value divided by planned value. Lastly, to better quantify how many days ahead of schedule we are, we looked at time variance. The time variance is 50.5 days, meaning that we have completed 170.5 days’ worth of work, according to the project plan, in just 120 days. This only considers the critical tasks, so the time variance tells us that we are on track to finish 50.5 days ahead of the project’s scheduled end date. The time variance plot in Appendix A, Figure 2 shows its progression across all assessments.

c. **Changes from 90 Day Milestone and Trending Forward.** Comparing the 90-day assessment with the 120-day, both SPI and CPI have grown, indicating that we have gotten further ahead of schedule and began using our resources more efficiently. The primary cause for the increasing of SPI and time variance between 90 and 120 days was beginning the “Demo” task within this window. Starting this expensive, critical task early put us on pace to finish earlier and increased earned value on a task with a planned value of zero. We are trending toward greater productivity, as evident by the upward trend in the cost and schedule variance plots in the dashboard. The CPI increase can be attributed to completing the entire “Remove Old Ductwork” task and over 50% of the “Install New Windows” task using much less cost than anticipated. This indicates a trend of more efficient usage of resources. This is most clearly illustrated in the dashboard by toggling between the 90 day and 120 days assessments for cost efficiency and schedule variance by task. It clearly shows the change from many tasks being overbudget to only a few, as well as the change from being behind on a critical task at 90 days to being on pace or ahead of all of them at 120 days.

4. **WAY AHEAD.** Based on the results from the three assessments, this section discusses our project team’s future focuses, concerns, and changes to the monitoring process.

a. **Resource Management.** The assessments show that at the beginning, many tasks were behind schedule, but by 120 days all current tasks were either on schedule or ahead, with the exception of scaffolding which remained behind. However, while cost variance by task improved significantly as time progressed, we still had six tasks at the 120-day mark whose actual cost exceeded earned value. This is alarming because it suggests that some working crews may be adding additional resources, or crashing the task, to get back on schedule and resulting in a cost deficit on the task. As a result, monitoring how labor crews manage scarce resources will be a key focus going forward.

b. **Change Control.** Given the high number of tasks which were overbudget at 120 days, we are concerned with not only the management and use of resources, but how potential changes in their management and use affect the scope of the project. To address this concern, we will institute a formal scope change request process by which any request to deviate from the project plan must be signed off by the PM. This process will be recorded in writing, and the project plan and schedule will be immediately updated to reflect the change if accepted by the PM.

c. **Recommendation for Next Assessment.** At the next assessment, we recommend adding a feature to the dashboard that identifies tasks which were drastically behind schedule and then made a rapid recovery. We would then recommend that the crews working on these tasks be consulted so that we can understand what measures they took to make such a recovery and determine if they assumed additional risk or an alternate course of action that does not align with the project plan. While our dashboard may indicate that we are under budget and ahead of schedule, we want to do the due diligence to ensure that we are not sacrificing quality to achieve these metrics.

5. The point of contact for this memorandum is the undersigned at eric.dailey@westpoint.edu

ERIC DAILEY
CDT LT, USCC G4
Project Manager

Encls
1. Appendix A – Dashboard
2. Appendix B – Earned Value Analysis Table

Appendix A - Dashboard

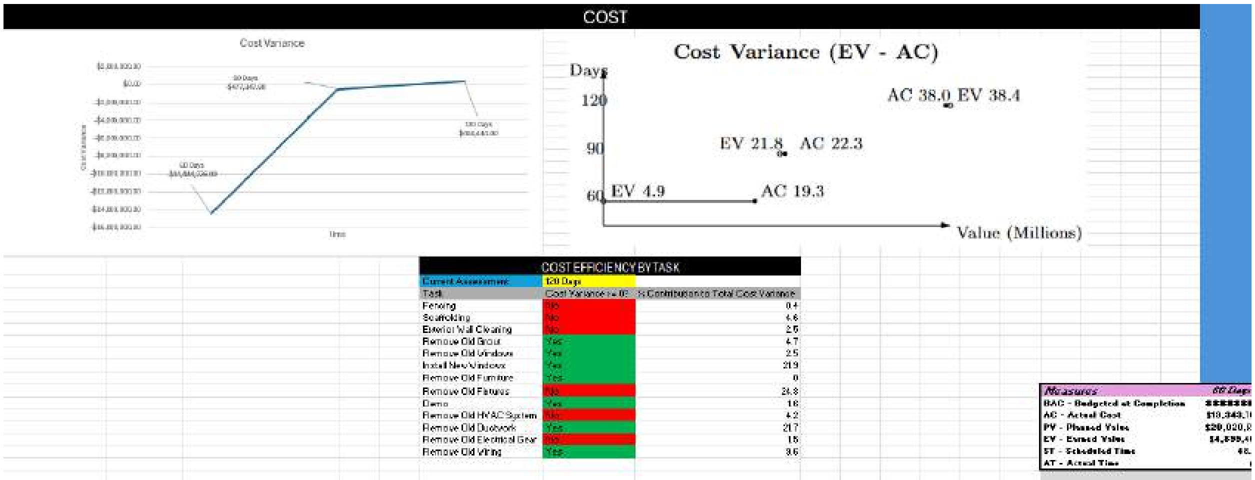


Figure 1: Dashboard (Left half)

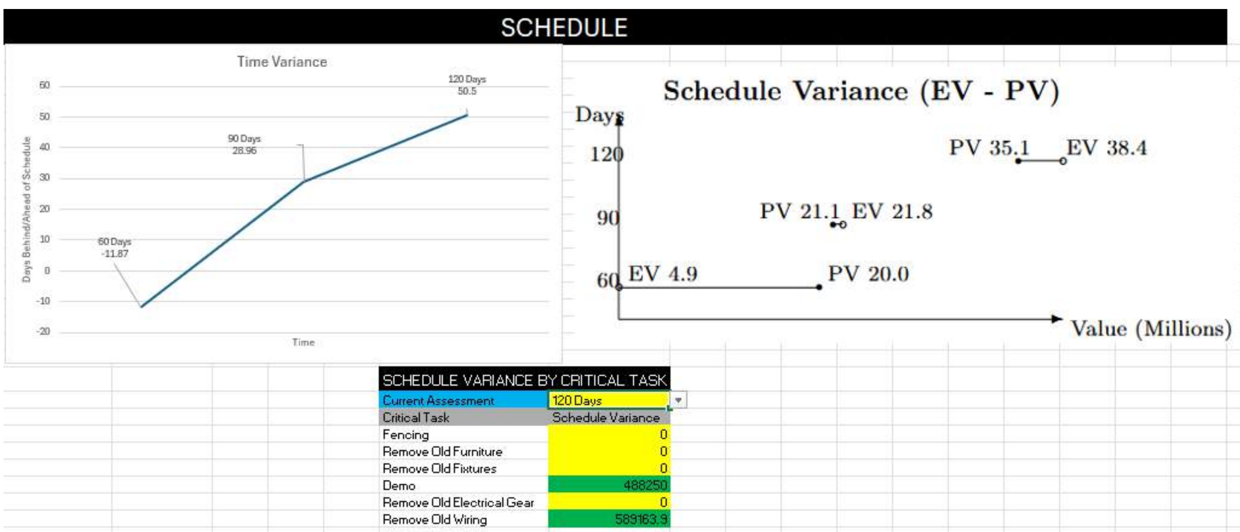


Figure 2: Dashboard (Right half)

Appendix B – Earned Value Analysis Table

Measures	*60 Days*	*90 Days*	120 Days
BAC - Budgeted at Completion	$260,717,320	$260,717,320	$260,717,320
AC - Actual Cost	$19,343,700	$22,301,350	$38,038,500
PV - Planned Value	$20,020,552	$21,048,709	$35,059,656
EV - Earned Value	$4,899,464	$21,824,002	$38,442,944
ST - Scheduled Time	48.13	118.96	170.5
AT - Actual Time	60	90	120

Metrics	*60 Days*	*90 Days*	120 Days
CV - Cost Variance = EV - AC	-14444236	-477347.8	404444
SV - Schedule Variance = EV - PV	-15121088	775293.3	3383287.9
TV - Time Variance = ST - AT	-11.87	28.96	50.5

Indices	*60 Days*	*90 Days*	120 Days
SPI - Schedule Performance Index = EV	0.24	1.04	1.10
CPI - Cost Performance Index = EV/AC	0.25	0.98	1.01
CSI - Cost Schedule Index = SPI*CPI	0.06	1.01	1.10

Completion Metrics	*60 Days*	*90 Days*	120 Days
ETC - Estimate to Complete = (BAC-EV)	$1,010,001,045	$244,118,537	$219,935,910
EAC - Estimate at Completion = ETC+A(	$1,029,344,745	$266,419,887	$257,974,410

Figure 3: Earned Value Analysis Table

Acknowledgement of Assistance

ChatGPT. Assistance given to the author, AI. I used ChatGPT to give me .tex code to make the two dumbbell plots in the dashboard. I prompted iteratively until I got the plot formatted appropriately. OpenAI, (https://chatgpt.com/share/6917fc1e-1c18-8000-9a1a-2fe8d53bd904). West Point, NY, 14NOV2025.

Course Project - Stage Gate #4

Instructions/Rubric/Student Example

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-1 24 Nov 2025

MEMORANDUM FOR RECORD

SUBJECT: Project Stage Gate #4 Instruction Sheet

1. **Purpose.** Project Stage Gate #4 aims to develop an Agile project plan and report on the progress of the newly renovated Thayer Hall room scheduling system.

2. **References.** No change from EM411 Course Instructional Memorandum.

3. **Summary.** ***This Deliverable is worth a total of 75 points. This is an individual event.*** Project Managers will submit their project deliverables **NLT 2359 (EDT) on 10 Dec 2025** via Canvas Assignments. Detailed submission requirements are outlined in paragraphs 5 and 6. One submission per Cadet.

4. **Background.** Imagine the Thayer Hall Renovation Project nearing completion. The Dean wants to ensure that the large new meeting spaces in Thayer Hall have a state-of-the-art scheduling system. Based on your previous experiences with Agile project methodology, OGRE has once again been hired to spearhead the Agile development of the project plan. Previous experience also suggests that with your team size and the type of project, you should be able to plan completion in one 4-week sprint. The OGRE team assigned to this project consists of three developers, and you serve as both the Product Owner and Scrum Master.

5. **Stage Gate Instructions.** PMs will conduct Agile Project Planning, Sprint Execution, report on Agile Metrics, and produce a ***Memorandum for Record*** detailing their work.

a. **Agile Project Planning.** Develop a product backlog with at least 4 user stories for the room scheduling software based on your understanding of the Dean's needs/wants/desires. At least 2 tasks for development should accompany each user story. Use an estimating technique (like the Bucket System) to determine the amount of story points for each user story. Plan and schedule your tasks (as many as are reasonable based on your team's 4-week sprint velocity of 100 points/sprint) using a Scrum Board. Assume a 4-week sprint that begins on *1 October 2027* (when Thayer Hall renovations are nearly complete). Prioritize and allocate the user stories to this sprint within your Backlog. Provide an initial estimate of the project timeline.

b. **Sprint Execution.** Update your Scrum Board for tracking tasks. Allocate user stories to your Scrum team using a Scrum Board. Document the completion of user stories during the sprint on your Scrum Board (e.g., moving Stories and associated Tasks to the 'Done' portion of the Board). Imagine you are now at the end of your planned Sprint (completing 4-weeks of

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-1 24 November 2025

MEMORANDUM FOR OGRE, United States Military Academy, West Point, NY 10996

SUBJECT: Interview with Dean on Thayer Hall Room Scheduling System

1. **Official Transcript of Interview:**

Interviewer: Project Manager
Interviewee: Dean of West Point

Interviewer: Good day, Dean. Thank you for taking the time to meet with us. We are here to discuss the development of a new scheduling software system for Thayer Hall. To begin, please provide an overview of the main objectives and requirements for this software system.

Dean: Certainly. The primary objective is to ensure efficient and effective utilization of Thayer Hall's resources while maintaining the highest standards of organization and security. The new scheduling software system should accomplish the following:

Centralized Scheduling: We need a system that can centrally manage and schedule classrooms, lecture halls, and other spaces within Thayer Hall. This is critical for both academic and administrative functions.

User-Friendly Interface: The system should be user-friendly, allowing staff and faculty to request and manage bookings easily. It should also provide a straightforward way for students to check room availability.

Access Control: Security is paramount. We require strict access control, allowing only authorized personnel to make reservations and ensuring that the system's data is kept secure.

Integration: The system should seamlessly integrate with other West Point systems, such as the academic calendar and faculty schedules, to avoid conflicts and ensure accurate scheduling.

Real-Time Updates: To minimize scheduling conflicts, real-time updates to room availability should be available to users.

Reporting and Analytics: The system should provide robust reporting and analytics capabilities, allowing us to evaluate room usage and make informed decisions for future resource allocation.

Interviewer: Thank you for outlining these key requirements. Could you also specify any specific challenges or pain points you've experienced with the current scheduling process?

Dean: The current scheduling process is time-consuming, manual, and often prone to errors. We frequently encounter scheduling conflicts, leading to disruptions in classes and events. Additionally, the lack of integration with other West Point systems results in inefficient resource allocation.

Interviewer: Understood. To wrap up, please highlight the top priorities or features that you consider most critical for this scheduling software system.

Dean: Certainly. Our top priorities are as follows:

Reliability: The system must be highly reliable, with minimal downtime or disruptions.
Security: Strict access control and data security measures are non-negotiable.
Integration: Seamless integration with our existing systems is a must.
User Experience: It should be intuitive and user-friendly to encourage adoption.
Scalability: The system should accommodate future growth and additional buildings on campus.

Interviewer: Thank you, Dean, for providing valuable insights into the requirements and priorities for the new scheduling software system. This information will guide the project team in creating user stories and ensuring that the system meets West Point's needs.

2. The point of contact for this memorandum is the undersigned at daniel.finch@westpoint.edu.

DANIEL FINCH
Senior Instructor
EM411 Course Director

work). Assume your team could only complete approximately 80% of your planned tasks due to unforeseen circumstances, a global supply chain (chip) disruption, and technical difficulties.

c. **Agile Metrics and Progress Tracking.** Create and maintain a simple burndown chart using Excel to track the progress of your team's efforts. Calculate and report the team's current sprint velocity. Assume the previous two 4-week sprints you and your team worked on before this project had total story points completed as 105 and 95, respectively.

d. **Memorandum.** Provide the Dean with an overview of your Agile project in a short memorandum (2-3 pages). Ensure you've summarized the key aspects of each portion of your plan and execution, as noted in Paragraph 5, and answer all questions discussed in Annex B. The memorandum should be written in the Army Memorandum Format (AR 25-50). Annex B is provided as a template for developing your memorandum.

6. **Submission requirements.** Review the assignment rubric posted on Canvas. Project Stage Gate #4 is **due NLT 2359 (EDT) on 10 Dec 2025 via Canvas** and will consist of the following deliverables:

a. Memorandum. See Annex B for details on the format and content requirements.

b. Product Backlog, Scrum Board, and Sprint Burndown Chart as an Annex or Appendix.

c. Ensure a cover sheet and works cited are submitted with your memorandum.

7. **List of Supporting Files.**

a. Annex A, Interview with Dean (docx)

b. Annex B, Memorandum Template (docx)

8. The Point of Contact for this memorandum is the undersigned.

DANIEL FINCH
Senior Instructor
EM411 Course Director

SG#4 Grading Rubric (1)

Agile Planning

Superior

Complete answer with all work shown. 4 user stories shown with at 2 or more tasks. Approximately 100 story points planned for the 4 week sprint. Stories and tasks written in correct format, and numbered accordingly. Complete product backlog submitted with all stories and tasks listed.

30 to >26 pts

Good

Complete answer with work shown. 4 user stories shown with at 2 or more tasks. Approximately 100 story points planned for the 4 week sprint. Stories and tasks written in correct format, and numbered accordingly. Complete product backlog submitted with all stories and tasks listed. Some minor errors in the above but largely good.

26 to >23 pts

Okay

Mostly complete answer with work shown. 4 user stories shown with at at least 2 tasks each. Approximately 100 story points planned for the 4 week sprint. Some larger errors in the above with formatting or understanding. Product backlog may have errors.

23 to >20 pts

Poor

Incomplete answer with little or no work shown. May not have all 4 stories and tasks. Estimation incomplete or incorrectly done. Stories and tasks written in wrong format, and numbered incorrectly or missing. Product backlog submitted with major errors. Some significant errors in the above.

20 to >0 pts

Sprint Execution

Superior

Complete answer with all work shown. Scrum task board present showing all tasks in an Appendix. ~80% of tasks should be shown in the 'Done' portion of board. Tasks named/labeled/numbered correctly.

15 to >13 pts

Good

Complete answer with most work shown. Scrum task board present showing all tasks in an Appendix. ~80% of tasks should be shown in the 'Done' portion of board. Tasks named/labeled/numbered correctly. Minor errors to the above.

13 to >12 pts

Okay

Mostly complete answer with work shown. Scrum task board present showing all tasks in an Appendix. ~80% of tasks should be shown in the 'Done' portion of board. Tasks named/labeled/numbered correctly. Some larger errors to aspects of the above,

12 to >10 pts

Poor

Incomplete complete answer with some or no work shown. Scrum task board missing or incomplete showing tasks. Tasks named/labeled/numbered incorrectly. Some major errors to aspects of the above,

10 to >0 pts

Agile Metrics and Tracking

Superior

Complete answer with all work shown and interpretation justified. Project burndown chart present with 3 sprints. Correctly identified total product backlog story points. Discussion of most updated sprint velocity.

15 to >13 pts

Good

Complete answer with most work shown and interpretation correct. Project burndown chart present with 3 sprints. Correctly identified total product backlog story points. Discussion of most updated sprint velocity. Some minor errors to above,

13 to >12 pts

Okay

Mostly complete answer with work shown and interpretation justified. Project burndown chart present with 3 sprints. Correctly identified total product backlog story points. Discussion of most updated sprint velocity. Some larger errors to aspects of the above.

12 to >10 pts

Poor

Incomplete answer with all some or no work shown and interpretation incorrect. Project burndown chart done incorrectly. Total product backlog story point misidentified. Missing or incomplete discussion of most updated sprint velocity. Major errors to the above.

10 to >0 pts

Memorandum

Superior

Each of the 4 areas in the MFR was correctly summarized and addressed. BLUF clearly stated and has an important takeaway for the Dean. Future sprint forecast correct and way ahead is clear.

10 to >9 pts

Good

Each of the 4 areas in the MFR was correctly summarized and addressed. BLUF stated and has an important takeaway for the Dean. The future sprint forecast is correct, and the way ahead is clear. Some minor errors to the memo or clarity.

9 to >7 pts

Okay

MFR was summarized and addressed. BLUF stated for the Dean. The future sprint forecast is present, and the way ahead is there. Some major errors to the memo or clarity.

7 to >5 pts

Poor

4 areas in the MFR was improperly summarized and addressed. BLUF missing or inaccurate. The future sprint forecast is incorrect or missing, and the way ahead is unclear. Some significant errors to the memo or clarity.

5 to >0 pts

Formatting

Superior

Neat and free of errors, Professional, Ready to present to the client. Cover sheet, works cited present. No spelling errors and free of major grammatical issues. AR25-50 format is flawless.

5 to >4 pts

Okay

Mostly neat and free of errors, Professional, Ready to present to the client. Cover sheet, works cited present. No spelling errors and free of most grammatical issues. Minor AR25-50 format issues.

4 to >3 pts

Poor

Multiple errors, not professional, Would not be ready to present to the client. The cover sheet and works cited may be missing. Spelling errors and grammatical issues throughout. Major AR25-50 format issues. Appendices or annexes improperly formatted, or files named incorrectly.

3 to >0 pts

DEPARTMENT OF THE ARMY
DEPARTMENT OF SYSTEMS ENGINEERING
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

EM411 AT 26-1 10 December 2025

MEMORANDUM FOR Dean, United States Military Academy, West Point, NY 10996

SUBJECT: Agile Project Report

1. **BLUF.** In order to implement the new scheduling system, a 4-week sprint with 8 tasks focusing on simplicity, adaptability, security, and functionality was created. When they were organized and planned throughout the 4-week time period, 84% of these tasks would be completed. After evaluating the performance, it was determined that it would take an additional sprint consisting of 5 days to complete this iteration of the development of the scheduling system.

2. **Agile Project Plan.** In the agile planning process, 4 user stories were created considering the needs of the Dean in his interview provided in Annex A. The user stories were all created from the perspective of an administrator, primarily focusing on simplicity, adaptability, security, and functionality. The specific stories can be found below in the Product Backlog (Appendix A). Each of these user stories are divided into 2 tasks, creating a total of 8 tasks for the entire sprint. The specific tasks for each story can also be found below in Appendix A.

a. There is a total of 100 points allocated to this sprint, they were balanced using the Fibonacci Sequence method. This method is done by weighing the tasks in terms of difficulty against each other. This resulted in the point values ranging from 18 to 6 points. The difficulty of tasks was determined by considering the level of expertise needed to perform the tasks and the additional materials needed. The story point estimates can be found below in Appendix A.

b. With the Product Backlog complete, a Scrum Board was created using the information from the Product Backlog. The Scrum Board is divided into four weeks, with the status of each of the eight tasks at the end of each week logged. At the end of the 1st and second weeks, no tasks were completed, but many were in review. By week 3, both tasks for story 1 were completed and all other tasks were either in progress or in review. At the end of week 4, and consequently the end of the sprint, all tasks were completed except for task 2.1 – Enable scheduling synchronization across system (16). This task still being in review puts the sprint completion at 84%, which was slightly greater than the projected 80% completion. This scrum board displays a spring that has a slow start, with many tasks starting but taking time to complete. However, by week 3, the completion rate increases significantly, enabling a majority of the work to get complete by the end of week 4. The full scrum board can be found below in Appendix B.

3. **Sprint Execution and Agile Metrics.** As stated previously, the Scrum Board (Appendix B) displays a project that is only 84% of the work by the end of the sprint. This 84% was the

result of task 2.1 (Enable scheduling and synchronization across system) still being in review. Task 2.1 is part of story 2 and is worth 16 story points. To determine how many additional days it will take to complete the sprint, a sprint burndown chart (Appendix C) was created using the data from this most recent sprint, and two previous sprints weighted at 105 and 95 story points – putting the total story points for the entire project at 300. The chart itself displayed a project that is overall right on schedule, primarily in sprints 1 and 2. However, productivity significantly decreases with sprint 3, as it only completed 84% of the expected story points. However, these points will likely be completed very early into the 4th sprint as the task is already in review.

a. The sprint burndown chart (Appendix C) calculated the sprint velocity at 94.667 points per sprint. This is significantly larger than the 16 points needed. Dividing 16 by the spring velocity, it was found that it will take 0.169 additional sprints to complete the final task. This means that it will take approximately 4-5 days to complete the final task.

4. **Way Ahead.** Going forward, the project will take approximately 4-5 more days to complete the final task. This means that the project needs one smaller sprint in order to finish. During this sprint the focus should be on completing task 2.1 and ensuring that there are no issues with the final product. There will be no changes to the plan as there is only one task that needs to be completed, and it is already in the final phase before completion. For the next milestone, the focus should be on ensuring that the final product is ready for presentation and implementation. Ideally the final phase would begin with training and end with the implementation phase.

5. The point of contact for this memorandum is the undersigned at georgia.brownfield@westpoint.edu.

Georgia Brownfield
Georgia Brownfield
Cadet LT, Company D3
Project Manager

EM411 AT 26-1
SUBJECT: Agile Project Report

Encls
1. Appendix A – Product Backlog

Product Backlog						
Story/Task ID	User Story or Task	Story Point Estimate	Priority	Status	Assigned to Sprint	Sprint #
1	As an administrator I want a simplistic scheduleing system so that room booking issues are minimized.	22	Medium	Not Started	Yes	3
1.1	Design a user-friendly interface for clear navigation.	10	Medium	Not Started		
1.2	Implement double-booking confliction detection.	12	Medium	Not Started		
2	an administrator I want an adaptive product that updates across all systems so that accuracy is maximiz	28	Medium	Not Started	Yes	3
2.1	Enable scheduling synchronization across the building's system.	16	Medium	Not Started		
2.2	Integrate automatic notifications for booking changes.	12	Medium	Not Started		
3	As an administrator I want a highly secure system so that no outside threats can alter it.	26	High	Not Started	Yes	3
3.1	Establish role-based access controls.	14	High	Not Started		
3.2	Encrypt scheduling data.	12	High	Not Started		
4	As an administrator I want to integrate this system so that there is continuity across all USMA buildings	24	Medium	Not Started	Yes	3
4.1	Merge building room directories into a central database.	18	Medium	Not Started		
4.2	Train staff in each building on system use.	6	Medium	Not Started		

2. Appendix B – Scrum Board

To Do	In Progress	In Review	Done
		[1.1] Design user-friendly interface (10)	
		[1.2] Implement double booking confliction detection (12)	
	[2.1] Enable scheduling synchronization across system (16)		
[2.2] Integrate automatic notification system (12)			
[3.1] Establish role-based access controls (14)			
		[3.2] Encrypt Data (12)	
	[4.1] Merge building directories into database (18)		
[4.2] Train staff in each building (6)			

To Do	In Progress	In Review	Done
		[1.1] Design user-friendly interface (10)	
		[1.2] Implement double booking confliction detection (12)	
	[2.1] Enable scheduling synchronization across system (16)		
	[2.2] Integrate automatic notification system (12)		
		[3.1] Establish role-based access controls (14)	
	[3.2] Encrypt Data (12)		
	[4.1] Merge building directories into database (18)		
[4.2] Train staff in each building (6)			

To Do	In Progress	In Review	Done
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			[1.2] Implement double booking confliction detection (12)
	[2.1] Enable scheduling synchronization across system (16)		
		[2.2] Integrate automatic notification system (12)	
		[3.1] Establish role-based access controls (14)	
		[3.2] Encrypt Data (12)	
		[4.1] Merge building directories into database (18)	
	[4.2] Train staff in each building (6)		

To Do	In Progress	In Review	Done
			[1.1] Design user-friendly interface (10)
			[1.2] Implement double booking confliction detection (12)
		[2.1] Enable scheduling synchronization across system (16)	
			[2.2] Integrate automatic notification system (12)
			[3.1] Establish role-based access controls (14)
			[3.2] Encrypt Data (12)
			[4.1] Merge building directories into database (18)
			[4.2] Train staff in each building (6)

3. Appendix C – Sprint Burndown Chart

Total Story Points in Product Backlog	**300**	
Actual Story Points CompletedWork	**284**	
Sprint Iteration	Story Points Completed	Story Points Remaining
1	105	195
2	95	100
3	84	16

Overall Sprint Velocity	94.66667
Required Sprints Remaining	0.169014